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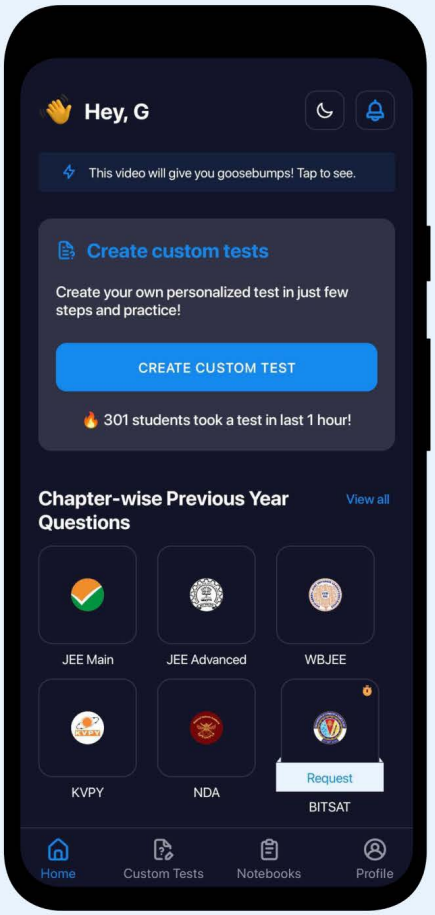

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RAY OPTICS

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1. INTRODUCTION

The branch of physics called **optics** deals with the behaviour of light and other electromagnetic waves. Light is the principal means by which we gain knowledge of the world. Consequently the nature of light has been the source of one of the longest debates in the history of science.

Electromagnetic radiation with wavelengths in the range of about 4000 Å to 7000 Å, to which eye is sensitive is called light.

Our investigation of light will revolve around two questions of fundamental importance (1) What is the nature of light and (2) How does it behave under various circumstances ? The answers to these two questions can be found in Maxwell's field equations (which is out of JEE syllabus). These equations predict the existence of electromagnetic waves that travel at the speed of light. They also describe how these waves behave. Interestingly, not all light phenomena can be explained by Maxwell's theory. Experiments performed at the beginning of this century showed that light also has corpuscular, or particle like properties.

We investigate the behaviour of a beam of light when it encounters simple optical devices like mirrors, lenses and apertures. Under many circumstances, the wavelength of light is negligible compared with the dimensions of the device as in the case of ordinary mirrors and lenses. A light beam can then be treated as a ray whose propagation is governed by simple geometric rules. The part of optics that deals with such phenomena is known as **geometric optics.** However, if the wavelength is not negligible compared with the dimensions of the device (for example a very narrow slit), the ray approximation becomes invalid and we have to examine the behaviour of light in terms of its wave properties. This study is known as **physical optics.**

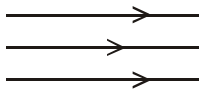
2. RAY

A ray can be defined as an imaginary line drawn in the direction in which light is travelling. Light behaves as a stream of energy propagated along the direction of rays. The rays are directed outward from the source of light in straight lines.

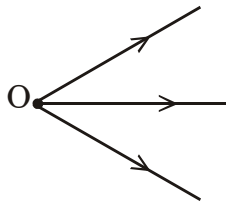
2.1 Beam of Light

A beam of light is a collection of these rays. There are mainly three types of beams.

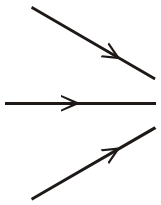
(i) **Parallel beam of light :** A search light and the headlight of a vehicle emit a parallel beam of light. The source of light at a very large distance like sun effectively gives a parallel beam.



(ii) **Divergent beam of light :** The rays going out from a point source generally form a divergent beam.



(iii) **Convergent beam of light :** A beam light that is going to meet (or convery) at a point is known as a convergent beam. A parallel beam of light of passing through a convex lens become convergent beam.

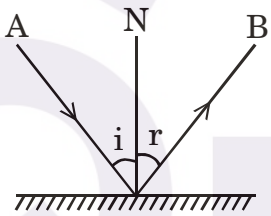


3. REFLECTION

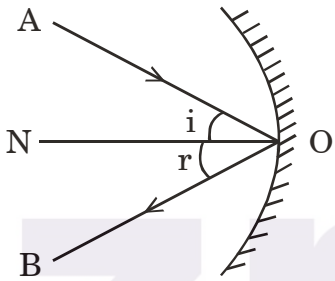
When a ray of light is incident at a point on the surface, the surface throws partly or wholly the incident energy back into the medium of incidence. This phenomenon is called reflection.

Surface that causes reflection are known as mirrors or reflectors.

3.1 Important terms related to reflection :



Plane Mirror

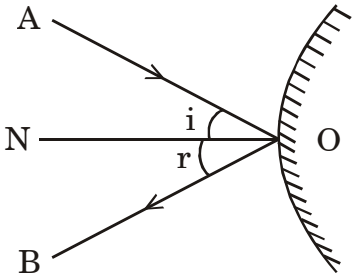


Concave Mirror

Angle of incidence : Angle which the incident ray makes with the normal at the point of incidence. (denoted by i)

Angle of reflection : The angle which the reflected ray makes with the plane normal of the point of incidence. (denoted by r).

Glancing Angle : The angle which the incident ray makes with the plane reflecting surface is called glancing angle. It is generally denoted by g. g = 90 – i.



Convex Mirror

3.2 Law of Reflection

(i) The incident ray, the reflected ray and the normal to the reflected surface at the point of incidence lie in same plane

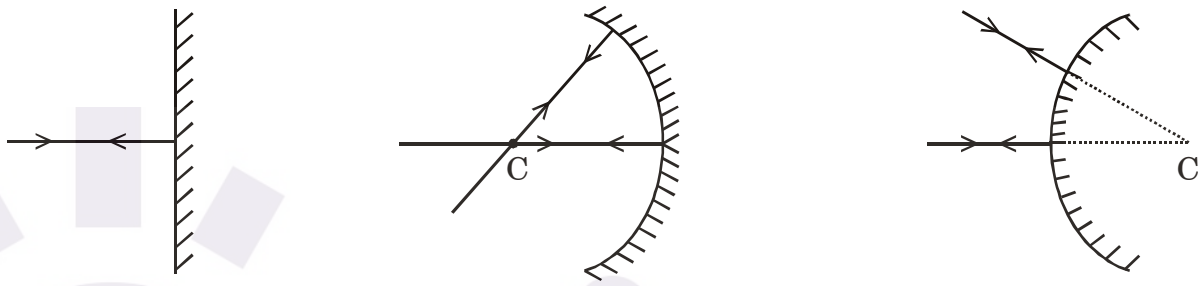
(ii) The angle of incidence is equal to the angle reflection, i.e. i = r. These laws hold for all reflecting surfaces either plane or curved.

Question: Do incident ray and reflected ray differ in terms of frequency, wavelength and speed ?

Solution : None of frequency, wavelength and speed changes due to reflection. However, intensity and have amplitude ($I \propto A^2$) usually decreases. However in ray optics we do not consider this change.

Question : What will happen if the ray falls normal to the surface of the mirror ?

Solution : In this case, when i = 0, the ray retraces its path. i.e. r = 0.



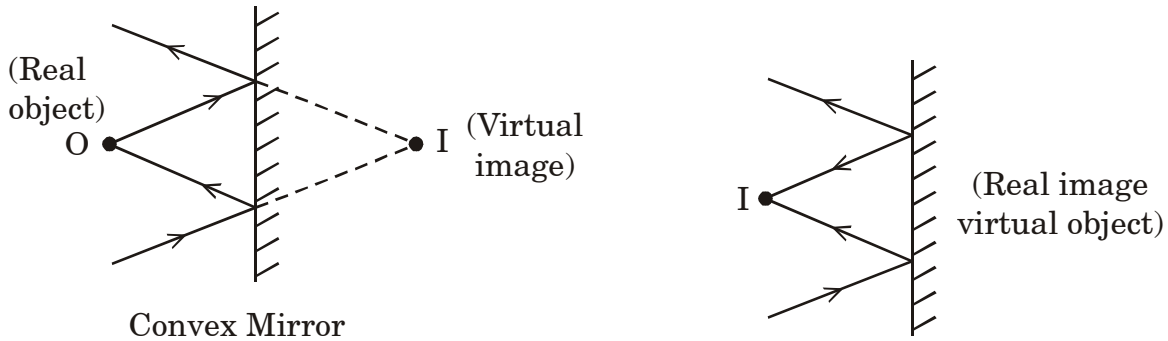
Plane Mirror Concave Mirror Convex Mirror

3.3 Real / Virtual Object (Image) :

When a group of incident rays strike a surface, the point from which they diverge or the point a which they try to converge is known as **object.** After reflection we get a group of reflected rays. The point where reflected rays converge or the point from which the reflected rays appear to diverge is known as the **image** of the object.

Objects and images may be real or virtual. If the light, after reflection, actually passes through the point where the image is located, the image is called as a **real image**; if the light only appears to come from the point where the image is located, the image is called a **virtual image.** A real image will appear on a screen placed at its position while a virtual image will not.

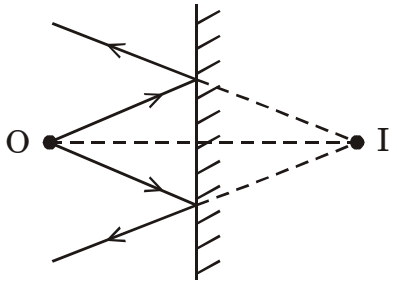
Similarly, if the object is a point where the light energy is actually present, it is a **real object.** If the light rays only tend to go towards the point to here the image is located, it is a **virtual object.**

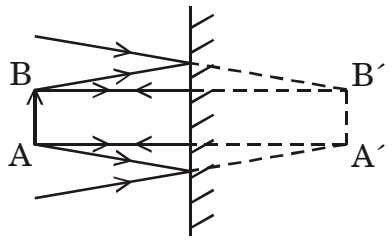


Convex Mirror

3.4 Reflection from a plane surface (or plane mirror)

Almost everybody is familiar with the image formed by a plane mirror. If the object is real, the image formed by a plane mirror is virtual, erect, of same size and at the same distance from the mirror. The ray diagram of the image of a point object and of an extended object is as shown below.

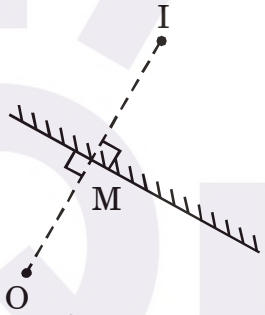




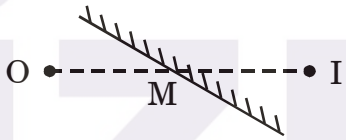
{Here AB is the object and A′B′ is the image formed.}

Important points :

1. To find the location of image of an object from an inclined plane mirror, you have to see the perpendicular distance of the object from the mirror.

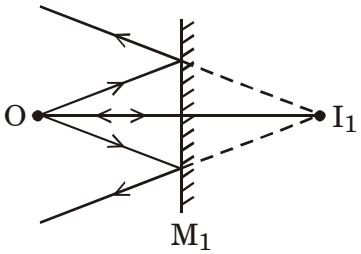


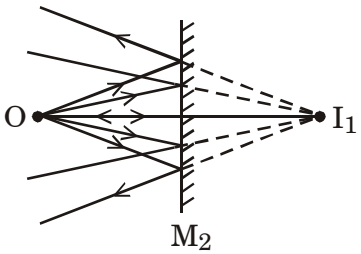
Correct
OM = MI



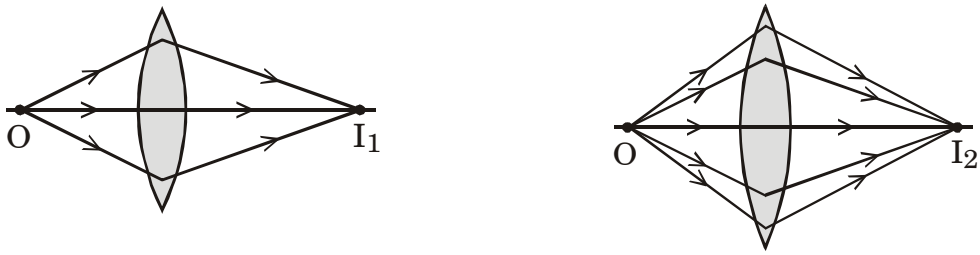
Wrong
OM = MI

2. Suppose an object O is placed in front of two plane mirrors M_1 and M_2. Size of M_2 is more than that of M_1. In this case the intensity of the images formed by M_2 (i.e., I_2) will be more than that formed by M_1 (i.e. I_1). This is because I_2 is formed from more number of reflected



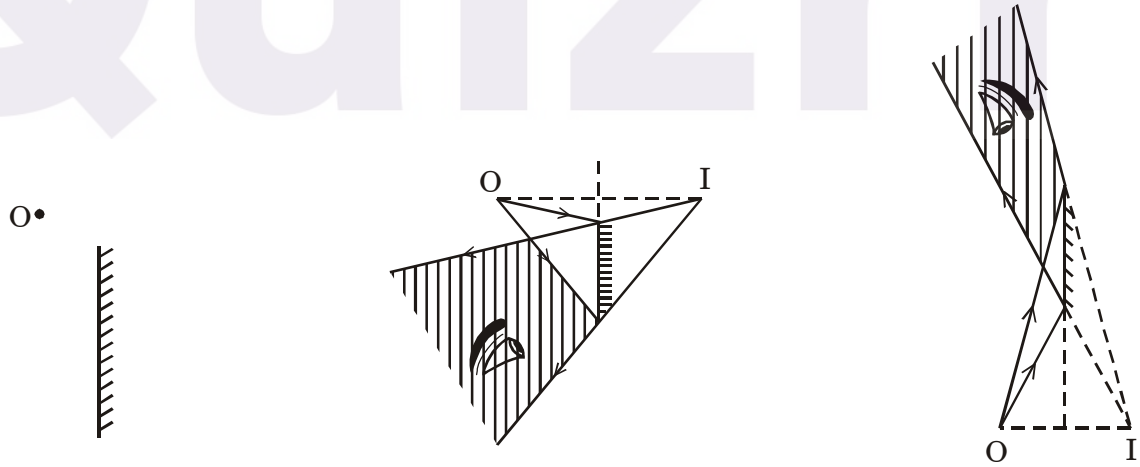


rays. Or we can say it is formed from more light. The same is the case with an image formed by a lens of large aperture.



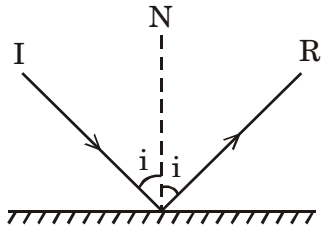
In the figure shown, I_2 will be more intense. This is the reason why we choose a lens of large aperture in telescopes to see the distant objects. You might have heard that the strength of an observatory is measured from the aperture of its lens. Now, you can think if this is the case then why don't we take a lens of an aperture as large as we wish. Actually some technical problems arise in casting a lens of large aperture. So we have some limitations in doing so.

3. **Field of view of an object for a given mirror :** Suppose a point object O is placed in front of a small mirror as shown in Fig. (a), then a question arises in mind whether this mirror will form the image of this object or not or suppose an elephant is standing in front of a small mirror, will the mirror form the image of the elephant or not. The answer is yes, it will form. A mirror whatever may be the size of it forms the images of all objects lying in front of it. But every object has its own field of view for the given mirror. The field of view is the region between the extreme reflected rays and depends on the location of the object in front of the mirror. If our eye lies in the field of view then only we can see the image of the object otherwise not. The field of view of an object placed at different locations in front of a plane mirror are shown in Fig. (b) and (c).

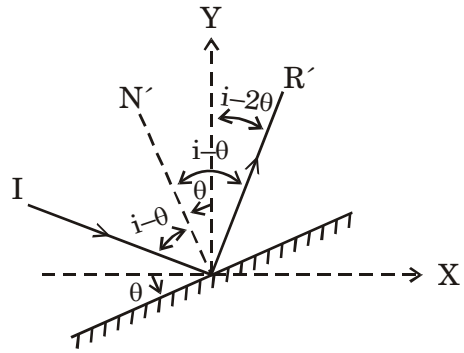


(a) (b) (c)

4. Suppose a mirror is rotated by an angle θ (say anticlockwise), keeping the incident ray fixed then the reflected ray rotates by 2θ along the same sense, i.e., anticlockwise.



(a)



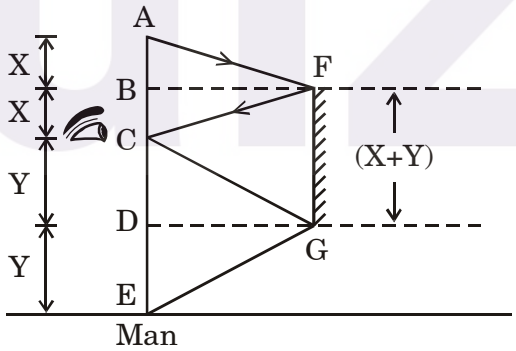
(b)

In figure (a), I is the incident ray, N the normal and R the reflected ray.

In figure (b), I remains as it is. N and R shift to N´ and R´.

From the two figures we can see that earlier the reflected ray makes an angle i with y-axis while after rotating the mirror it makes the angle i – 2θ. Thus, we may conclude that the reflected ray has been rotated by angle 2θ.

5. The minimum length of a plane mirror to see one's full height in it is $\frac{H}{2}$, where H is the height of man. But the mirror (F) reaches the eye at C. Similarly the ray starting from the foot (E) after reflecting from the lower end (G) also reaches the eye at C. In two similar triangles ABF and BFC, AB = BC = x (say), Similarly in triangles CDG and DGE, CD = DE = y (say)



Now, we can see that height of the man is 2 (x + y) and that of mirror is (x + y), i.e., height of the mirror is half the height of the man.

Note : The mirror can be placed anywhere between the centre line BF (of AC) and DG (of CE).

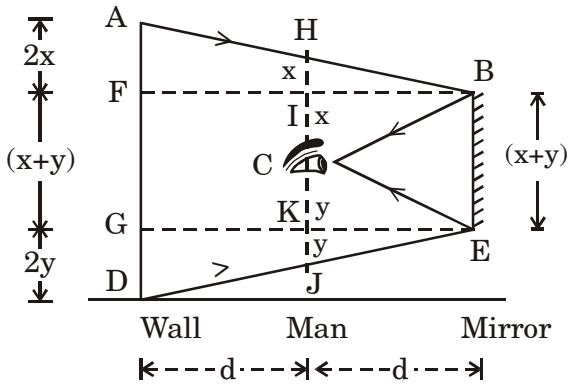
6. A man is standing exactly at midway between a wall and a mirror and he want to see the full height of the wall (behind him) in a plane mirror (in front of him). The minimum length of mirror in this case should be $\frac{H}{3}$, where H is the height of wall. The ray diagram in this case is drawn in Fig. below.

In triangles HBI and IBC, HI = IC = x (say), Now, in triangles HBI and ABF,

$$\frac{AF}{HI} = \frac{FB}{BI}$$

or $\frac{AF}{x} = \frac{2d}{d}$

or AF = 2x



Similarly we can prove that DG = 2y if, CK = KJ = y

Now, we can see that height of the wall is 3 (x + y) while that of the mirror is (x + y).

7. Number of images formed between two mirrors inclined at an angle θ with each other is given by :

$$m = \frac{360}{\theta}$$

Case I : if m is even then n (number of images) = m – 1.

Case II : if m is odd, then n (number of images) = m – 1. {if object is on angle bisector}

OR

n = m {other wise}

Case III : if m is rational (fractional) then, n = [m] {where [•] is the greatest integer function}

Example 1

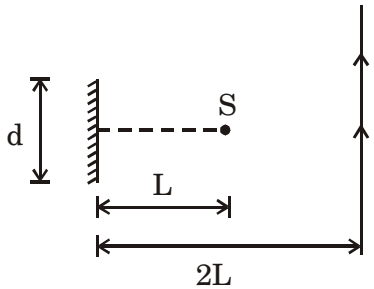
A point source of light S, placed at a distance L in front of the centre of a mirror of width d, hangs vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance 2L from it as shown. The greatest distance over which he can see the image of the light source in the mirror is

(a) d/**2**

(b) d

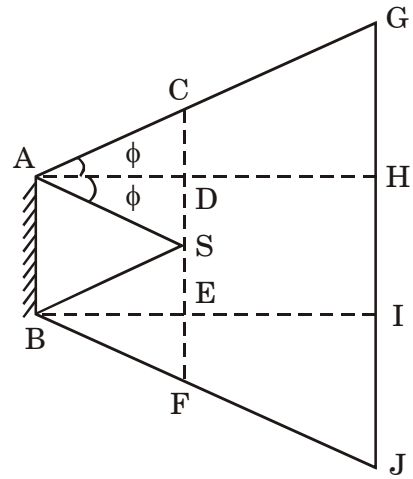
(c) **2**d

(d) **3**d



Solution :

(d) The ray diagram will be as shown in Fig.



$$HI = AB = d$$

$$DS = CD = \frac{d}{2}$$

Since, $AH = 2AD$

$\therefore$ $GH = 2CD = 2\frac{d}{2} = d$

Similarly $IJ = d$

$\therefore$ $GJ = GH + HI + IJ$

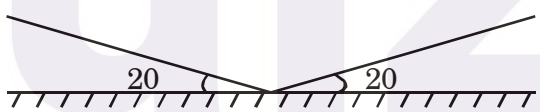
$= d + d + d = 3d$

Example 2

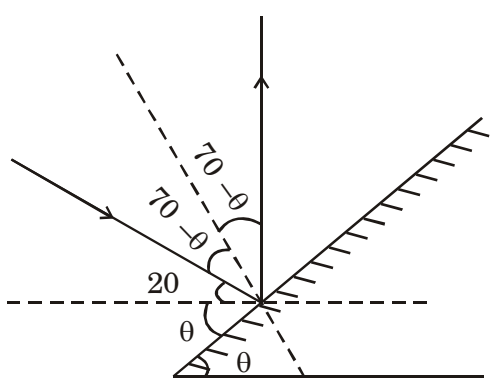
At ray of light is travelling at an angle of 20 below horizontal. At what angle with horizontal must a plane mirror be placed in its path so that it becomes vertically upwards after reflection ?

Solution :

Let us first place the mirror horizontally. The reflected ray now goes at an angle of 20 above horizontal as shown.



Now, for the reflected rays to be vertical, let us assume that the mirror makes an angle θ with the horizontal.



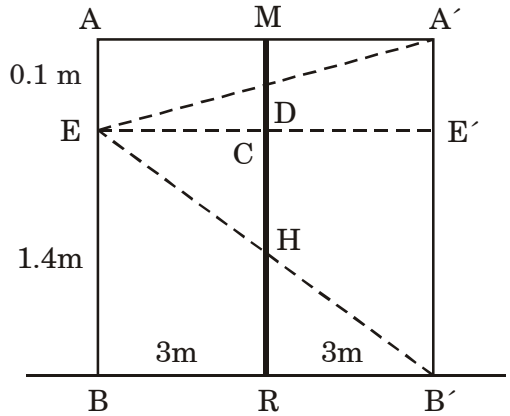
Now, from the figure, if reflected ray is vertical, then $20 + (70 - \theta) + (70 - \theta) = 90$

$\Rightarrow 70 = 2\theta \quad \Rightarrow \quad \theta = 35$

Example 3

A boy is 1.5 m tall and just sees his image in a vertical plane mirror 3 m away. His eyes are 1.4 m from the floor level. Determine the vertical dimension of the mirror.

Solution :

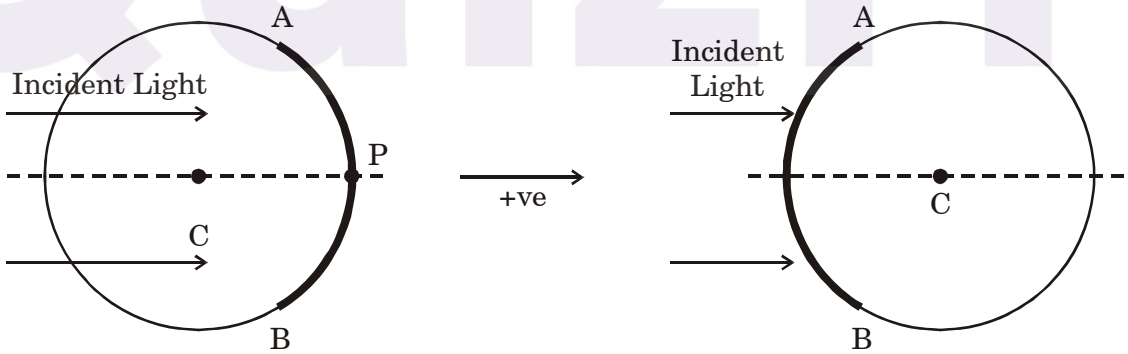


Since a plane mirror forms an image at a distance same as that of object, hence image will be formed at 3m behind the mirror. Now let AB be the height of boy, the image formed is AB'. Now, triangles $AA'E$ and AEE' are congruent, also MD = DC = 0.05 m. Similarly, CH = HR = $\frac{1.4}{2}$ = 0.70 m. Hence, height vertical dimension is DC + CH = 75 cm.

3.5 Reflection at Spherical Mirrors

(a) **Some important definitions :**

(i) **Spherical mirrors :** A spherical mirror is a part of follow sphere or a spherical surface. They are classified as **concave** or **convex** according to the reflecting surface being concave or convex respectively.



(a) Concave Mirror (b) Convex mirror

(ii) **Centre of currature :** The centre C of the sphere of which mirror is a part is called the centre of curvature of the mirror.

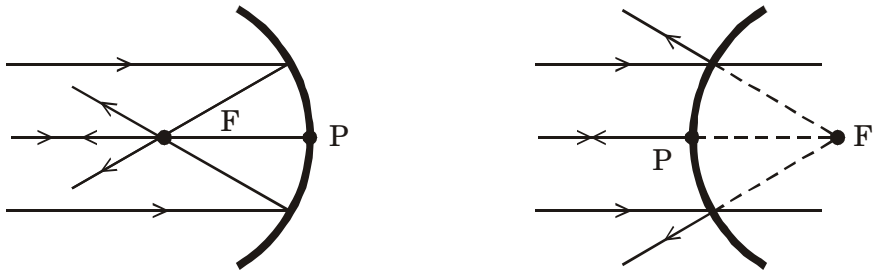
(iii) **Pole or Vertex :** P, the centre of the mirror surface is the pole.

(iv) **Radius of curvature (R) :** The distance CP is called the radius of curvature.

Note : All distances are measured from point P. We can see using the rules of sign convention given earlier that R is positive for convex mirror and negative for concave mirror.

(v) **Principal axis :** The line CP joining the pole and the centre of curvature of the spherical mirror is called the **principal axis.**

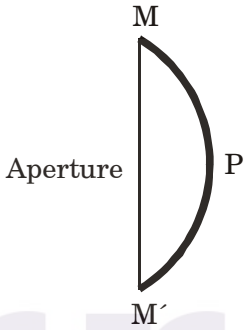
(vi) **Focus (F) :** If a parallel beam of rays, parallel to the principal axis and close to it, is incident on a spherical mirror; the reflected rays converge to a point F (in case of a concave mirror) or appear to diverge from a point F (in case of convex mirror) on the principal axis. The point F is called the flows of the spherical mirror.



Concave mirror Convex mirror

(vii) **Focal length (F) :** Focal length is the distance PF between the pole P and flows F along the principal axis.

(viii) **Aperture :** The line joining the end points of a spherical mirror is called the operature or linear operature.



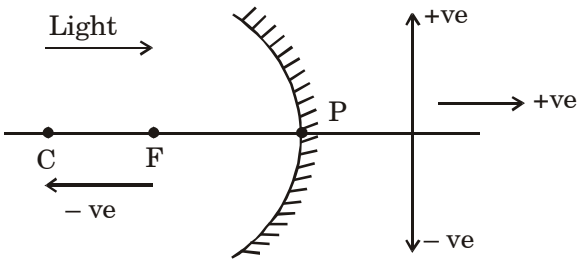
SIGN CONVENTION

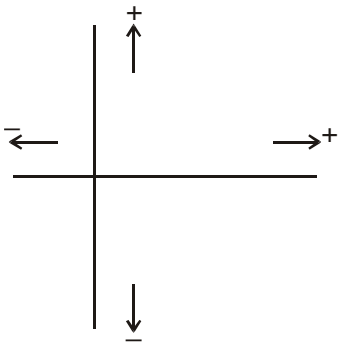
Although a number of books follow different sign conventions, but we will be using the following sign conventions :

(1) Whenever and wherever possible, the ray of light is taken to travel from left to right.

(2) The distances above principal axis are taken to be positive while below it is negative.

(3) Along the principal axis, distances are measured from the pole and in the direction of light are taken to be –ve if in front of the mirror and +ve if behind it.





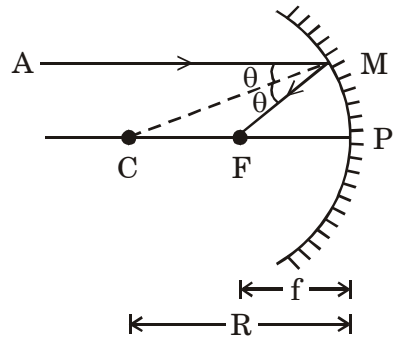
3.5.1 Paraxial Rays :

Rays which are close to the principal axis and make small angles with it, i.e., they are nearly parallel to the axis, are called paraxial rays.

Our treatment of spherical mirrors will be restricted to such rays which means we shall consider only mirrors of small aperture. In diagrams, however, they will be made larger for clarity.

And rays far away from principal axis are called Marginal Rays.

3.5.2 Relation between f and R



A ray AM parallel to the principal axis of a concave mirror of small aperture is reflected through the principal focus F. If C is the centre of curvature, CM is the normal to the mirror at M because the radius of a spherical surface is perpendicular to the surface. From first law of reflection,

$$\angle i = \angle r$$

or $$\angle AMC = \angle CMF \setminus \theta \text{ (say)}$$

But $$\angle AMC = \angle MCF \text{ (alternate angles)}$$

$\therefore$ $$\angle CMF = \angle MCF$$

Therefore, D FCM is thus isosceles with FC = FM. The rays are paraxial and so M is very close to P. Therefore,

$$FM \approx FP$$

$\therefore$ $$FC = FP$$

or $$FP = \frac{1}{2} CP$$

$$\boxed{f = \frac{R}{2}}$$

3.5.3 Ray Tracing :

(i) Ray parallel to principal axis after reflection passes through focus.

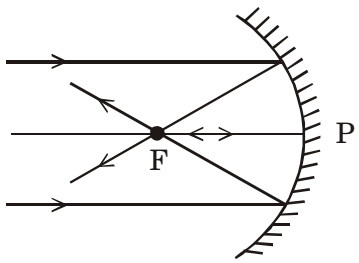
(ii) Rays parallel to each other but not parallel to principal axis form image on the focal plane.

(iii) Incident Rays passing through C retrace their path.

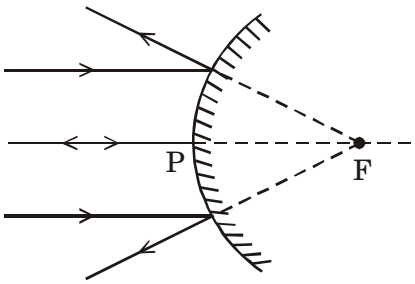
(iv) The incident and reflected ray make equal angle with principal axis if the ray is incident at 0. {this is used to locate the position of mirror}.

3.5.4 Rules for Image Formation :

(i) A ray parallel to principal axis passes (or appear to pass) through focus after reflection.

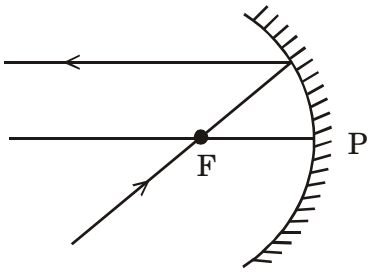


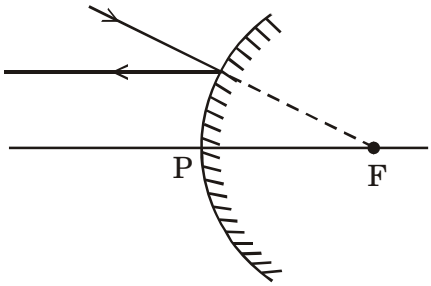
Concave mirror



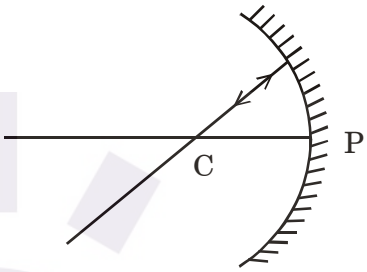
Convex mirror

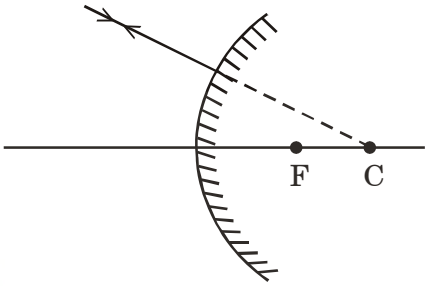
(ii) A ray passing through or directed towards focus after reflection from the spherical mirror becomes parallel to the principal axis.



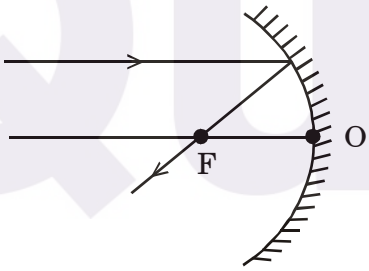


(iii) A ray passing through or directed towards the centre of curvature, after reflection from the spherical mirror, retraces its path.

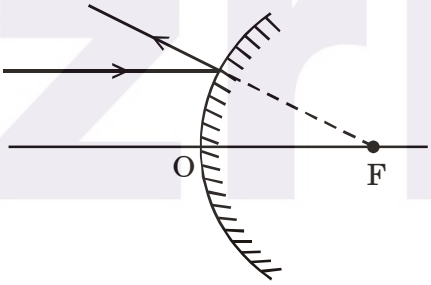




- Above rule holds, if incident ray is travelling from left to right else directions just reverses.



negetive focus



positive focus

- Distances measured above principal axis are taken to be positive while distances measured below principal axis are taken to be negative.

notations used :

u: → Distance of the object from pole of spherical mirror.

v: → distance of the image from the pole of the spherical mirror.

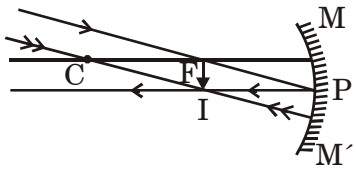
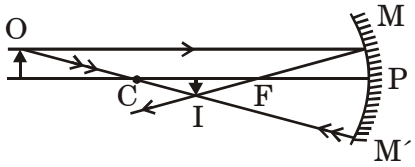
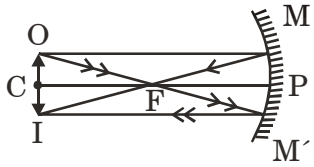
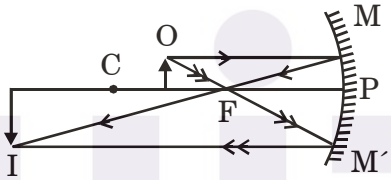
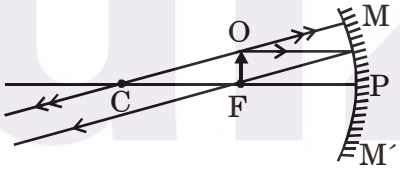
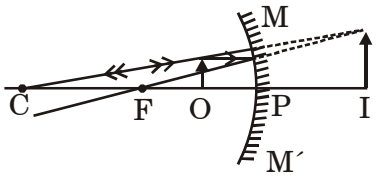
f: → focal length of the spherical mirror.

R: → Radius of curvature of the spherical mirror.

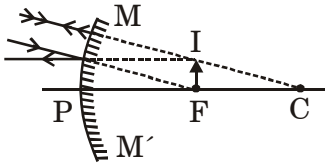
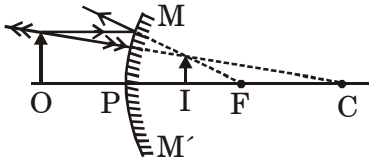
- All the symbols used are assigned values with proper sign convention. i.e. for concave mirror f is always negative and so on.

3.5.6. Image Formation by Spherical Mirrors

(a) For concave mirror

S.No.	Position of object	Ray-diagram	Details of image
1.	At infinity		Real, inverted, very small (m <<–1), at F
2.	Between ∞ and C		Real, inverted diminished (m < – 1), between F and C
3.	At C		Real, inverted, equal (m = –1), at C
4.	Between F and C		Real, inverted, enlarged (m > – 1), between C and ∞
5.	At F		Real, inverted, very large (m >> – 1), at infinity
6.	Between F and P		Virtual, erect, enlarged (m > + 1), behind the mirror

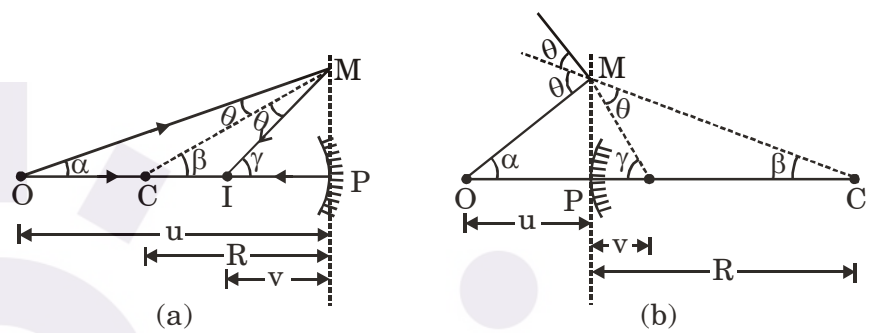
(b) For convex mirror

1.	At infinity		Virtual, erect, very small (m << + 1), at F
2.	In front of mirror		Virtual, erect, diminished (m < + 1), between P and F

* In case of image formation unless stated object is taken to be real and extended and is represented by an arrow. Also the mirror is of small size but large radius of curvature and only rays near the axis are considered (otherwise the image will be distorted).

3.5.7. The Mirror Formula

In figures (a) and (b) a ray OM from a point object O on the principal axis is reflected at M so that the angle θ, made by the incident and reflected rays with the normal CM are equal. A ray OP strikes the mirror normally and is reflected back along PO. The intersection I of the reflected rays MI and PO in figure (a) gives a **real** point image of O and in figure (b) gives a **virtual** point image of O. Let α, β and γ be the angles as shown. As the rays are paraxial, these angles are small, we can take $\alpha \approx \tan\alpha = \frac{MP}{OP}, \beta = \frac{MP}{CP}$ and $\gamma = \frac{MP}{IP}$.



Now, let us take the two figures simultaneously

Concave	Convex
In triangle CMO, $\beta = \alpha + \theta$ (the exterior angle)	In triangle CMO, $\theta = \alpha + \beta$ ... (iv) (the exterior angle)
or $\theta = \beta - \alpha$ ... (i)	In ΔCMI $\gamma = \theta + \beta$
In ΔCMI, $\gamma = \beta + \theta$	or $\theta = \gamma - \beta$ ... (v)
∴ $\theta = \gamma - \beta$ ... (ii)	From Eqs. (iv) and (v)
From Eqs. (i) and (ii)	$2\beta = \gamma - \alpha$ ... (vi)
$2\beta = \gamma + \alpha$ ... (iii)	Substituting the values of α, β and γ, we get
Substituting the values of α, β and γ, we get	$\frac{2}{CP} = \frac{1}{IP} + \frac{1}{OP}$ ... (B)
$\frac{2}{CP} = \frac{1}{IP} + \frac{1}{OP}$ ... (A)	If we now substitute the values with sign, i.e., CP = – R, IP = – v and OP = – u, we get
If we now substitute the values with sign, i.e., CP = – R, IP = – v and OP = – u	$\frac{2}{R} = \frac{1}{v} + \frac{1}{u}$
we get, $\frac{2}{R} = \frac{1}{v} + \frac{1}{u}$	or $\boxed{\frac{1}{v} + \frac{1}{u} = \frac{1}{f}}$ as $\left(f = \frac{R}{2}\right)$
or $\boxed{\frac{1}{v} + \frac{1}{u} = \frac{1}{f}}$ as $\left(f = \frac{R}{2}\right)$	

Note : Out of u, v and f values of two will be known to us and we will be asked to find the third. In such type of problems two cases are possible. Case 1 is when signs of all three will be known to us from the given information. Under this condition substitute all three with sign, then answer (i.e., the third quantity) will come without sign. Only numerical value of the unknown comes. Case 2 is when the sign of third unknown quantity is not known to us. Under such situation substitute only the known quantities with sign. Sign with numerical value will automatically come in the answer.

Power of Mirror

It is defined as,

$$P = -\frac{1}{f\,(\text{in m})}$$ The unit of power is dioptre.

3.5.8. Linear Magnification

For linear objects, the ratio of the image size (I) to the object size (O) is called linear magnification or transverse magnification or lateral magnification. If is generally denoted by m.

$$m = \frac{\text{Height of the image}}{\text{Height of the object}} = \frac{I}{O}$$

$$m = -\frac{v}{u} = \frac{f}{f-u} = \frac{(f-v)}{f}$$

Note : (a) m will be negative for a real-inverted image of a real object.

(b) m will be positive for a virtual erect image of a real object.

Example 4

An object 5 mm high is placed 30 cm from a convex mirror whose focal length is 20 cm. Find the position (in cm), size (in mm) and nature of the image.

Solution :

Given

$u = -30$ cm, $f = +20$ cm

we know $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

or $v = \frac{uf}{u-f} = \frac{(-30)(+20)}{-30-20} = +12$ cm

The image is formed 12 cm behind the mirror, it is virtual.

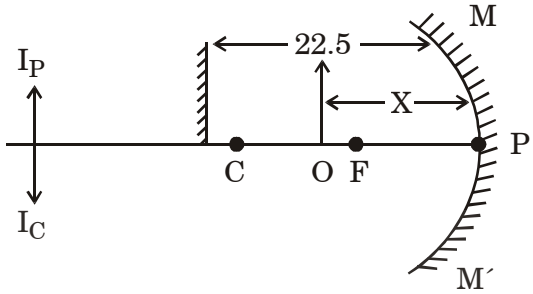
Now, $m = \frac{I}{O} = \frac{-v}{u} = -\frac{12}{-30}$

or, $I = \frac{2}{5} \times O = \frac{2}{5} \times 5 = +2$ mm

He the height of the image = 2 mm. The positive sign indicates that the image is erect.

Example 5

A plane mirror is placed 22.5 cm in front a concave mirror of focal length 10 cm. Find where an object can be placed between the two mirrors, so that the first image in both the mirrors coincide.



Solution :

As shown in figure, if the object is placed at a distance x from the concave mirror, its distance from the place mirror will be (22.5 – x). So, plane mirror will form equal and erect image of object at a distance (22.5 – x) behind the mirror.

Now as according to given problem the image formed by concave mirror coincides with the image formed by concave mirror, therefore for concave mirror.

$$v = -[22.5 + (22.5 - x)] = -(45 - x) \quad \text{and} \quad u = -x$$

So, $\frac{1}{-(45-x)} + \frac{1}{x} = \frac{1}{-10}$ or $\frac{45}{45x - x^2} = \frac{1}{10}$

i.e., $x^2 - 45x + 450 = 0$ or $(x - 30)(x - 15) = 0$

i.e. $x = 30$ cm or $x = 15$ cm

But as the distance between two mirrors is 22.5 cm, x = 30 cm is not admissible. So the object must be at a distance of 15 cm from concave mirror.

Example 6

Find the distance of object from a concave mirror of focal length 10 cm so that image size is four times the size of the object.

Solution :

Concave mirror can form real as well as virtual image. Here nature of image is not given in the question. So we will consider two possible cases.

When image is real : Real image is formed on the same side of object, i.e., u, v and f all are negative. So let,

$u = -x$, also $m = -4$ for image to be real

then $v = -4x$ as $\left|\frac{v}{u}\right| = |m| = 4$ and $f = -10$ cm

Substituting in, $$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

We have $$\frac{1}{-4x} - \frac{1}{x} = \frac{1}{-10}$$

or $$\frac{5}{4x} = \frac{1}{10}$$

$\therefore$ $x = 12.5$ cm **Ans.**

Note : $|x| > |y|$ and we know that in case of a concave mirror, image is real when object lies beyond F.

When image is virtual : In case of a mirror image is virtual when it is formed behind the mirror, i.e., u and f are negative while v is positive. So let

$u = -y$, also $m = +4$ for image to be virtual

then $v = +4y$ and $f = -10$ cm

Substituting in, $$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

We have $$\frac{1}{4y} - \frac{1}{y} = \frac{1}{-10}$$

or $$\frac{3}{4y} = \frac{1}{10}$$

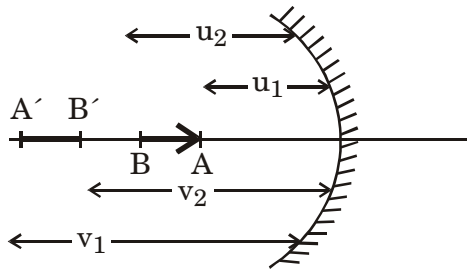
$\therefore$ $y = 7.5$ cm **Ans.**

Note : Here $|y| < |f|$ as we know that image is virtual when the object lies between F and P.

3.5.9 Longitudinal Magnification

When an object is kept parallel to the principle axis, then let u_1, and u_2 be the distance of the end points of the object and let v_1, v_2 be the ends points of the image formed.

Here, AB is the object and $A'B'$ is the image formed.



Note : Please observe that B′ will be closer to AB as compared to A′.

The point can be seen, as image of the objects closer to the mirror are created farther as compared to the images of the objects farther from the mirror.

Here, $$m = -\frac{(v_2 - v_1)}{(u_2 - u_1)}$$

also, $$m = -\frac{dv}{du} \quad \text{\{for short objects\}}$$

Some Important Points :

1. For spherical mirrors

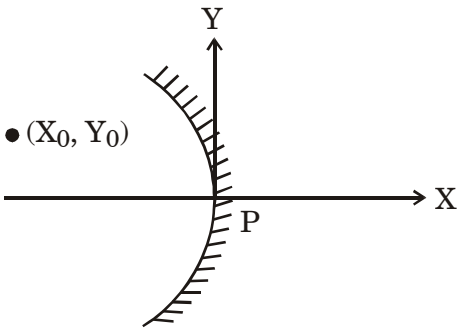
$$m = -\frac{v}{u}$$

Positive value of m means v and u are of opposite sign. So is u is negative then v is positive and vice-versa. Thus, if m = + 2 for a real object, it means image is virtual, erect and two times greater in size. Similarly $m = -\frac{1}{2}$ means image is real, inverted and $\frac{1}{2}$ times in size (that of object).

2. **Method of finding coordinates of image of a point object if the coordinates of object are known :** Suppose coordinates of a point object (x_0, y_0) with respect to the coordinate axes shown in figure are known to us. The coordinates of image (x_i, y_i) can be obtained using

$$\frac{1}{x_0} + \frac{1}{x_i} = \frac{1}{f} \qquad \left(\frac{1}{v} + \frac{1}{u} = \frac{1}{f}\right)$$

or $$\boxed{x_i = \frac{fx_0}{x_0 - f}} \qquad \text{... (i)}$$



Similarly $$m = \frac{y_i}{y_0} = -\frac{v}{u} = -\frac{x_i}{x_0}$$

or $$\boxed{y_i = \frac{fy_0}{f - x_0}} \qquad \text{... (iii)}$$

3. For Concave Mirror

S.No.	Position of object	Details of image
1.	At ∞	At F, real, inverted $\lvert m \rvert << 1$
2.	Between C and ∞	Between F and C, real, inverted, $\lvert m \rvert < 1$
3.	At C	At C, real, inverted, $\lvert m \rvert = 1$
4.	Between F and C	Between C and ∞, real, inverted, $\lvert m \rvert > 1$
5.	At F	At infinity, real, inverted, $\lvert m \rvert >>1$
6.	Between F and P	behind the mirror, virtual, erect $\lvert m \rvert > 1$

For Convex Mirror

S.No.	Position of object	Details of images
1.	At inifinity	At F, virtual, erect, $\lvert m \rvert << 1$
2.	In front of mirror	Between P and F, virtual, erect, $\lvert m \rvert < 1$

4. If an object is placed with its length along the principal axis, then so called **longitudianal magnification** becomes,

$$m_L = \frac{I}{O} = -\left(\frac{v_2 - v_1}{u_2 - u_1}\right) = -\frac{dv}{du} \qquad \text{(for small objects)}$$

From, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ we have

$$-v^{-2}\,dv - u^{-2}\,du = 0$$

or $$\frac{dv}{dy} = -\left(\frac{v}{u}\right)^2$$

or $$\boxed{m_L = -\frac{dv}{du} = \left(\frac{v}{u}\right)^2 = m^2}$$

5. If we differentiate the mirror formula

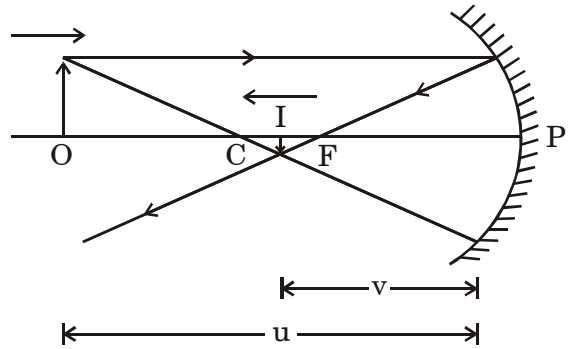
$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

with respect to time, we get

$$-v^{-2}.\frac{dv}{dt} - u^{-2}\frac{du}{dt} = 0 \qquad \text{(as f = constant)}$$

or $$\boxed{\frac{dv}{dt} = -\left(\frac{v^2}{u^2}\right)\frac{du}{dt}}$$

Here $\frac{du}{dt}$ is the rate by which u is changing. Or it is the object speed if mirror is stationary. Similarly, $\frac{dv}{dt}$ is the rate by which v (distance between image and mirror) is changing. Or it is image speed if mirror is stationary. So if at a known values of v and u, the object speed is given, we can find the image speed from the above formula.



Let us take an example for a concave mirror.

Suppose the object is moved from infinity towards focus.

As u is decreasing therefore,

$$\left(-\frac{du}{dt}\right) = \text{rate of decrease of u} \qquad \text{(speed of object)}$$

Therefore, $\left(\frac{dv}{dt}\right)$ = rate of increase of v (speed of image)

Further v < u (when the object lies between ∞ and C)

$$\therefore \quad \left(\frac{dv}{dt}\right) < \left(-\frac{du}{dt}\right)$$

Hence, as the object is moved towards mirror the image (which is real) will recede from the mirror with speed less than the speed of object.

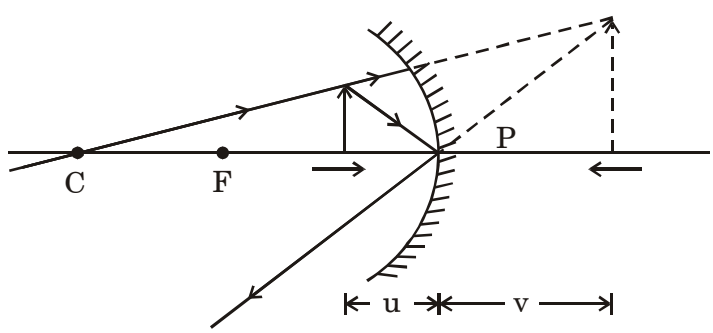
When the object is at C, image is also at C, i.e., v = u or $\left(\frac{dv}{dt}\right) = \left(-\frac{du}{dt}\right)$. Hence, speed of image is equal to the speed of object. When the object lies between C and F. v > u, i.e., image speed is more than the object speed when object comes inside F image becomes virtual i.e., u and f are negative while v is positive.

Hence, $\frac{1}{v} - \frac{1}{u} = -\frac{1}{f}$

or $\frac{1}{u} - \frac{1}{v} = \frac{1}{f}$

or $-u^{-2}\left(\frac{du}{dt}\right) - v^{-2}\left(\frac{dv}{dt}\right) = 0$

$$\boxed{\left(-\frac{dv}{dt}\right) = \left(\frac{v^2}{u^2}\right)\left(-\frac{du}{dt}\right)}$$



Now as u is further decreased, v also decreases to keep $\frac{1}{f}$ constant. So, $-\frac{du}{dt}$ is rate with which object is approaching towards mirror and $\left(-\frac{dv}{dt}\right)$ is rate by which image is approaching towards mirror. Further in this case we know that image is always enlarged or v > u. Therefore, image speed is more than the object speed. Thus, we may conclude the above discussion as under :

When an object is moved from $-\propto$ to F, the image (real) moves from F to $-\propto$ and the when the object is further moved from F to P image (now virtual moves from $+\propto$ to P.

Note : when the object is either at centre of curvature C or at pole P the two speeds are equal. When the object is at pole it can be assumed as if the image is forming by a plane mirror due to the small aperture of the mirror.

6. Graph between $\frac{1}{v}$ versus $\frac{1}{u}$: Let us first take the case of a concave mirror. Here, two cases are possible.

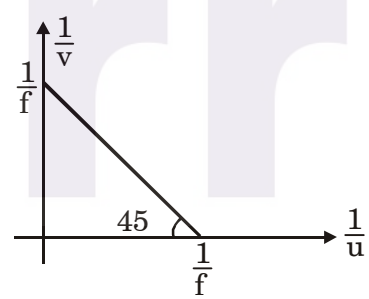
Case 1. When the image is real, i.e., object lies between F and inifinity. In such a situation u, v and f are negative. Hence, the mirror formula

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f} \text{ becomes}$$

$$-\frac{1}{v}-\frac{1}{u}=-\frac{1}{f}$$

or again $$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

or $$\frac{1}{v}=-\frac{1}{u}+\frac{1}{f}$$



Comparing with y = – x + c, the desired graph will be a straight line with slope –1 and slope equal to $\frac{1}{f}$.

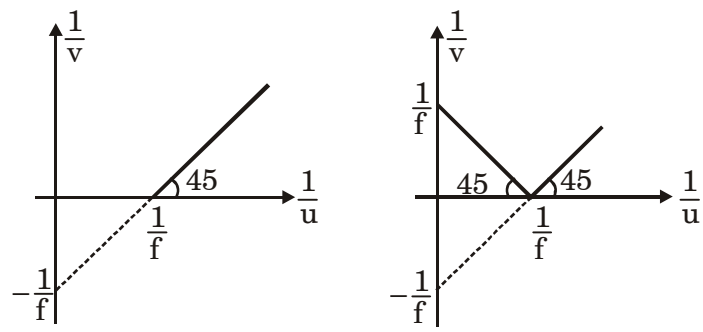
The corresponding graph is shown in figure

Case 2. When the image is virtual, i.e., object lies between F and P. Under such situation u and f are negative while v is positive. The mirror formula thus becomes

$$\frac{1}{v}=\frac{1}{u}-\frac{1}{f}$$

Comparing it with y = x – c the desired graph is a straight line with slope + 1 and intercept $-\frac{1}{f}$.

The graph is thus shown in Fig. (a) The two graphs can be drawn in one single graph as in Fig. (b)



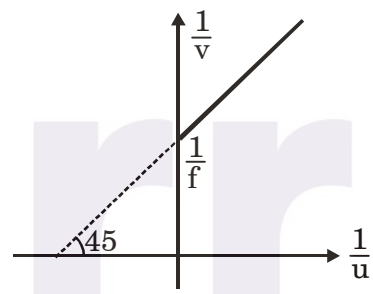
(a) (b)

Note : Here $\frac{1}{u}$ and $\frac{1}{v}$ are really the magnitudes of $\frac{1}{u}$ and $\frac{1}{v}$ (i.e., without sign)

For a **convex mirror** image is always virtual, i.e., u is negative while v and f are positive. Hence, the mirror formula becomes.

$$\frac{1}{v} = \frac{1}{u} + \frac{1}{f}$$

Comparing with y = x + c, the desired graph is a straight line of slope + 1 and intercept $\frac{1}{f}$.



Example 7

A thin rod of length (f/3) is placed along the principal axis of a concave mirror of focal length f such that its image which is real and elongated, just touches the rod. What is the magnification?

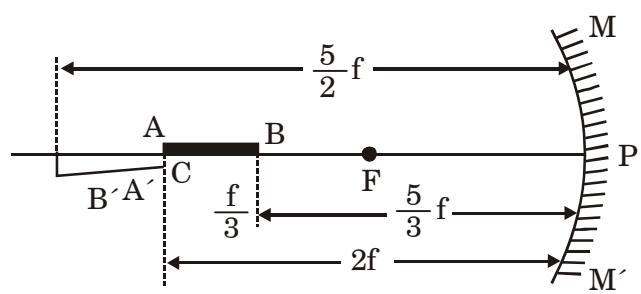
Solution :

As the image is real and enlarged, the object must be between C and F. Also as the image of one end coincides with the end itself, i.e., $v_A = u_A$,

$$\frac{1}{v_A} + \frac{1}{v_A} = \frac{1}{-f}$$

i.e., $v_A = u_A = -2f$

i.e., the end A is at C.



Now as the length of the rod is (f/3), its other end B will be at a distance [2f – (f/3)] = (5/3) f from P. So if the distance of image of end B from P is V_B,

$$\frac{1}{v_B} + \frac{1}{-(5/3)f} = \frac{1}{-f} \quad \text{i.e.} \quad v_B = -\frac{5}{2}f$$

So the size of the image

$$|v_B| - |v_A| = \frac{5}{2}f - 2f = \frac{1}{2}f$$

and $$m = \frac{(v_B - v_A)}{(u_B - u_A)} = \frac{(1/2)\,f}{(1/3)\,f} = -\frac{3}{2}$$ **Ans.**

Negative sign implies that image is inverted with respect to object and so is real.

Example 8

If a luminous point is moving at a speed V_0 towards a spherical mirror, along its axis, show that the speed at which the image of this object is moving will be given by

$$V_i = -\left[\frac{f}{u-f}\right]^2 V_0$$

Solution :

In case of spherical mirrors

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \qquad \text{... (1)}$$

or $$-\frac{1}{v^2}\frac{dv}{dt} - \frac{1}{u^2}\frac{du}{dt} = 0$$ [as f is constant]

or $$V_i = -\left[\frac{v}{u}\right]^2 V_0 \qquad \left[\text{as } \frac{dv}{dt} = V_i \text{ and } \frac{du}{dt} = V_0\right]$$

But from Eq. (1) $$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} \quad \text{i.e.} \quad v = \frac{uf}{(u-f)}$$

So $$V_i = -\left[\frac{f}{u-f}\right]^2 V_0$$ **Ans.**

which is the desired result.

Note : From this result it is clear that in case of concave mirror, i.e., with $f \to -f$ if :

$u = -\infty$, $v = -f$ and $V_i = 0$

$u = -2f$, $v = -2f$ and $V_i = V_0$

And $u = -f$, $v = -\infty$ and $V_i = -\infty$

i.e., if the object moves with constant speed V_0 towards the mirror from ∞ to F, the image moves away from the mirror from F to ∞ with speed V_i increasing from 0 to ∞. The two speeds are equal when both cross each other at C.

Example 9

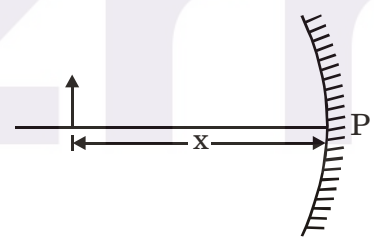
An object is placed at right angles to the principal axis of a concave mirror at such a position that the image formed is half the size of object. The object is now shifted to another position, displaced relative to the earlier position by Δx = 10 cm, and the size of image now becomes one fourth of the object. Find initial distance of object from the pole of mirror and also the focal length of mirror.

Solution :

Initially, let the distance of object from the pole be 'x'.

$\therefore \quad u = -x$

Since the given mirror is concave and the image is diminished, it is also real and **inverted** so that magnification has a negative sign and $m = \frac{I}{O} = -\frac{1}{2}$. Further, the description of image implies that the object is between C and ∞ so that the image is between F and C.



$$m = -\frac{v}{u} = -\frac{1}{2}$$

$$\therefore \quad v = -\frac{u}{2} = \frac{-x}{2} \qquad [\because u = -x]$$

Applying $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$$-\frac{2}{x} - \frac{1}{x} = \frac{1}{f}$$

$$\therefore \quad -\frac{3}{x} = \frac{1}{f} \qquad \text{... (1)}$$

On shifting the position of object by 10 cm, size of image becomes one fourth of the object while in the earlier position, it was half. It implies that the object has been shifted by Δx = 10 cm away from

the pole (As mentioned above, object is between C and ∞ so that the image is between F and C. If the object is **shifted away from C towards** ∞, the image is shifted between F and C to towards F. Thus u increases and v decreases resulting in a reduced value of $|m| = \frac{v}{u}$ as in the present case). Now, distance of object from pole = x + 10

$\therefore$ $u = -(x + 10)$

and $m = -\frac{v}{u} = -\frac{1}{4}$ (diminished image in case of concave mirror also implies **real** and **inverted)**

$\therefore$ $v = \frac{u}{4} = \frac{-(x+10)}{4}$

Using $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$$-\frac{4}{x+10} - \frac{1}{x+10} = \frac{1}{f}$$

$\therefore$ $\frac{-5}{x+10} = \frac{1}{f}$ ..... (2)

From Eq. (1) and Eq. (2),

$$\frac{-3}{x} = \frac{-5}{x+10}$$

$\therefore$ $3x + 30 = 5x$

or $x = 15$ cm **Ans.**

Thus the object, initially, is at a distance 15 cm from the pole.

From Eq. (2),

$$\frac{-5}{x+10} = \frac{1}{f}$$

$\therefore$ $\frac{-5}{15+10} = \frac{1}{f}$

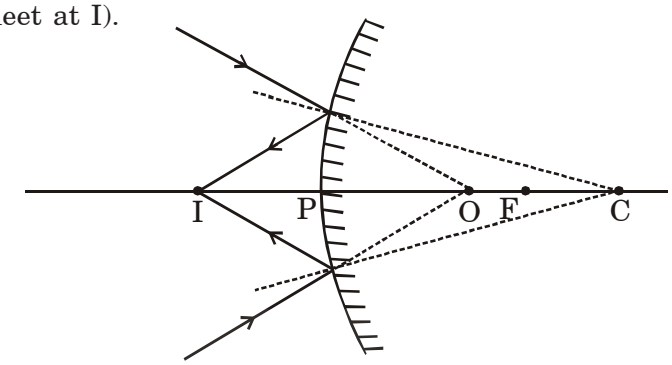
$\therefore$ $f = -5$ cm **Ans.**

Example 10

Show that for a virtual object, a convex mirror gives (a) real and erect image for u < f **and (b) virtual and inverted image for** u > f**.**

Solution :

(a) Fig. show rays converging towards the point O and intercepted by a convex mirror. O acts as a virtual object and, as shown, its distance from pole u < f. Rays are reflected from the mirror and from **real image at I** (since they actually meet at I).



Using

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{u-f}{uf}$$

$$\therefore \quad v=\frac{uf}{u-f} \qquad \text{... (1)}$$

Here u is positive (measured from P to O)

f is positive.

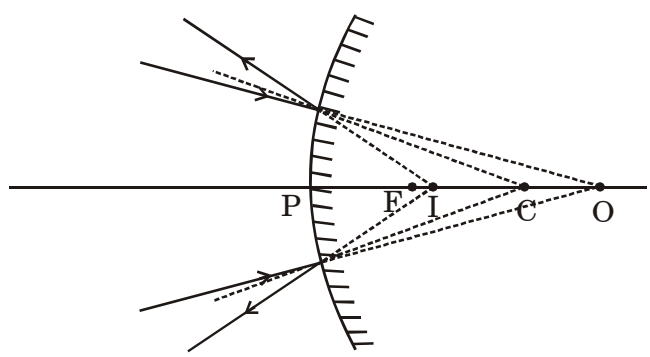
Since u < f, v from Eq. (1) is negative.

u being positive and v negative, magnification, $m=-\frac{v}{u}$ will have a positive value.

Hence the image will be erect.

Thus we get a real and erect image.

(b) Fig. shows O as a virtual object with u > f. Here the reflected rays appear to diverge from I so that I acts as a virtual image of virtual object O. Obviously, distance of object from pole 'u' (measured from P to O) is positive and u > f.



Using $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

$$v=\frac{uf}{u-f}$$

Both u and f are positive and u > f

$\therefore$ v is positive

$m = -\frac{v}{u}$, both v and u being positive, m will be negative. Hence the image is inverted.

Example 11

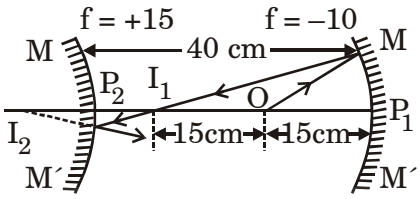
A concave mirror of focal length 10 cm and a convex mirror of focal length 15 cm are placed facing each other 40 cm apart. A point object is placed between the mirrors, on their common axis and 15 cm from the concave mirror. Find the position and nature of the image produced by the successive reflections, first at concave mirror and then at convex mirror.

Solution :

According to given problem, for concave mirror,

$u = -15$ cm and $f = -10$ cm

So $\frac{1}{v} + \frac{1}{-15} = \frac{1}{-10}$, i.e., $v = -30$ cm



i.e., concave mirror will form real, inverted and enlarged image I_1 of object O at a distance 30 cm from it, i.e., at a distance 40 – 30 = 10 cm from convex mirror.

For convex mirror the image I_1 will act as an object and so for it $u = -10$ cm and $f = +15$ cm.

$\therefore \frac{1}{v} + \frac{1}{-10} = \frac{1}{15}$, i.e., $v = +6$ cm **Ans.**

So final image I_2 is formed at a distance 6 cm behind the convex mirror and is virtual as shown in Fig.

Note : As here $m_1 = -\left[\frac{-30}{-15}\right] = -2$ and $m_2 = -\left[\frac{+6}{-10}\right] = +0.6$ so $m = m_1 \quad m_2 = (-2)(+0.6) = -1.2$, i.e., final image is inverted with respect to object, virtual, 1.2 times of the object size and is at a distance 6 cm behind the convex mirror.

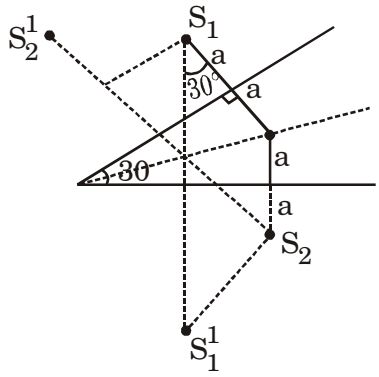
Example 12

Two plane mirrors intersect at an angle θ. An object is placed on the bisector of the angle between them. Show the location of the closest four images if :-

(a) θ = 30 and (b) θ = 120

Solution :

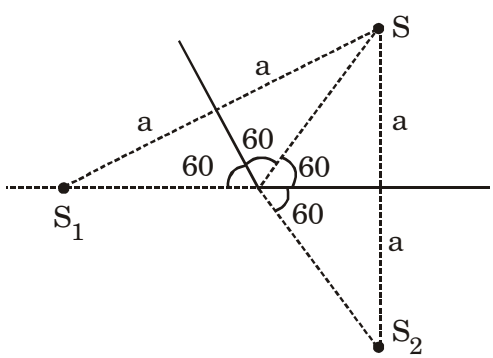
(a)



Here, S_1 is the image of source S in the top mirror and is distance a behind the mirror. Now, this S_1 becomes a source for the bottom mirror, giving image S_1'.

The situation is repeated in the other mirror to give S_2 and S_2'.

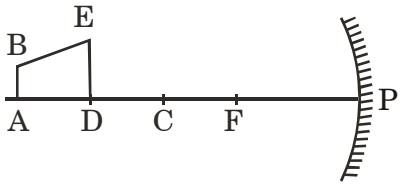
(b)



In the case, there is only one image located behind each mirror. No other images appear because the initial images lie in the planes of the respective mirrors.

Example 13

An object ABED is placed in front of a concave mirror beyond centre of curvature C as shown in figure. State the shape of the image.

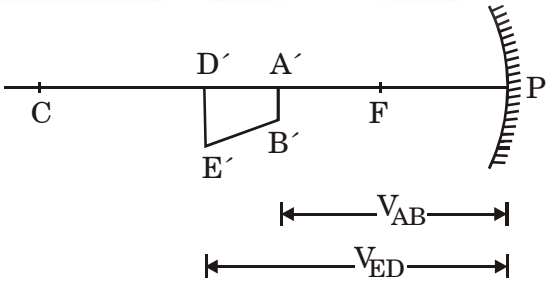


Solution :

Object is placed beyond C. Hence, the image will be real and it will lie between C and F. Further u, v and f all are negative, hence the mirror formula will become

$$-\frac{1}{v}-\frac{1}{u}=-\frac{1}{f}$$

or $$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{u-f}{uf}$$



or $$v=\frac{f}{1-\frac{f}{u}}$$

Now $u_{AB} > u_{ED}$

$\therefore$ $v_{AB} < v_{ED}$

and $|m_{AB}| < |m_{ED}|$ $\left(\text{as } m=-\frac{v}{u}\right)$

Therefore, shape of the image will be as shown in figure.

Also note that $v_{AB} < u_{AB}$ and $v_{ED} < u_{ED}$,

So $|m_{AB}| < 1$ and $|m_{ED}| < 1$

Example 14

It is desired to cost the image of a lamp, magnified 5 times, upon a wall 1.2 m distant from the lamp. What kind of spherical mirror is required and what is its position ?

Solution :

Concept : To cast the image on a wall, it must be real.

Hence, for a real image v is positive, and u is also positive. Thus the image will be inverted.

Now, $-\frac{v}{u} = -5$ and we also know that $|v - u| = -1.2$ {distance between lamp and wall}

or $5u - u = -1.2$

$\Rightarrow$ $\boxed{u = -0.3 \text{ m},}$ $\boxed{v = -1.5 \text{ m}}$

The mirror is thus 0.3 m from the lamp and 1.5 m from the wall.

Also, $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$ $\Rightarrow$ $\boxed{f = -0.25 \text{ m}}$

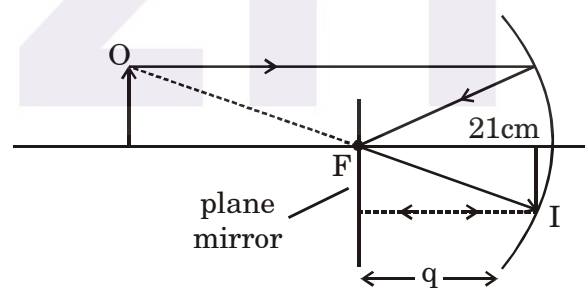
Thus, the mirror is CONCAVE of radius 0.5 m.

Example 15

An object is placed 42 cm in front of a concave mirror of focal length 21 cm. Light from the concave mirror is reflected onto a small plane mirror 21 cm in front of the concave mirror. where is the final image ?

Solution :

Given : $u = -42$ m, $f = -21$ cm.



Now, $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$

$\Rightarrow$ $\frac{1}{v} = \frac{1}{-21} - \frac{1}{-42}$

$\Rightarrow$ $v = -42$ cm

{or we could have directly seen the final image would be at 42 cm from the fact that the object is at centre of curvature, hence the image will also be a centre of curvature}.

Now, use the mirror formula for the plane mirror, with $f = \infty$ and $u = -42 - (-21) = -21$ cm, since the image produced by the concave mirror alone is the object for the plane mirror.

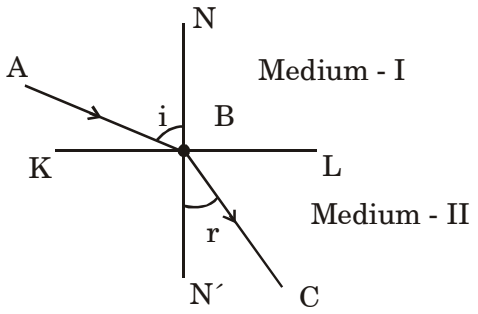
$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f} \quad \Rightarrow \quad \frac{1}{-21} + \frac{1}{q} = \frac{1}{\infty}$$

$\Rightarrow$ $q = 21$ cm.

Hence, the position of the final image I is 21 cm in front of the plane mirror.

4. REFRACTION OF LIGHT

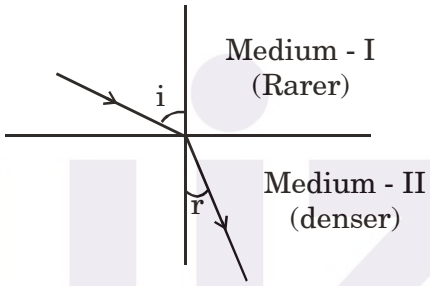
Whenever a ray of light comes from one transparent medium to another, it gets deviated from its original path while crossing the interface of the two media (except in case of normal incidence). This phenomenon of deviation or bending of light rays from their original path while passing from one medium to another is called **refraction.**



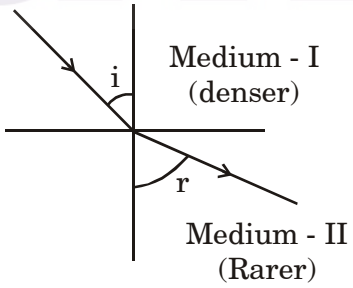
AB → incident ray BC → refracted ray NBN´ → Normal

i → angle of incidence r → angle of refraction KL → interface

- If the refracted ray bends towards the normal with respect to the incident ray, then the second medium is said to be optically denser as compared to the first medium.



- If the refracted ray bends away from the normal, then the second medium is said to be (optically) rarer as compared to the first medium.



4.1 Laws of Refraction

(i) The incident ray, there fracted ray and the normal to the refracting surface at the point of incidence, all lie in the same plane.

(ii) The ratio of sine of the angle of incidence to the sine of the angle of refraction is constant for any two given media.

i.e. $$\frac{\sin i}{\sin r} = \text{costant} = {}_1\mu_2 = \text{Refractive index}$$

This is called **Snell's law.**

4.2 Refractive Index (μ or n)

(A) Absolute Refractive Index

It is defined as the ratio of speed of light in free space c to that in a given medium v,

i.e., μ or $n = \frac{c}{v}$ ...(1)

Regarding refractive index it is worth that :

(1) It is a scalar and has no units and dimensions.

(2) If ϵ_0 and μ_0 are electric permitivity and magnetic permeability respectively of free space while ϵ and μ those of a given medium, then according to electromagnetic theory

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \text{ and } v = \frac{1}{\sqrt{\epsilon \mu}}$$

$$n = \frac{c}{v} = \sqrt{\frac{\epsilon \mu}{\epsilon_0 \mu_0}} = \sqrt{\epsilon_R \mu_R} \quad \text{... (2)}$$

(3) As in vacuum or free space, speed of light of all wavelengths is maximum and equal to c so for all wavelengths the refractive index of free space is minium and is

$$\mu = \frac{c}{v} = \frac{c}{c} = 1$$

(4) For a given light, denser is the medium, less will be the speed of light and so greater will be the refractive index, e.g.,

as $v_{glass} < v_{water}$, $\mu_G > \mu_W$

i.e., for a given light refractive index depends on nature of medium [i.e., $\mu \propto (1/v)$].

(5) For a given medium (other than free space), the speed of light of different wavelengths is different, i.e. $v \propto \lambda$ and $\mu = (c/v)$, $\mu \propto (1/\lambda)$, i.e., greater the wavelength of light lesser will be the refractive index, e.g. $\lambda_R > \lambda_B$, so in water or glass $\mu_R < \mu_B$, i.e., for a given medium (other than free space) refractive index depends on wavelength of light.

(6) As for light in free space $c = f \lambda_0$ and in a medium $v = f \lambda$

$$\mu = \frac{c}{v} = \frac{\lambda_0}{\lambda} \quad \text{...(3)}$$

i.e., for a given light and medium refractive index is equal to the ratio of wavelength of light in free space to that in the medium.

(B) Relative Refractive Index

When light passes from one medium to the other, the refractive index of medium 2 relative to 1 is written as ${}_1\mu_2$ and is defined as,

$$ {}_1\mu_2 = \frac{\mu_2}{\mu_1} = \frac{(c/v_2)}{(c/v_1)} = \frac{v_1}{v_2} \qquad ...(4) $$

while using the concept of relative μ, it must be kept in mind that :

(1) $ {}_2\mu_1 = \frac{\mu_1}{\mu_2} = \frac{v_2}{v_1} $

So that $ ({}_1\mu_2)({}_2\mu_1) = \frac{v_1}{v_2} \times \frac{v_2}{v_1} = 1 $

i.e. $ {}_1\mu_2 = \frac{1}{{}_2\mu_1} \qquad ...(5) $

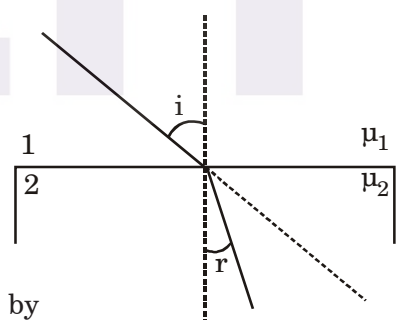
(2) Usually 'μ' is used for relative refractive index and it implies the refractive index of denser medium relative to rarer one, i.e.

$$ \mu = \frac{\mu_D}{\mu_R} = \frac{v_R}{v_D} > 1 $$

(3) In lens theory μ is used for the refractive index of material of lens relative to the medium, i.e.,

$$ \mu = \frac{\mu_{lens}}{\mu_{medium}} $$

and can be greater than, less than or equal to unity.



(4) As according to Snell's law

$\mu_1 \sin i = \mu_2 \sin r$

the refractive index of medium 2 relative to 1 will be given by

$$ {}_1\mu_2 = \frac{\mu_2}{\mu_1} = \frac{\sin i}{\sin r} \qquad ...(6) $$

4.3 Principle of Reversibility of Light Rays

A ray travelling along the path of the refracted ray is reflected along the path of the incident ray.

In the same way, a refracted ray reversed to travel back along its path will get refracted along the path of the incident ray. Thus, the incident ray and the refracted ray are mutually reversible. This is called the **reversibility of light.**

$$_1\mu_2 = \frac{\sin i}{\sin r} \quad \text{... (1)}$$

similarly, when path is reversed, then

$$_2\mu_1 = \frac{\sin r}{\sin i} \quad \text{... (2)}$$

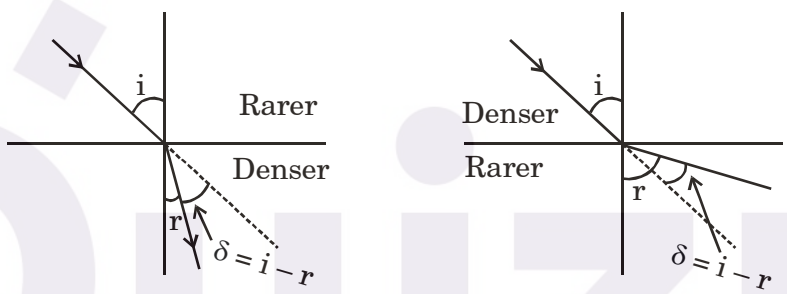
Multiply (1) and (2), we get,

$$_1\mu_2 \quad _2\mu_1 = 1$$

$$\text{or } _1\mu_2 = \frac{1}{_2\mu_1}$$

Deviation of ray due to Refraction

Hence, from the figure we can see that the deviation (δ) = $|i - r|$



4.4 Critical Angle

When a ray of light goes from denser medium to a racer medium, the angle of refraction is greater than the angle of incidence.

From this statement, we can also observe that, in some cases the angle of refraction can increase to become 90 or more than 90 .

The angle of incidence for which the angle refraction becomes 90 is called the article angle for that interface.

It is generally denoted by c.

Let μ_R be the refractive index of the rarer medium and μ_D be the refractive index of the denser medium. Obviously, $\mu_r < \mu_D$.

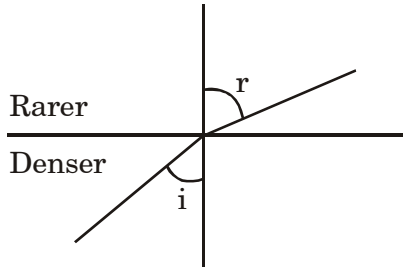
Hence, from Snell's law $\frac{\sin c}{\sin 90°} = \frac{\mu_r}{\mu_D}$

$$\text{or} \quad \sin c = \frac{1}{\mu}$$

where $\mu = \frac{\mu_D}{\mu_R}$ is the refractive index of the denser medium with respect to the rarer medium.

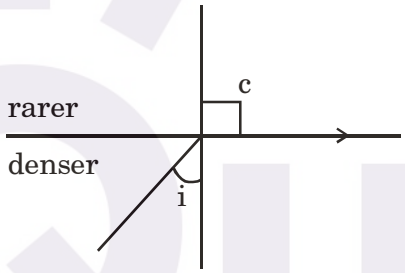
4.5 Total Internal Reflection

While a light ray goes from a denser medium to a rarer medium, the angle of refraction is greater than the angle of incidence. If the angle of incidence is increased, the angle of refraction may eventually become 90 .

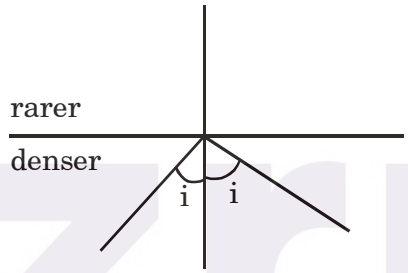


i < c
refraction

If a ray of light travelling in a denser medium strikes a rarer medium at an angle of incidence i which is greater than the critical angle C, it gets totally reflected back into the same medium. This phenomenon is called as **total internal reflection.**



i = c



i > c
total internal
reflection

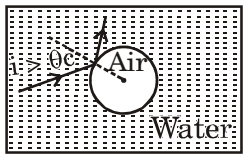
Some Illustrations of Total Internal Reflection

(1) Shining of air bubble

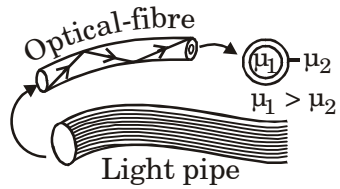
An air bubble in water shines due to total internal reflection at its outer surface. Here light is propagating from denser medium (water) to rarer medium (air) and if $i > \theta_C$(49) TIR will take place at the surface of the bubble. This is shown in figure (A)

(2) Sparkling of diamond

The sparkling of diamond is due to total internal reflection inside it. As refractive index for diamond is 2.5 so θ_C = 24 . Now the cuttings of diamond are such that $i > i_C$, so TIR will take place again and again inside it. The light which beams out from a few places in some specific directions makes it sparkle. Recently developed synthetic rutile (titanium dioxide) displays even greater brilliance as for it $\mu \simeq 2.9$ and so $\theta_C \simeq 20$.



(A)



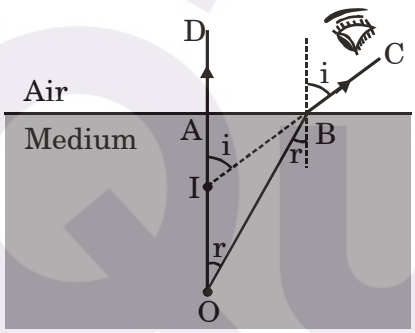
(B)

(3) Optical fibre

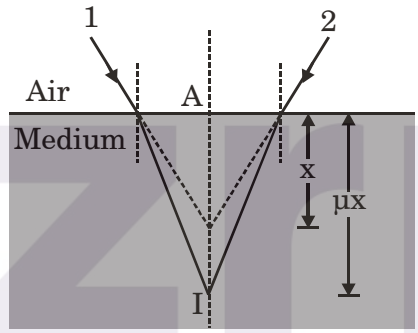
Now a days total internal reflection is used in fibre optics. In it light through multiple total internal reflections is propagated along the axis of a glass fibre of radius of few microns in which index of refraction of core μ_1 (= 1.7) is greater than that of surroundings (called cladding) μ_2 (= 1.5) as shown in Fig. (B). At present fibre optics is used in telecommunications for propagating signals from one place to the other as through a single fibre about 300 signals can be transmitted simultaneously without any interference whereas an electric wire can transmit only one. This is also used by doctors to visualise internal sites of the body.

4.6 Single Refraction from a Plane Surface

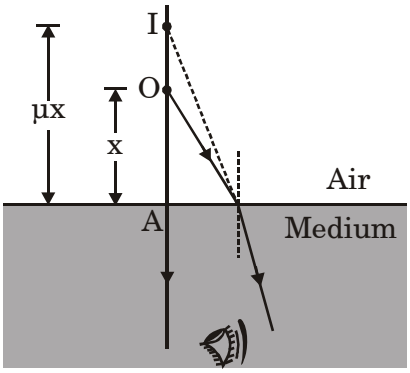
Following four results can be drawn after refraction from a plane surface.



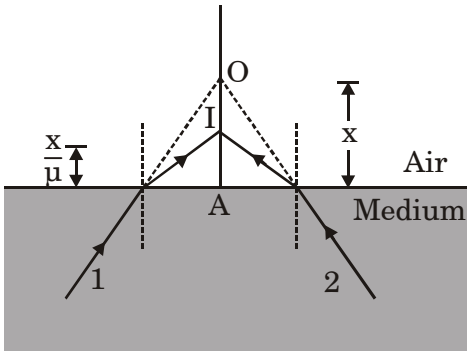
(a)



(b)



(c)



(d)

In all the four figures, single refraction is taking place through a plane surface. Refractive index of medium (may be glass, water etc.) is μ. In figures (a) and (d) the ray of light is travelling from denser

to rare medium and hence, it bends away from the normal. In figures (b) and (c) the ray of light is travelling from a rare to a denser medium and hence, it bends towards the normal. Now, let us take the four figures individually.

(a) Object O is placed at a distance x from A. Ray OA, which falls normally on the plane surface, passes undeviated as AB. Ray OB, which falls at angle r (with the normal) on the plane surface, bends away from the normal and passes as BC in air. Rays AD and BC meet at I after extending these two rays backwards. Let BC makes an angle i (> r) with normal.

$$\sin i \approx \tan i = \frac{AB}{AI} \qquad \text{... (i)}$$

and

$$\sin r \approx \tan r = \frac{AB}{AO} \qquad \text{... (ii)}$$

Dividing Eq. (i) by (ii), we have $\frac{\sin i}{\sin r} = \frac{AO}{AI}$

or

$$\mu = \frac{AO}{AI} \qquad \left(\text{as } \frac{\sin i}{\sin r} = \mu\right)$$

$$\therefore \qquad AI = \frac{AO}{\mu} = \frac{x}{\mu}$$

If point O is at a depth of d from a water surface, then the above result is also sometimes written as,

$$\boxed{d_{apparent} = \frac{d_{actual}}{\mu}}$$

or the apparent depth is μ times less than the actual depth.

(b) In the absence of the plane refracting surface the two rays 1 and 2 would had met at O. Proceeding in the similar manner we can prove that after refraction from the plane surface they will now meet a point I where,

$$AI = \mu x \qquad \text{(if } AO = x\text{)}$$

(c) In this case object is at O, a distance AO = x from the plane surface. When seen from inside the medium it will appear at I, where

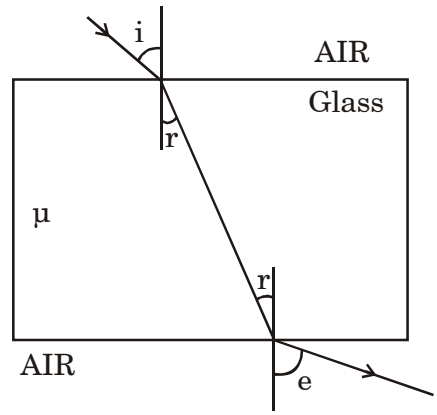
$$AI = \mu x$$

(d) The two rays 1 and 2 meeting at O will not meet at I after refraction from the plane surface.

Where $AI = \frac{AO}{\mu} = \frac{x}{\mu}$.

Note : In all the four cases the change in the value of x is μ times whether it is increasing or decreasing. All the relations can be derived for small angles of incidence as done in part (a).

4.6 Refraction Through A Glass Slab



When a light ray passes through a glass slab having parallel faces, it gets refracted twice before finally emerging out of it.

First refraction takes place from air to glass. So,

$$\mu = \frac{\sin i}{\sin r} \qquad ...(i)$$

The second refraction takes place from glass to air.

So, $$\frac{1}{\mu} = \frac{\sin r}{\sin e} \qquad ... (ii)$$

From eq. (i) and (ii), we get

$$\frac{\sin i}{\sin r} = \frac{\sin e}{\sin r} \quad \Rightarrow \quad i = e$$

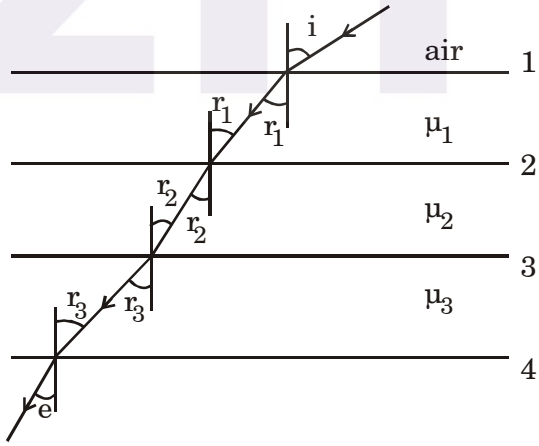
Thus, emergent ray is parallel to incident ray.

If there exists a region of index through which ray has to pass; such that there exists a medium of same refractive index on the entry and exit of the ray then the ray gets laterally shifted by distance d

Illustration

In the adjacent system what is the angle between incident ray and emergent ray ?

Solution :



Applying Snell's law at first interface

$\sin i = \sin r_1$ ..... (i)

Applying Snell's law at 2, 3 & 4

We get

$\mu_1 \sin r_1 = \mu_2 \sin r_2$ ..... (ii)

$\mu_2 \sin r_2 = \mu_3 \sin r_3$ ..... (iii)

$\mu_3 \sin r_3 = \sin e$ ..... (iv)

Combining all the form equation we get

$\sin i = \mu_1 \sin r_1 = \mu_2 \sin r_2 = \mu_3 \sin r_3 = \sin e$

or $\sin i = \sin e$

$\rightarrow i = e$

$\therefore$ emergent ray is parallel to incident ray. This result is independent of no. of slabs used and holds as long as incident ray and emergent ray are in same medium.

4.7 Shift due to Glass Slab

4.7.1 Lateral shift

The perpendicular distance between the incident ray and the emergent ray and the light is incident obliquely on a parallel sided refracting glass slab is called lateral shift ?

In right-angled triangle OBK, we have

$\angle BOK = i - r$

$\therefore \quad \sin(i - r) = \dfrac{d}{OB}$

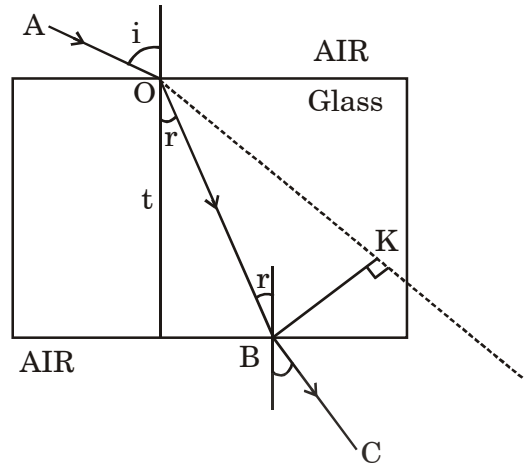
or $\quad d = OB \sin(i - r) \quad$ ... (i)

In right angled triangle ON´B, we have

$\cos r = \dfrac{ON'}{OB}$ or $OB = \dfrac{t}{\cos r} \quad$ ... (ii)

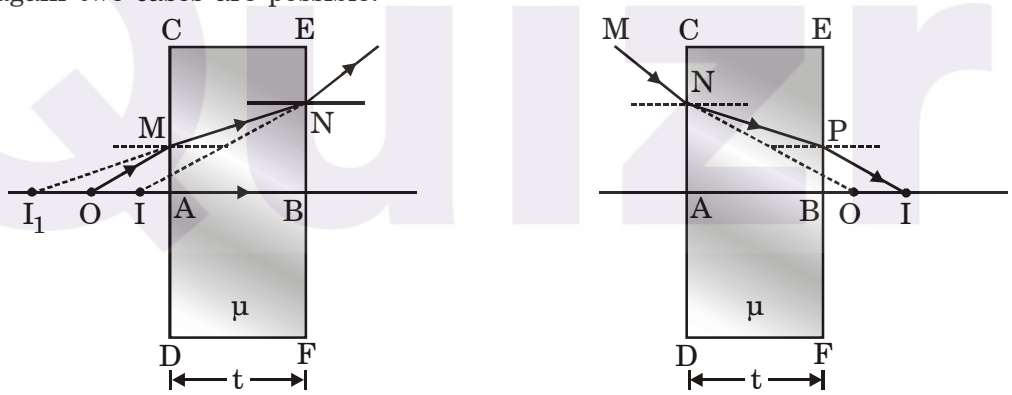
From (i) and (ii) we get

$$\boxed{d = \frac{t}{\cos r}\sin(i - r)}$$



4.7.2 Normal Shift

Here again two cases are possible.



(a) (b)

(a) An object is placed at O. Plane surface CD forms its image (virtual) at I1. This image acts as object for EF which finally forms the image (virtual) at I. Distance OI is called the normal shift and its value is,

$$\boxed{OI = \left(I - \frac{1}{\mu}\right)t}$$

This can be proved as under :

Let $\quad OA = x$

Let $AI_1 = \mu x$ (Refraction from CD)

then

$$BI_1 = \mu x + t$$

$$BI = \frac{BI_1}{\mu} = x + \frac{t}{\mu}$$ (Refraction from EF)

$\therefore$ $OI = (AB + OA) - BI$

$$= (t + x) - \left(x + \frac{t}{\mu}\right)$$

$$= \left(1 - \frac{1}{\mu}\right)t$$ **Hence Proved.**

(b) The ray of light which would had met line AB at O will now meet this line at I after two times refraction from the slab. Here $OI = \left(1 - \frac{1}{\mu}\right)t$

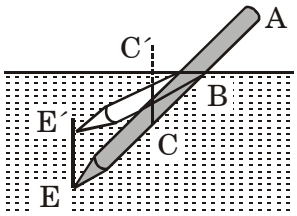
Some Illustrations of Refraction

(1) Bending of an object

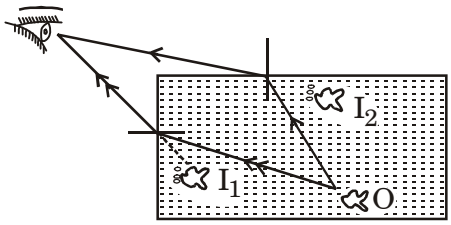
When a point object in a denser medium is seen from a rarer medium it appears to be at a depth (d/μ).So if a linear object is dipped inclined to the surface of a liquid, (say water) actual depth will be different for its different points and so apparent depth. Due to this the object appears to be inclined from its actual position BE as shown in Fig. (A).

(2) Visibility of two images of an object

When an object is in a glass container and is seen from a level higher than that of liquid in the container as shown in Fig. (B), two images I_1 and I_2 of object O can be seen simultaneously—one due to refraction at the upper surface while the other at the side surface.



(A)



(B)

(3) Twinkling of stars

Stars are self-luminous distant objects, so only a few rays of light reach the eye through the atmosphere. However, due to fluctuations in refractive index of atmosphere the refraction becomes irregular and the light sometimes reaches the eye and sometimes it does not. This gives rise to twinkling of stars. If from moon or free space we look at a star this effect will not take place and star light will reach the eye continuously.

Example 16

Monochromatic light falls at an angle of incidence 'i' on a slab of a transparent material, refractive index of this material being 'μ' for the given light. What should the relation between i and μ be so that the reflected and the refracted rays are mutually perpendicular ?

Solution :

In Fig., r is the angle of reflection and r´ the angle of refraction.

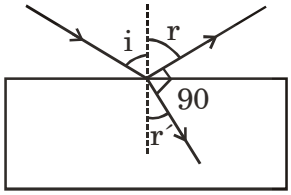
According to the given condition, considering the reflected and the refracted rays to be perpendicular to each other,

$r + 90 + r' = 180$

$\therefore \quad r' = 90$

or $\quad r' = 90 - i$ [i = r, law of reflection]

According to Snell's law,



$1 \sin i = \mu \sin r'$

$\sin i = \mu \sin (90 - i)$

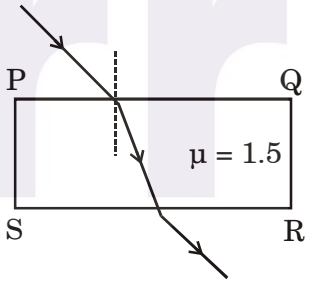
$\sin i = \mu \cos i$

$\mu = \tan i$

$i = \tan^{-1} (\mu)$ **Ans.**

Example 17

In Fig. a ray of light falls on a glass plate of thickness 6 cm and refractive index 1.5 such that it deviates through 30 due to refraction at the boundary PQ. Find the lateral displacement of the ray as it emerges out of the plate.



Solution :

In Fig. (B)

$i = r + 30$ (opposite angles)

$r = i - 30$

Applying Snell's law at the boundary PQ.

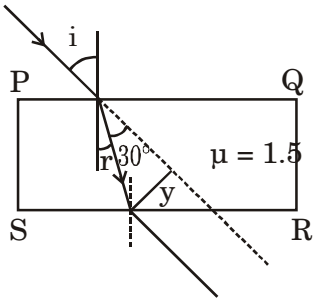
$1 \sin i = \mu \sin r$

$\sin i = 1.5 \sin (i - 30)$

$\sin i = 1.5 (\sin i \cos 30)$

$\cos i \sin 30)$

$$= 1.5\left(\frac{\sqrt{3}}{2}\sin i - \frac{\cos i}{2}\right)$$



$$\sin i = \frac{1.5 \times \sqrt{3}}{2} \sin i - \frac{1.5}{2} \cos i$$

or $\sin i = 1.3 \sin i - \cos i$

$\therefore$ $0.3 \sin i = 0.75 \cos i$

$\tan i = 2.5$

$i = \tan^{-1}(2.5) = 68.2$

Lateral displacement $y = t \sin i \left(1 - \frac{\cos i}{\sqrt{\mu^2 - \sin^2 i}}\right)$

Here $\mu = 1.5$, $t = 6$ cm, $i = 68.2$

$\therefore$ $y = 6 \sin 68.2 \left(1 - \frac{\cos 68.2°}{\sqrt{(1.5)^2 - \sin^2 68.2°}}\right)$

$y = 3.83$ cm **Ans.**

Example 18

In a tank, a 4 cm thick layer of water $\left(\mu = \frac{4}{3}\right)$ floats on a 6 cm thick layer of an organic liquid (μ = 1.48). Viewing at a normal incidence, how far below the water surface does the bottom of tank appear to be ?

Solution :

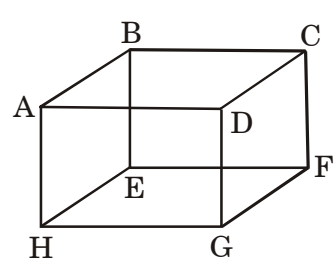
$$d_{AP} = \frac{d_1}{\mu_1} + \frac{d_2}{\mu_2}$$

$$= \frac{6}{1.48} + \frac{4}{4/3}$$

$D_{AP} = 7.05$ cm **Ans.**

Example 19

An air bubble is trapped inside a glass cube of edge 30 cm. Looking through the face ABEH, the bubble appears to be at normal distance 12 cm from this face and when seen from the opposite face CDGE, it appears to be at normal distance 8 cm from CDGF. Find refractive index of glass and also the actual position of the bubble.



Solution : In Fig. below :

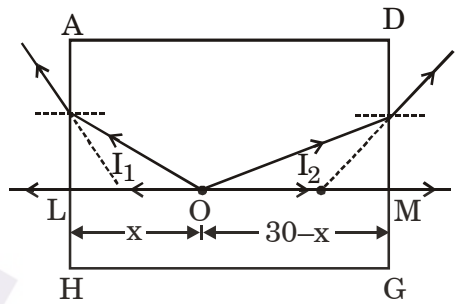
AH represents face ABEH

DG represents face CDGF

AD represents face ABCD

and HG represents face EFGH

All observations are made from outside the cube. Let O be the actual position of bubble at distance x from AH. Obviously, its real distance from DG is 30 – x. I_1 is the apparent position of bubble as seen from the side of AH and I_2 its apparent position seen from the side of DG.



Objecting being in the denser medium and observer in the rarer.

$$\mu = \frac{\text{Actual distance of bubble from refracting surface}}{\text{Apparent distance of bubble from refracting surface}}$$

here μ is refractive index of denser medium (glass) relative to the surrounding rarer maximum (air).

For refraction at AH

$$\mu = \frac{LO}{LI_1}$$

$$LI_1 = \frac{LO}{\mu} = \frac{x}{\mu}$$

but $LI_1 = 12$ cm (given)

$$\therefore \quad \frac{x}{\mu} = 12 \quad \text{... (1)}$$

$$\mu = \frac{MO}{MI_2}$$

$$MI_1 = \frac{MO}{\mu} = \frac{30-x}{\mu}$$

but $MI_2 = 8$ cm (given)

$$\therefore \quad \frac{30-x}{\mu} = 8 \quad \text{... (2)}$$

Adding Eqs. (1) and (2)

$$\frac{x}{\mu} + \frac{30-x}{\mu} = 20$$

$$\therefore \quad \mu = 1.5$$

Ans.

From Eq. (1)

$$\frac{x}{\mu} = 12$$

or $$\frac{x}{1.5} = 12$$

$\therefore$ $$x = 18 \text{ cm}$$ **Ans.**

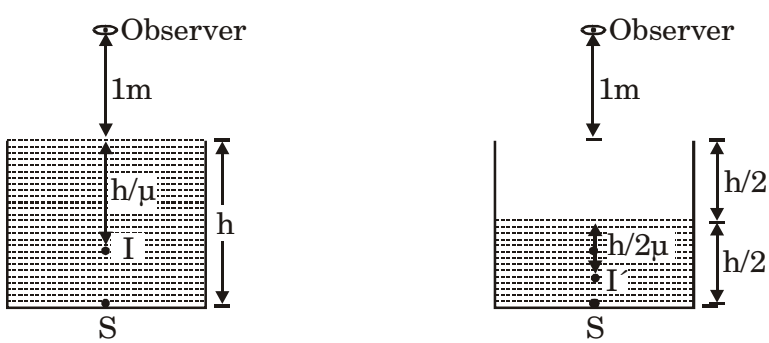
Thus actual position of bubble is at normal distance 18 cm from AH (face ABEH) or at a distance 12 cm from DG (face CDGF).

Example 20

A container completely filled with water (μ = 4/3) has a scratch at its bottom. An observer looks at the scratch from a height 1 m above the surface of water. A small hole is now made in a wall of the container and very close to its bottom so that water flows out. When the level of water reduces to half, the hole is closed. It appears to the observer that the scratch has moved away by a distance 25 cm relative to its initial position. Find the initial level (height) of water in the container.

Solution :

Initially, let water be filled to a height 'h' m. Actual depth of scratch from the surface is 'h'. However, due to refraction at water-air boundary, it will appear to be $\frac{h}{\mu}$ so that the observer finds the scratch to be at a distance $1 + \frac{h}{\mu}$. When the container is half filled, as shown in Fig. actual depth of scratch below the surface of water is h/2 but, due to refraction, it appears to be $\frac{h}{2\mu}$ to the observer located in air. In this case, the scratch appears to the observer to be at distance $1 + \frac{h}{2} + \frac{h}{2\mu}$.



(A) (B)

According to the given condition,

$$1+\frac{h}{2}+\frac{h}{2\mu} = 1+\frac{h}{\mu}+0.25 \quad [25 \text{ cm} = 0.25 \text{ m}]$$

$$\therefore \quad \frac{h}{2}-\frac{h}{2\mu} = 0.25$$

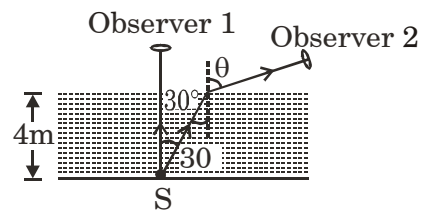
since $\mu = 4/3$

$$\frac{h}{2}-\frac{3h}{8} = 0.25$$

$$h = 2 \text{ m}$$ **Ans.**

Example 21

A point source of light is placed at the bottom of a tank filled with water $\left(\mu = \frac{4}{3}\right)$ to a depth 4 m. Find apparent depth of the source below the surface as viewed by (a) observer 1 and (b) observer 2.



Solution :

(a) Observer 1 receives light rays from the source which fall on the water air boundary at normal incidence ($i \to 0$) and we can use the relation

$$\mu = \frac{\text{Actual depth}}{\text{Apparent depth}}$$

$$\text{Apparent depth} = \frac{\text{Actual depth}}{\mu} = \frac{4}{4/3} = 3\text{m}$$ **Ans.**

(b) The relation used in (a) is not valid for observer 2 since incidence ($i = 30$) is much away from the normal.

In this situation, apparent depth can be expressed as,

$$d_{AP} = \frac{d_{AC}}{\mu}\left[\frac{\mu\cos\theta}{\sqrt{\mu^2-\sin^2\theta}}\right]$$

Here θ is the angle of vision at the surface, i.e. the angle of refraction as light passes from water to air.

Applying Snell's law,

$$1 \sin\theta = \mu \sin 30$$

$$\sin\theta = \frac{4}{3} \times \frac{1}{2} = \frac{2}{3}$$

$$\therefore \quad \cos\theta = \sqrt{1 - 4/9} = \sqrt{5/9}$$

$$d_{AP} = \frac{4}{(4/3)}\left[\frac{4/3 \times \sqrt{5/9}}{(4/3)^2 - (2/3)^2}\right]^3$$

$$d_{AP} = 1.91 \text{ m} \qquad \textbf{Ans.}$$

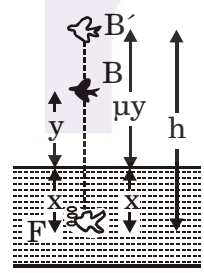
Example 22

A fish rising vertically to the surface of water in a lake uniformly at the rate of 3 m/s observes a king-fisher (bird) diving vertically towards the water at a rate of 9 m/s vertically above it. If the refractive index of water is (4/3), find the actual velocity of the dive of the bird.

Solution :

If at any instant, the fish is at a depth x below the water surface while the bird is at a height y above the surface, then the apparent height of the bird above the surface as seen by fish will be

$$\frac{d_{AC}}{d_{AP}} = \frac{\mu_1}{\mu_2}, \text{ i.e., } \frac{y}{d_{AP}} = \frac{1}{\mu} \text{ or } d_{AP} = \mu y$$



So the total total apparent distance of the bird as seen by fish in water at depth x will be

$$h = x + \mu y$$

$$\text{or} \quad \frac{dh}{dt} = \frac{dx}{dt} + \mu\frac{dy}{dt}, \text{ i.e., } 9 = 3 + \mu\frac{dy}{dt}$$

$$\frac{dy}{dt} = \frac{6}{(4/3)} = 4.5 \text{ m/s} \qquad \textbf{Ans.}$$

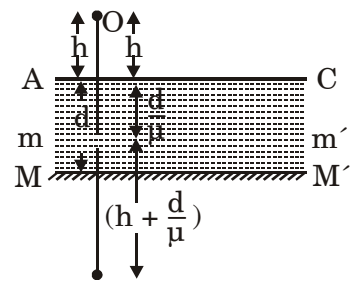
Example 23

A vessel having perfectly reflecting plane bottom is filled with water (μ = 4/3) to a depth d**. A point source of light is placed at a height** h **above the surface of water. Find the distance of final image from water surface.**

Solution :

As shown in Fig., water will form the image of bottom, i.e., mirror MM´ at a depth (d/μ) from its surface [as $d_{AC}/d_{AP}) = (\mu_1/\mu_2)$]. So the distance of object O from virtual mirror mm´ will be $h + (d/\mu)$.

Now as a plane mirror forms image behind the mirror at the same distance as the object is in front of it, the distance of image I from mm´ will be h + (d/μ) and as the distance of virtual mirror from the surface of water is (d/μ), the distance of image I from the surface of water will be



$$\left[h + \frac{d}{\mu}\right] + \frac{d}{\mu} = h + \frac{2d}{\mu} = h + \frac{3}{2}d \left[\text{as } \mu = \frac{4}{3}\right]$$ **Ans.**

Example 24

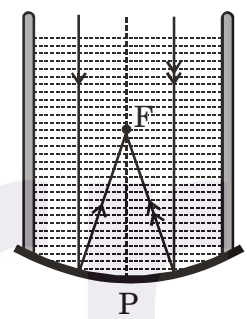
A concave mirror of radius of curvature one metre is placed at the bottom of a tank of water. The mirror forms an image of the sun when it is directly overhead. Calculate the distance of the images from the mirror for 80 cm of water in the tank.

Solution :

The focal length of the mirror is f = (R/2) = (100/2) = 50 cm and the sun is at infinity, i.e. μ = ∞; so from mirror formula we have

$$\frac{1}{v} + \frac{1}{-\infty} = \frac{1}{-50} \text{ i.e., } v = \mathbf{-50\ cm}$$

So when the length of water column in the tank is 80 cm, the image of sun will be 50 cm from the mirror as shown in Fig.



Example 25

A rectangular block of glass is placed on a printed page lying on a horizontal surface. Find the minimum value of the refractive index of glass for which the letters on the page are not visible from any of the vertical faces of the block.

Solution :

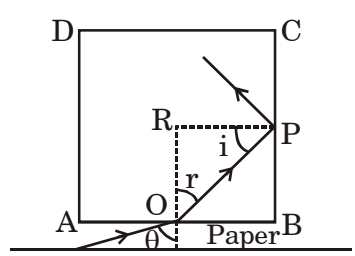
The situation is depicted in Fig. Light will not emerge out from the vertical face BC if at it

$i > \theta_C$

or $\sin i > \sin \theta_C$

or $\sin i > \frac{1}{\mu}$ ... (1)

$$\left[\text{as } \sin\theta_C = \frac{1}{\mu}\right]$$



But from Snell's law at O

$$1 \quad \sin\theta = \mu \sin r$$

And in ΔOPR,

$$r + i = 90 \text{ , i.e., } r = 90 - i$$

So $$\sin\theta = \mu \sin(90 - i),$$

i.e., $$\cos i = \frac{\sin\theta}{\mu}$$

so $$\sin i = \sqrt{1-\cos^2 i} = \sqrt{1-(\sin\theta/\mu)^2} \qquad \text{... (2)}$$

So substituting the value of sin i from Eq. (2) in (1),

$$\sqrt{1-\frac{\sin^2\theta}{\mu^2}} > \frac{1}{\mu}, \text{ i.e., } \mu^2 > 1 + \sin^2\theta$$

Now as $(\sin^2\theta)_{max} = 1$

$$\mu^2 > 2, \text{ i.e., } \mu > \sqrt{2}$$

so $$\mu_{min} = \sqrt{2}$$

Alternative Solution : According to theory of prism, light does not emerge from a prism if angle of prism A > $2\theta_C$, whatever be the angle of incidence. Now as the angle of prism is the angle between the faces on which light is incident and from which it emerges, so here A = 90 . Hence light will not emerge from vertical faces for all values of θ, if 90 > $2\theta_C$, i.e., 45 > θ_C

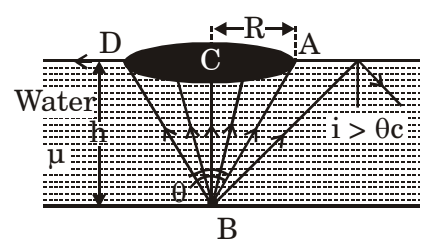
or $\sin 45 > \sin\theta_C$, i.e. $\frac{1}{\sqrt{2}} > \frac{1}{\mu}$

$$\mu > \sqrt{2}, \text{ i.e., } \mu_{min} = \sqrt{2}$$

Example 26

A bulb is placed at a depth of $2\sqrt{7}$ m in water and a floating opaque disc is placed over the bulb so that the bulb is not visible from the surface. What is the minimum diameter of the disc ?

Solution :



As shown in Fig., light from bulb will not emerge out of water if at the edge of disc, i > θ_C

or sin i > sin θ_C ..... (1)

Now if R is the radius of disc and h is the depth of bulb from it.

$$\sin i = \frac{R}{\sqrt{R^2 + h^2}} \text{ and } \sin\theta_C = \frac{1}{\mu}$$

So Eq. (1) becomes

$$\frac{R}{\sqrt{R^2 + h^2}} > \frac{1}{\mu} \text{ or } R > \frac{h}{\sqrt{\mu^2 - 1}} \qquad \text{... (2)}$$

Here $h = 2\sqrt{7}$ m and μ = (4/3)

So $$R_{min} = \frac{2\sqrt{7}}{\sqrt{(16/9) - 1}} = 6 \text{ m}$$

So diameter of disc = 2R = 2 6 = 12 m **Ans.**

Note : Here it is worthy to note that :

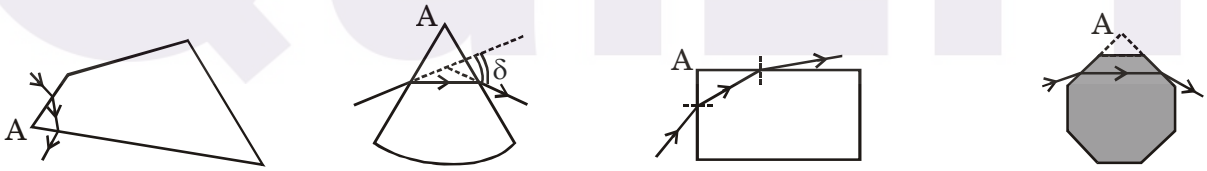
(i) If there had been no opaque disc, only that light will emerge out of the water which was initially confined in the cone of angle of vertex $\theta = 2\theta_C$.

(ii) Further in accordance with principle of reversibility, a fish or diver inside the water will see the whole outer would confined in a circular patch whose angular size at the eye's of observer is $2\theta_C = 2\sin^{-1}(1/\mu)$ while linear size $2R = \left(2h/\sqrt{\mu^2 - 1}\right)$.

5. PRISM-THEORY

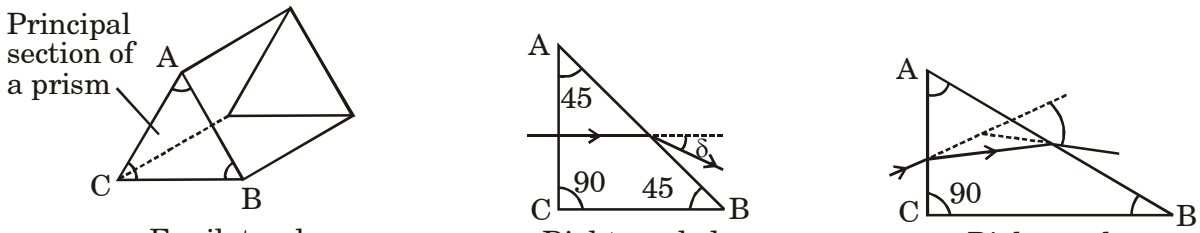
5.1 Prism

Prism is a transparent medium bounded by any number of



(A) (B) (C) (D)

surfaces in such a way that the surface on which light is incident and the surface from which light emerges are plane and non-parallel.



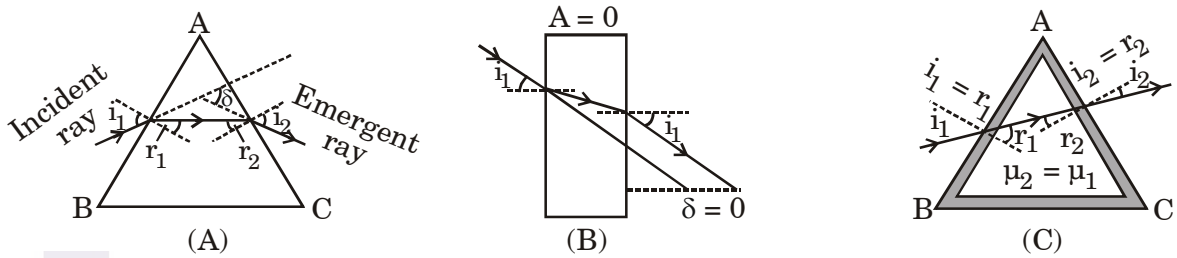
Equilateral prism (A) Right-angled isosceles prism (B) Right-angle prism (C)

Generally equilateral, right-angled isosceles or right-angled prisms are used.

With reference to prism here it is worth noting that :

(1) Angle of prism or refracting angle of prism means the angle between the faces on which light is incident and from which it emerges. In Fig. 1, A is the angle of prism.

(2) Angle of deviation means the angle between emergent and incident rays, i.e., the angle through which incident ray turns in passing through a prism. It is represented by ρ in Figs. (1) (B), (2) (B, C) and (3) (A).



(3) If the faces of a prism on which light is incident and from which it emerges are parallel, then the angle of prism will be zero and as incident ray will emerge parallel to itself, deviation will also be zero, i.e. the prism will act as a transparent plate.

(4) If μ of the material of the prism is equal to that of surroundings, no refraction at its faces will take place and light will pass through it undeviated, i.e. ρ = 0 [Fig. 3 (C)].

5.2 Condition of No emergence

The light will not emerge out of a prism for all values of angle of incidence if at face AB for i_1 = max = 90 , at face AC

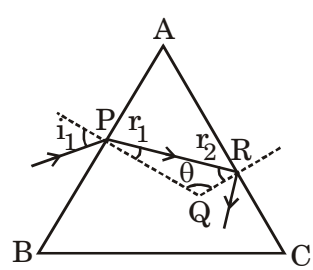
$$r_2 > \theta_C \quad \text{... (1)}$$

Now from Snell's law at face AB, we have

$$1 \quad \sin 90 = \mu \sin r_1$$

i.e. $r_1 = \sin^{-1}(1/\mu)$

or $r_1 = \theta_C$ ...(2)



So from Eqs. (1) and (2)

$$r_1 + r_2 > 2\theta_C \quad \text{... (3)}$$

However, in a prism

$$r_1 + r_2 = A^* \quad \text{... (4)}$$

So from Eqs. (3) and (4)

$$A > 2\theta_C \quad \text{... (5)}$$

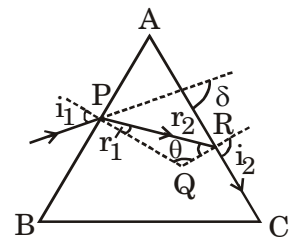
or $\sin(A/2) > \sin\theta_C$

i.e. $\mu > [\text{cosec}\,(A/2)]$ ... (6)

i.e., a ray of light will not emerge out of a prism (whatever be the angle of incidence) if $A > 2\theta_C$, i.e. if $\mu > \text{cosec}\,(A/2)$.

5.3 Condition of Grazing Emergence

If a ray can emerge out of a prism, the value of angle of incidence i_1 for which angle of emergence $i_2 = 90$ is called condition of grazing emergence. In this situations as the ray emerges out of face AC, i.e. TIR does not take place at it,



$$r_2 < \theta_C \qquad ...(1)$$

But as in a prism

$$r_1 + r_2 = A$$

So $$r_1 = A - (< \theta_C)$$

i.e., $$r_1 > A - \theta_C \qquad ...(2)$$

Now from snell's law at face AB, we have

$$1 \sin i_1 = \mu \sin r_1$$

which in the light of Eq. (2) gives

$$\sin i_1 > \mu \sin (A - \theta_C)$$

i.e., $$\sin i_1 > \mu [\sin A \cos \theta_C - \cos A \sin \theta_C]$$

i.e., $$\sin i_1 > \mu\left[(\sin A)\sqrt{\left(1 - \sin^2 \theta_C\right)} - \cos A \sin \theta_C\right]$$

i.e., $$\sin i_1 > \left[\sqrt{(\mu^2 - 1)} \sin A - \cos A\right] \qquad [\text{as } \sin \theta_C = (1/\mu)]$$

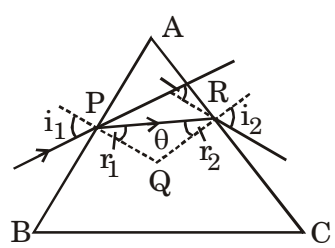
or $$i_1 > \sin^{-1}\left[\sqrt{\left(\mu^2 - 1\right)} \sin A - \cos A\right]$$

or $$(i_1)_{min} = \sin^{-1}\left[\sqrt{\left(\mu^2 - 1\right)} \sin A - \cos A\right] \qquad ...(3)$$

i.e., light will emerge out of a prism only if angle of incidence is greater than $(i_1)_{min}$ given by Eq. (3). In this situation deviation will be given by $\delta = (i_1 + 90 - A)$ with i_1 given by Eq. (3).

5.4 Deviation Produced by Prism

If light is incident on face AB of a prism of prism angle A, at an angle of incidence i_1 and emerges from face AC at an angle of emergence i_2 as shown in Fig., the angle of deviation δ will be



$$\delta = (i_1 - r_1) + (i_2 - r_2)$$

i.e. $$\delta = (i_1 - r_2) - (i_1 - r_2) \qquad ...(1)$$

Now as in ΔPQR

$$r_1 + r_2 + \theta = 180$$

and in quadrilateral PQRA

$$A + 90 + \theta + 90 = 360$$

i.e., $$A + \theta = 180$$

So $$r_1 + r_2 + \theta = A + \theta, \text{ i.e., } r_1 + r_2 = A \quad ...(2)$$

Substituting the value of $(r_1 + r_2)$ from Eq. (2) in (1), we have

$$\delta = [i_1 + i_2 - A] \text{ with } (r_1 + r_2) = A \quad ...(3)$$

This is the required result and holds good if emergent ray exists.

The following few situations are worth considering :

(a) Deviation by a prism of small angle

If angle of prism A is small, r_1 and r_2 (as $r_1 + r_2$) and hence i_1 and and i_2 will also be small. Since for small angles in $\sin\theta = \theta$. Snell's law at first and second surfaces of prism gives respectively :

$$i_1 = \mu r_1 \text{ and } \mu r_2 = i_2$$

So that, $$(i_1 + i_2) = \mu(r_1 + r_2)$$

$$= \mu A \text{ [as } r_1 + r_2 = A \quad ...(4)$$

So for small angle of prism, Eq. (3) reduces to

$$\delta = \mu A - A, \text{ i.e. } \delta = (\mu - 1)A \quad ...(5)$$

Note : Assuming the prism to be in the position of minimum deviation has been established that for small angle of prism $\delta_m = (\mu - 1)$ A. However, the formula $\delta = (\mu - 1)$ A is valid for all positions of the prism provided the angle of prism A is small (say < 10).

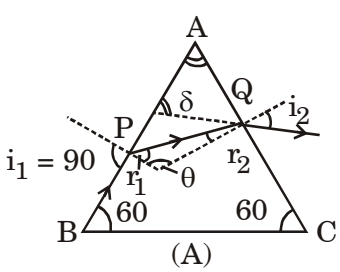
(b) Maximum deviation

Deviation will be maximum when angle of incidence i_1 is maximum, i.e. $i_1 = 90$; so Eq. (3) reduces to :

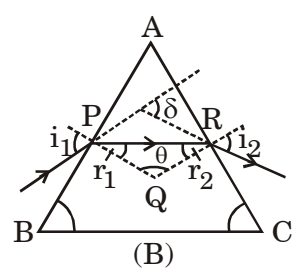
$$\delta_{max} = 90 + i_2 - A \quad ...(6)$$

However, when $$i_1 = 90 \text{ , } 1 \sin 90 = \mu \sin r_1$$

i.e., $$r_1 = \sin^{-1}(1/\mu) \text{ or } r_1 = \theta_C$$



(A)
Maximum-deviation



(B)
Minimum - deviation

and as in a prism

$$(r_1 + r_2) = A, \; r_2 = (A - \theta_C)$$

So at surface AC,

$$\mu \sin r_2 = 1 \sin i_2$$

i.e., $$\sin i_2 = \mu \sin (A - \theta_C)$$

or $$i_2 = \sin^{-1} [\mu \sin (A - \theta_C) \quad \text{... (7)}$$

So Eq. (6) gives maximum deviation while (7) angle of emergence in condition of maximum deviation i.e., $i_1 = 90$. This situation is reverse of grazing emergence, i.e., case (C), and may also be viewed as deviation at grazing incidence.

(c) Minimum deviation

$$\delta = (i_1 + i_2) - A$$

Theory and experiment show that ρ will be minimum when

$$i_1 + i_2 = i \text{ (say)} \quad \text{... (8)}$$

But when $i_1 + i_2 = i$, Snell's law at faces AB and AC respectively gives

$$1 \sin i = \sin r_1 \text{ and } \mu \sin r_2 = 1 \sin i$$

i.e., $$\sin r_1 = \sin r_2 \text{ or } r_1 = r_2 \text{ (say)} \quad \text{...(9)}$$

So in the light of Eqs. (8) and (9), (3) reduces to

$$\delta_{min} = (2i - A) \text{ with } r = (A/2) \quad \text{...(10)}$$

In this situation angle of emergence (or incidence) will be obtained by applying Snell's law, i.e.,

$$\mu \sin r = \sin i$$

or $$i = \sin^{-1} [\mu \sin (A/2)] \text{ [as } r = (A/2)] \quad \text{...(11)}$$

Here it is worth noting that in the situation of minimum deviation :

(1) Angle of incidence is equal to angle of emergence and is given by

$$i = \sin^{-1} [\mu \sin (A/2)]$$

(2) Angle of refraction inside the prism is equal to half the angle of prism, i.e.,

$$r = (A/2) = (A/2)$$

(3) As $r_1 = r_2$, $(90 - r_1) = (90 - r_2)$, so that AP = AR, and if the prism is equilateral [or isosceles with AB = AC], the ray inside the prism will be parallel to its base.

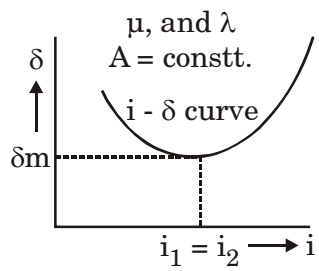
(4) As from Eq. (10), $i = (\delta_m + A/2)$ with $r = (A/2)$

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin\left[(A + \delta_m)/2\right]}{\sin(A/2)} \quad \text{...(12)}$$

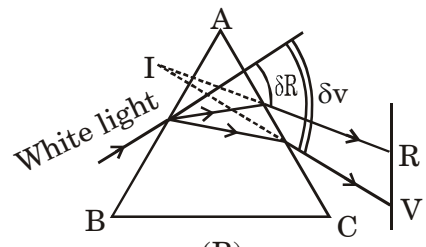
Important Points

The deviation produced by a prism depends on angle of incidence, i_1, angle of prism A, μ of the material of the prism and wavelength of light λ. So

(1) For a given A, material of prism and light, as angle of incidence increases, angle of deviation first decreases, reaches a minimum and then increases, as shown in Fig. (A).



(A)



(B)

(2) For a given material of prism, light and angle of incidence, as angle of prism increases, angle of deviation also increases,

i.e. $\delta \propto A$

e.g., deviation in 60 prism will be more than in 30 prism.

(3) For a given angle of incidence, angle of prism and light, as material of prism is changed, deviation will also change. Greater the μ or the prism, greater will be the deviation as,

$$\delta \propto (\mu - 1)$$

e.g., μ of flint glass is more than that of crown glass, so $\delta_F > \delta_C$, for same i_1, A and λ.

(4) For a given angle of incidence, angle of prism and material of prism, if wavelength of light is changed, deviation will also change as μ of a prism is different for different wavelengths. With increases in wavelength deviation decreases, i.e., deviation for red is least while maximum for violet [Fig. (B)]. (This is discussed in detail in the next section).

5.5 Dispersion of Light

When white light passes through a prism it splits up into constituent colours. This phenomenon is called **dispersion** and arises due to the fact that refractive index of a prism is different for different wavelengths. So different wavelengths in passing through a prism are deviated through different angles and as $\delta \propto (\mu - 1)$, violet is deviated most while red least giving rise to display of colours known as **spectrum** [Fig. (B)]. Rainbow, the most colourful phenomenon in nature, is primarily due to the dispersion of sunlight by rain-drops suspended in air.

In case of dispersion of light the angle between the extreme rays is a measure of dispersion and is called angular dispersion. So

Angular dispersion $\theta = (\delta_V - \delta_R)$ ...(13)

and the ratio of dispersion to mean deviation is called dispersive power ω i.e.,

$$\omega = \frac{\theta}{\delta} = \frac{\delta_V - \delta_R}{\delta} \quad ...(14)$$

But as for a prism $\delta = (\mu - 1)A$,

$$\omega = \frac{\mu_V - \mu_R}{(\mu - 1)} = \frac{d\mu}{(\mu - 1)} \quad \left\{\mu = \mu_Y = \frac{(\mu_V + \mu_R)}{2}\right\} \quad ...(15)$$

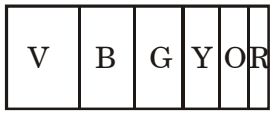
From expressions (14) and (15) it is clear that :

(1) Dispersive power like refractive index has no units and dimensions and depends on the material of the prism.

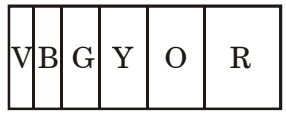
(2) As for a given prism dispersive power is constant,

$$\theta \propto \delta$$

i.e., dispersion of different wavelengths will be different and will be maximum for violet and minimum for red



Prism - spectra
(A)



Grating - spectra
(B)

(as deviation is maximum for violet and minimum for red), i.e., in the spectrum of white light produced by a prism spread of violet is most while least for red, (However, in grating spectra spread of violet is least while of red most).

(3) As for a given prism

$$\theta \propto \delta$$

a single prism produces both deviation and dispersion simultaneously, i.e., a single prism cannot give deviation without dispersion or dispersion without deviation.

However, if two prisms are combined together we can get deviation without dispersion or vice-versa. So if two prisms of prism angles Aand A´ and refractive indices μ and μ´ are placed together,

$$\delta = \delta_1 + \delta_2 = (\mu - 1)\,A + (\mu' - 1)\,A'$$

and $$\theta = \theta_1 + \theta_2 = (\mu_V - \mu_R)\,A + (\mu'_V - \mu'_R)\,A'$$

(a) For deviation without dispersion

$$\theta = 0\ ,\ \text{i.e.,}\ A' = \frac{(\mu_V - \mu_R)}{(\mu'_V - \mu'_R)}A \qquad ...(16)$$

i.e., the two prisms must be placed with their prism angles in opposite directions so that condition given by Eq. (16) may be satisfied. In this situation

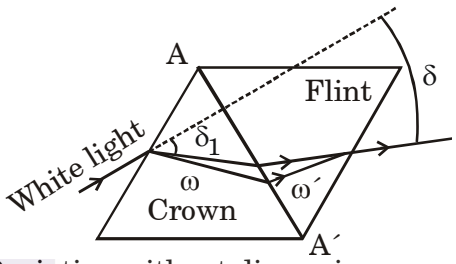
$$\delta = (\mu - 1)\,A - (\mu' - 1)\,\frac{(\mu_V - \mu_R)}{(\mu'_V - \mu'_R)}A$$

i.e. $$\delta = (\mu - 1)\,A\left[1 - \frac{(\mu_V - \mu_R)}{(\mu - 1)} \times \frac{(\mu' - 1)}{(\mu'_V - \mu'_R)}\right]$$

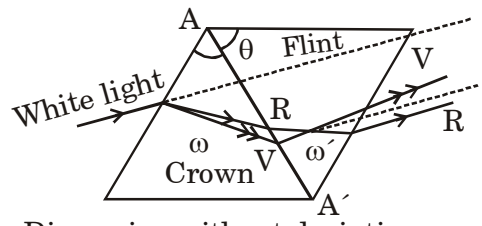
But as $\delta_1 = (\mu - 1)\,A$, $\omega = \dfrac{(\mu_V - \mu_R)}{(\mu - 1)}A$

and $\omega_1 = \dfrac{(\mu'_V - \mu'_R)}{(\mu' - 1)}$, $\delta = \delta_1[1 - (\omega/\omega')]$ ...(17)

usually $\omega' > \omega$, so δ is in the same direction as produced by the first prism [Fig. (A)].



Deviation without dispersion
(A)



Dispersion without deviation
(B)

The prism which produces deviation without dispersion is called **'achromatic prism'.**

(b) For dispersion without deviation

$$\delta = 0, \text{ i.e. } A' = \frac{(\mu - 1)}{(\mu' - 1)}A \qquad ...(18)$$

In this situation,

$$\theta = (\mu_V - \mu_R)\,A - (\mu'_V - \mu'_R)\,\frac{(\mu - 1)}{(\mu' - 1)}A$$

i.e. $$\theta = (\mu - 1)A\left[\frac{(\mu_V - \mu_R)}{(\mu - 1)} - \frac{(\mu'_V - \mu'_R)}{(\mu' - 1)}\right]$$

or $$\theta = \delta\,[\omega - \omega'] \qquad ...(19)$$

If $\omega' > \omega$, the resultant dispersion is negative, i.e., opposite to that produced by the first prism. The prism which produces dispersion without deviation is called **'direct vision prism'.**

Example 27

Find the minimum and maximum angle of deviation for a prism with angle A = 60 and μ = 1.5.

Solution : Minimum deviation :

The angle of minimum deviation occurs when i = e and $r_1 = r_2$ and is given by

$$\mu = \frac{\sin\left(\dfrac{A + \delta m}{2}\right)}{\sin A/_2} \Rightarrow \delta_m = 2\sin^{-1}(\mu \sin A/2) - A$$

Substituting $\mu = 1.5$ and $A = 60$, we get

$$\delta_m = 2 \sin^{-1}(0.75) - 60 = 37$$

Maximum Deviation : The deviation is maximum when $i = 90$ or $e = 90$ that is at grazing incidence or grazing emergence.

Let $i = 90$

$$\Rightarrow \quad r_1 = C = \sin^{-1}\left(\frac{1}{\mu}\right) \qquad \Rightarrow \qquad r = \sin^{-1}\left({}^{2}/_{3}\right) = 42°$$

$$\Rightarrow \quad r_2 = A - r_1 = 60 - 42 = 18$$

Using $\frac{\sin r_2}{\sin e} = \frac{1}{\mu}$, we have

$$\Rightarrow \quad \sin e = \mu \sin r_2 = 1.5 \quad \sin 18 = 0.462$$

$$\Rightarrow \quad e = 28$$

$$\therefore \quad \text{Deviation} = \delta_{max} = (i + e) - A = (90 + 28) - 60 = 58$$

Example 28

Calculate the dispersive power of crown and flint glass-prism from the following data :

For crown glass $\mu_V = 1.522$; $\mu_R = 1.514$ and for flint glass $\mu_V = 1.522$; $\mu_R = 1.514$

Solution :

For crown glass $\mu_V = 1.522$; $\mu_R = 1.514$

$$\mu_y = \frac{\mu_V + \mu_R}{2} = \frac{1.522 + 1.514}{2}$$

$$\therefore \quad \mu_y = 1.518$$

Hence, the dispersive power of crown glass

$$\frac{\mu_V + \mu_R}{\mu_y} = \frac{1.522 + 1.514}{1.518 - 1}$$

$$W = 0.01544$$

For flint glass $\mu_V = 1.662$; $\mu_R = 1.644$

$$\therefore \quad \mu = \frac{\mu_V + \mu_R}{2} = \frac{1.662 + 1.644}{2} = 1.6$$

Example 29

Find the angle of a prism of dispersive power 0.021 and refractive index 1.53 to form on achromatic combination with the prism of angle 4.2 and dispersive power 0.045 having refractive index 1.65. Also calculate the resultant deviation.

Solution :

$W = 0.021;\ \mu = 1.53;\ W' = 0.045;\ \mu' = 1.65$

$A' = 4.2$

For no dispersion

$$W^{\delta} + W'^{\delta'} = 0$$

or $\quad W(\mu - 1)A + W(\mu' - 1)A' = 0$

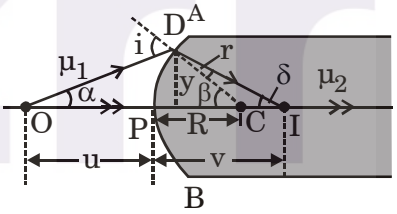
or $$A = -\frac{W'A'(\mu' - 1)}{(\mu - 1)} = \frac{0.045 \times 4.2 \times (1.65 - 1)}{0.021 \times (1.53 - 1)} = -11.4$$

Net deviation $= \delta + \delta' = (\mu - 1)A + (\mu' - 1)A'$

$= -11.04\ (1.53 - 1) + 4.2\ (1.65 - 1) = -3.12$

6. Refraction from Curved Surfaces

If the boundary between two transparent media is not plane and an object O in medium of refractive index μ_1 forms an image I in medium of refractive index μ_2 as shown in Fig., then from Snell's law at the boundary AB



$$\mu_1 \sin i = \mu_2 \sin r$$

and as for small angles,

$$\sin\theta = \tan\theta = \theta$$

the above equation reduces to

$$\mu_1\ i = \mu_2\ r \qquad ...(1)$$

But in ΔOCD $\quad i = (\alpha + \beta)$

while in ΔICD, $\quad \beta = (r + \delta)$

So substituting the values of i and r from these in Eq. (1), we get

$$\mu_1(\alpha + \beta) = \mu_2(\beta - \delta)$$

or $$\mu_1\alpha + \mu_2\delta = (\mu_2 - \mu_1)\beta$$

or $$\mu_1 \times \frac{y}{(-\mu)} + \mu_2 \times \frac{y}{(+v)} = (\mu_2 - \mu_1) \times \frac{y}{(+R)}$$

or $$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{(\mu_2 - \mu_1)}{R} \qquad ...(2)$$

This is the desired result. If we compare it with mirror formula,

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

We find that here

$$v \rightarrow (v/\mu_2) \text{ and } u \rightarrow (-\,u/\mu_1)$$

So transverse magnification in this case

$$m=\frac{I}{O}=-\frac{(v/\mu_2)}{(-\,u/\mu_1)}=\frac{\mu_1}{\mu_2}.\frac{v}{u} \qquad ...(3)$$

While using Eqs. (2) and (3) keep in mind that :

(1) These are valid for all single refracting surfaces-convex, concave or plane. In case of plane refracting surface

$R \rightarrow \infty$, so Eq. (2) reduces to

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=0 \quad \text{i.e.,} \quad \frac{u}{v}=\frac{\mu_1}{\mu_2}$$

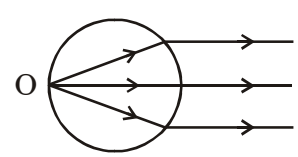
or $$\frac{d_{AC}}{d_{AP}}=\frac{\mu_1}{\mu_2}$$

(2) The rules for signs for a single reflecting surface are same as for spherical mirrors.

(3) If object or image itself is present at a refracting surface, refraction at that surface is not considered.

Example 30

An object O is stuck on the surface of a transparent solid sphere of radius 20 cm. Find refraction index of the sphere such that rays from the object after refraction from the opposite side emerge as a parallel beam, as shown. Also prove that refractive index has the same value, in the given situation, for any value of radius of sphere.



Solution :

For refraction at a spherical surface

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$$

u is the distance of object from the pole of refracting surface; it may be inferred as the normal distance of object from the refracting surface.

v is the distance of image from the pole of refracting surface; it may be inferred as the normal distance of image from the refracting surface.

μ_1 is the refractive index of the medium in which the object is located so that light is refracted while passing from medium of refractive index μ_1 to medium of refractive index μ_2.

R is the radius of curvature of the refracting surface.

In this problem,

$$\mu_2 = 1 \text{ (air)}$$

$$u = -40 \text{ cm} \qquad \text{(object is at distance 2R from the refracting surface)}$$

and $$v = \infty$$

$$\therefore \quad \frac{1}{\infty} - \frac{\mu_1}{(-40)} = \frac{1-\mu_1}{-20}$$

$$\therefore \quad \mu_1 = 2$$

Thus refractive index of the sphere is 2.

In general, for any radius of the sphere, say R.

$$\frac{1}{\infty} - \frac{\mu}{(-2R)} = \frac{1-\mu_1}{-R}$$

$$\therefore \quad \frac{\mu_1}{2R} = \frac{\mu_1 - 1}{R}$$

or $$\mu_1 = 2$$ **Ans.**

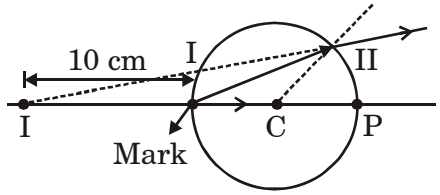
Therefore, in the given situation, refraction index of sphere will be 2 for any value of radius.

Example 31

A mark on the surface of a glass sphere (μ = 1.5) is viewed from a diametrically opposite position. It appears to be at a distance 10 cm from its actual position. Find the radius of sphere.

Solution :

As the mark is viewed from the diametrically opposite position, refraction takes place at side II of the surface (the mark being on side I as shown).



Here $\mu_1 = 1.5,\ \mu_2 = 1$

$u = -2R$

Using $$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1}{v} - \frac{1.5}{(-2R)} = \frac{1 - 1.5}{R}$$

$\therefore$ $$\frac{1}{v} = \frac{0.5}{R} - \frac{1.5}{2R} = -\frac{0.5}{2R}$$

or $v = -4R$

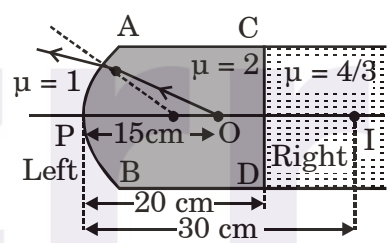
negative sign indicates that the image is formed to the left of refracting surface as shown in Fig. Further, it is given that the image of mark is at a distance 10 cm from the object.

Hence $4R = 2R + 10$

$R = 5$ cm **Ans.**

Example 32

The slab of a material of refractive index 2 has a curved surface APB of radius of curvature 10 cm and a plane surface CD. On the left of APB is air and on the right of CD is water with refractive indices as given in the figure. An object O is placed at a distance of 15 cm from the pole P as shown. What is the distance of final image of O from P as viewed from the left ?



Solution :

In case of refraction from a curved surface, we have

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{(\mu_2 - \mu_1)}{R}$$

Here $\mu_1 = 2;\quad \mu_2 = 1$

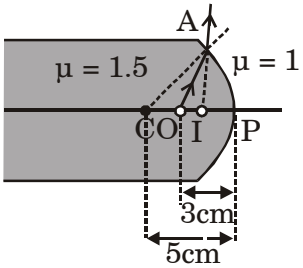
$R = -10$ cm and $u = -15$ cm

So $$\frac{1}{v} - \frac{2}{(-15)} = \frac{1-2}{-10},\ \text{i.e.,}\quad v = -30 \text{ cm}$$

i.e., the curved surface will form virtual image I at a distance of 30 cm from P. Since, the image is virtual there will be no refraction at the plane surface CD (as the rays are not actually passing through the boundary), the distance of final image I from P will remain 30 cm.

Example 33

An air bubble in glass (μ = 1.5) is situated at a distance 3 cm from a convex surface of diameter 10 cm as shown in Fig. 3.6. At what distance from the surface will the bubble appear ?



Solution :

In case of refraction from curved surface

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{(\mu_2 - \mu_1)}{R}$$

Here $\mu_1 = 1.5$; $\mu_2 = 1$; $R = -5$ cm and $u = -3$ cm

So $\frac{1}{v} - \frac{(1.5)}{(-3)} = \frac{1-1.5}{(-5)}$, i.e., $v = -2.5$ cm **Ans.**

i.e., the bubble will appear at a distance 2.5 cm from the curved surface inside the glass.

Note : If the surface had been plane, i.e, $R \to \infty$, it would yield

$$\frac{1}{v} - \frac{(1.5)}{(-3)} = \frac{(1-1.5)}{\infty}, \text{ i.e. } v = -2 \text{ cm.}$$

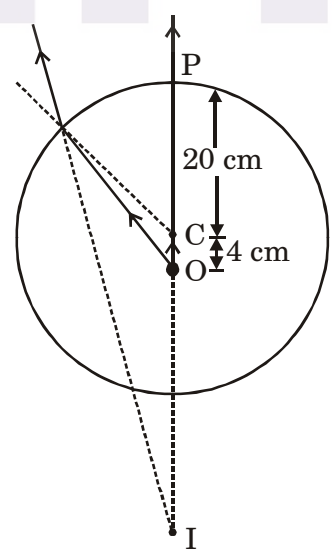
Example 34

A glass sphere (μ = 1.5) of radius 20 cm has a small air bubble 4 cm below its centre. The sphere is viewed from outside and along a vertical line through the bubble. Find the apparent depth of the bubble below the surface of sphere.

Solution : Here $u = -24$ cm

$\mu_1 = 1.5$

$\mu_2 = 1$



Using $$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1}{v} - \frac{1.5}{-24} = \frac{1-1.5}{-20}$$

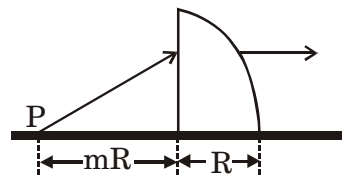
$$\frac{1}{v} = \frac{1}{40} - \frac{15}{24}$$

$\therefore$ $v = -26.67$cm

negative sign have indicates that the direction from P to I is opposite to the direction of incident light as shown. Hence, the bubble appears to be at a depth 26.67 cm from the surface of sphere.

Example 35

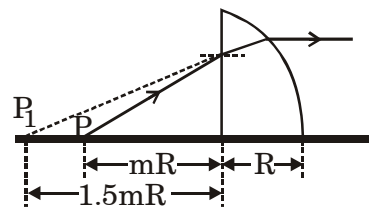
A quarter cylinder of radius R and refractive index 1.5 is placed on a table. A point object P is kept at a distance of mR from it. Find the value of m for which a ray from P will emerge parallel to the table as shown in the figure.



Solution :

For refraction at plane surface,

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$



Here $u = -mR$,

$\mu_2 = 1.5$,

$\mu_1 = 1$, $R = \infty$

$\therefore$ $v = 1.5\ mR$

For curved surface, P_1 acts as an object. For this surface,

given $\mu_1 = 1.5$, $\mu_2 = 1.0$, $u = -(1.5\ mR + R)$, $v = \infty$

$$\therefore \frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1}{\infty} + \frac{1.5}{(1.5m + 1)R} = \frac{1 - 1.5}{-R}$$

$$\therefore \frac{1.5}{1.5m + 1} = \frac{1}{2} \text{ or } 3 = 1.5\ m + 1$$

$$m = \frac{4}{3}$$ **Ans.**

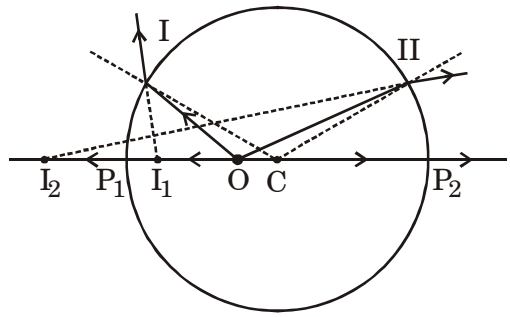
Example 36

A glass sphere (μ = 1.5) of diameter 50 cm has a small air bubble. Looking from outside along a diameter, the bubble appears to be at distance 10 cm from the surface. Find the apparent position of the bubble when it is viewed from the diametrically opposite position. Also find the actual position of the bubble.

Solution :

Looking from outside, say from the left, and along the diameter P_1P_2, the bubble which is actually at O, appears to be at I_1 such that $P_1I_1 = 10$ cm.

Using $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$



For refraction at side I of the surface

$$\mu_1 = 1.5$$
$$v = -10 \text{ cm}$$
$$R = -25 \text{ cm (diameter of sphere is 50 cm)}$$

$$\therefore \quad -\frac{1}{10} - \frac{1.5}{u} = \frac{1-1.5}{-25}$$

$$\text{or} \quad \frac{1.5}{u} = -\frac{1}{10} - \frac{1}{50}$$

$$\therefore \quad u = -12.5 \text{ cm}$$

Hence the bubble is actually at a distance 12.5 cm from P_1.

For refraction at side II of the surface

Looking from outside, this time from the right, let us apply the relation for refraction at side II. Distance of bubble (O) from P_2 = 50 – 12.5 = 37.5 cm

$$\mu_1 = -37.5 \text{ cm}$$
$$R = -25 \text{ cm}$$
$$\mu_1 = 1.5$$
$$\mu_2 = 1$$

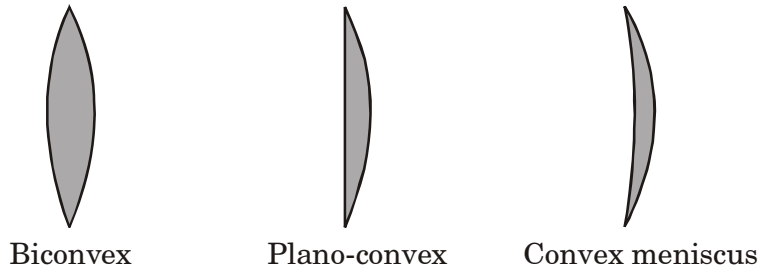
$$\frac{1}{v} - \frac{1.5}{(-37.5)} = \frac{1-1.5}{-25} \quad \text{or} \quad \frac{1}{v} = \frac{1}{50} - \frac{1.5}{37.5}$$

$$\therefore \quad v = -50 \text{ cm} \qquad \textbf{Ans.}$$

Thus the image will be formed 50 cm left of P_2 which implies that it forms at P_1 (diameter = 50 cm, given). Hence I_2 in the figure will coincide with P_1.

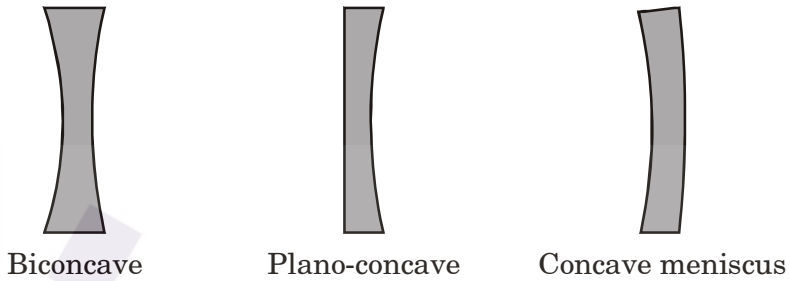
7. THIN LENSES

A lens is one of the most familiar optical devices for a human being. A lens is an optical system with two refracting surfaces. The simplest lens has two spherical surfaces close enough together that we can neglect the distance between them (the thickness of the lens). We call this a **thin lens.**



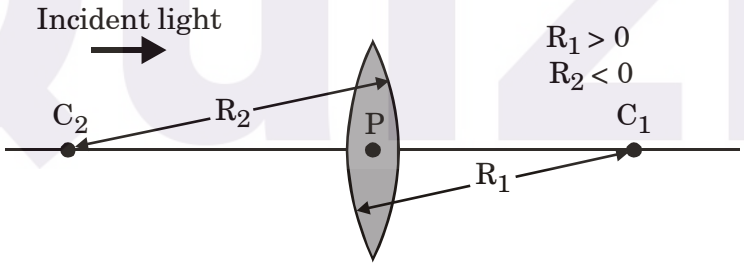
Lenses are of two basic types **convex** which are thicker in the middle than at the edges and **concave** for which the reverse holds.

Figure shows examples of both types bounded by spherical or plane surfaces.

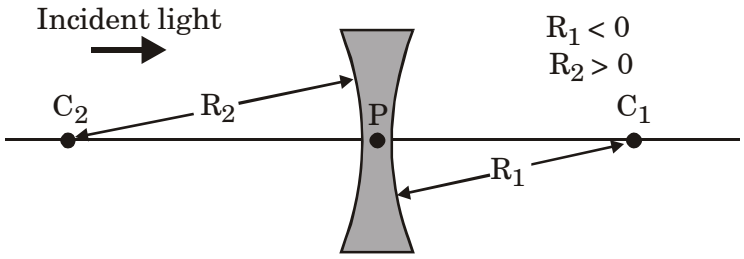


As there are two spherical surfaces, there are two centres of curvature C_1 and C_2 and correspondingly two radii of curvature R_1 and R_2.

The line joining C_1 and C_2 is called the **principal axis** of the lens. The centre P of the thin lens which lies on the principal axis, is called the **optical centre.**



(a) Converging thin lens

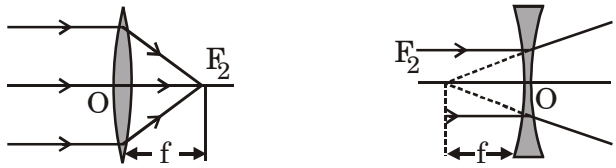


(b) a diverging thin lens

Principal Focus : A lens has two surfaces and hence two focal points. First focal point is an object point on the principal axis for which image is formed at infinity

(A) (B)

while second focal point is an image point on the principal axis for which object lies at infinity.



(A) (B)

Focal length (f) : f is defined as the distance between optical centre of a lens and the point where the parallel beam of light converges or appears to converge.

Aperture : In reference to a lens, aperture means the effective diameter of its transmitting area. So the brightness, i.e., intensity of image formed by a lens which depends on the light passing through the lens will depend on the square of aperture, i.e.

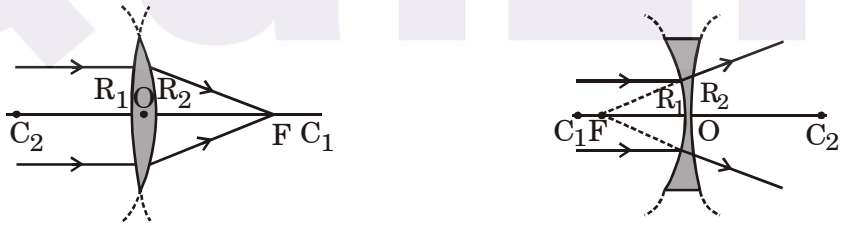
$$I \propto (\text{Aperture})^2$$

7.1 Sign Convention

It is same as explained in case of mirror, i.e.,

(1) Whenever and where ever possible, rays of light are taken to travel from left to right.

Convex lens Concave lens



R_1 (= OC_1) = Positive
R_2 (= OC_2) = Negative
f (= OF) = Positive
(A)

R_1 (= OC_1) = Negative
R_2 (= OC_2) = Positivve
f (= OF) = Nagative
(B)

(2) Transverse distances measured above the principal axis are taken to be positive while those below it negative.

(3) Longitudinal distances are measured from optical centre are taken to be positive if in the direction of light propagation and negative if opposite to it, i.e., according to our convention in case of A :

(i) To calculate an unknown quantity the known quantities are substituted with sign in a given formula.

(ii) In the result sign must be interpreted as there are number of sign conventions and same sign has different meaning in different conventions.

7.2 Rules for Image Formation

In order to locate the image formed by a lens graphically following rules are adopted :

(1) A ray passing through optical centre proceeds undeviated through the lens.

(2) A ray passing through first focus or directed towards it, after refraction from the lens, becomes parallel to the principal axis.

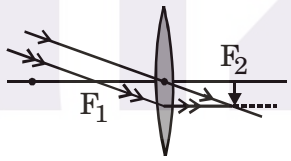
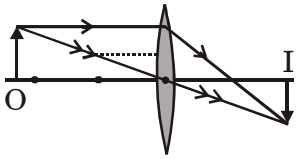
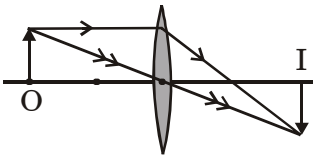
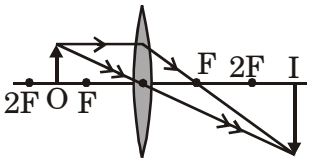
(3) A ray passing parallel to the principal axis after refraction through the lens passes or appears to pass through F_2.

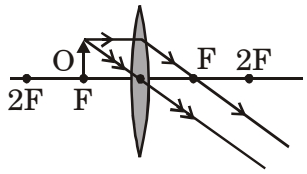
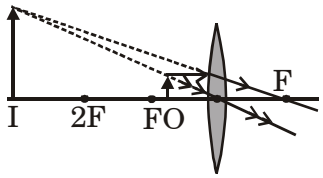
Only two rays from the same point of an object are needed for image formation and the point where the rays after refraction through the lens intersect or appear to intersect is the image of the object. If they actually intersect each other the images is real and if they are near to intersect the image is said to be virtual.

7.3 Image Formation by Lens

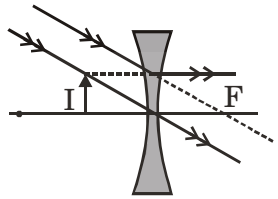
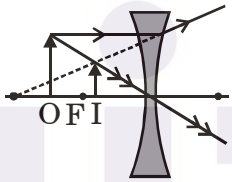
The image formed by a lens depends on the position of object and nature of lens. This all in a tabular form is given below :

(a) For Convergent or Convex Lens

S.No.	Position of Object	Ray-Diagram	Details of Image
1.	At infinity		Real, inverted Diminished (m << 1) At F
2.	Between ∞ and 2 F		Real, inverted Diminished (m < – 1) Between F and 2 F
3.	At 2 F		Real, inverted Equal (m = – 1) At 2F
4.	Between 2F and F		Real, inverted Enlarged (m > – 1) Between 2F and ∞

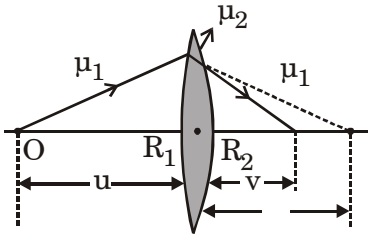
5.	At F		Read. inverted Enlarged, (m >> – 1) At infinity
6.	Between F and O		Virtual, erect Enlarged (m > + 1) Between ∞ and object on same side

(b) For Divergent or Concave Lens

S.No.	Position of Object	Ray-Diagram	Details of Image
1.	At infinity		Virtual, erect Diminished (m << + 1)) At F
2.	In front of lens		Virtual, erect Diminished (m < + 1) Between F and optical centre

7.4 Formulae

In case of image formation by a lens, the incident ray is refracted twice, at first and second surface respectively. The image formed by the first surface acts as object for the second. So from the formula for refraction at curved surface.



$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{(\mu_2 - \mu_1)}{R}$$

For first surface

$$\frac{\mu_L}{v_1} - \frac{\mu_M}{u} = \frac{(\mu_L - \mu_M)}{R_1} \quad ...(1)$$

while for second surface as,

$$u \rightarrow v_1;\ \mu_1 = \mu_L \text{ and } \mu_2 = \mu_M$$

$$\frac{\mu_M}{v} - \frac{\mu_L}{v_1} = \frac{(\mu_M - \mu_L)}{R_2} \quad ...(2)$$

So adding Eqs. (1) and (2), we have

$$\mu_M\left[\frac{1}{v}-\frac{1}{u}\right] = (\mu_L-\mu_M)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$$

or
$$\frac{1}{v}-\frac{1}{u} = (\mu-1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$$

with
$$\mu = \frac{\mu_L}{\mu_M} \qquad ...(3)$$

Now if object is at infinity, image will be formed at the focus, i.e., for $u = -\infty$, $v = f$ so that above equation becomes

$$\frac{1}{f} = (\mu-1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right] \qquad ...(4)$$

which is known as **Lens-maker's formula** and in its light Eq. (3) for a lens becomes

$$\frac{1}{v}-\frac{1}{u} = \frac{1}{f} \qquad ...(5)$$

which is known as the **Lens-formula.**

In case of lenses, transverse magnification is given by

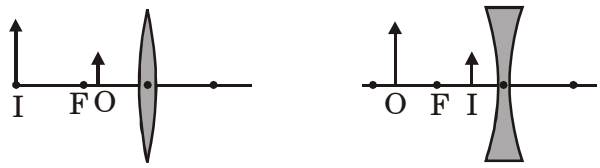
$$m = \frac{I}{O} = \frac{v}{u} \qquad ...(6)$$

while power*
$$P = \frac{1}{(f \text{ in m})}D$$

$$= \frac{100}{(f \text{ in cm})} \text{ dioptre} \qquad ...(7)$$

7.5 Important Points

(1) For real extended objects if the image formed by a single lens is erect (i..e., m is positive) it is always virtual. In this situation if the image is enlarged the lens is converging (i.e., convex) with object between focus and optical centre [Fig. (A)] and if diminished the lens is diverging (i.e., concave) with image between focus and optical centre [Fig.(B)].



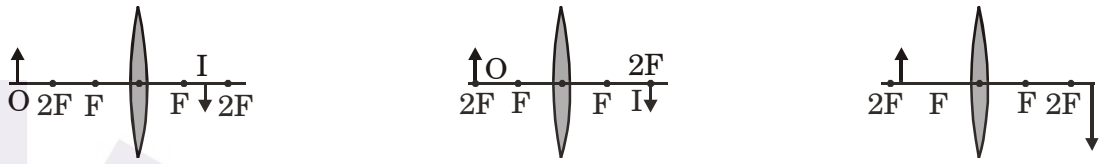
(A) (B)

(2) For real extended object, if the image formed by a single lens is inverted (i.e., m is negative) it is always real and the lens is convergent, i.e. convex. In this situation if the size of image is :

smaller than object object is between ∞ and 2F image is between F and 2F

equal to object object is at 2F image is at 2F

larger than object object is between C and F and image between C and ∞



(A) (B) (C)

(3) In case of inverted image formed by a lens the inversion is true, i.e., left is turned right and up, down as shown in Fig. (A).

(A) (B)

(4) As every part of a lens forms complete image, if a portion (say lower half) is obstructed (say covered with black paper) full image will be formed but brightness, i.e. intensity will be reduced (to half) [Fig. (B)].

(5) If an object is moved at constant speed towards a convex lens from infinity to focus, the image will move slower in the beginning and faster later on, away from the lens. This is because in the time the object moves from infinity to 2 F, the image will move from F to 2F and when the object moves from 2F to F, the image will move from 2F to infinity. At 2F the speed of object and image will be equal. In case of lens, speed of image

$$V_i = V_0\left[\frac{f}{u+f}\right]^2$$

where V_0 is the speed of object (u and f are to be substituted with proper sign).

(6) In case of sun-goggles, the radii of curvature of two surfaces are equal with centre on same side, i.e.,

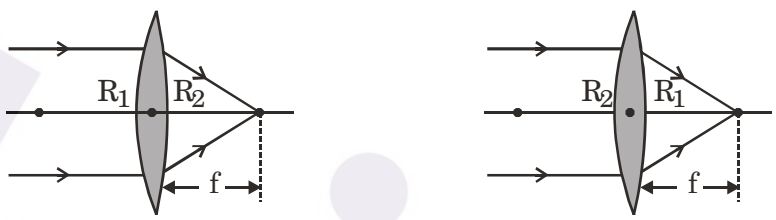
$$R_1 = R_2 = R$$

So $$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R} - \frac{1}{R}\right) = 0$$

i.e., $$f = \infty \text{ and } P = (1/f) = 0$$

This is why sun-goggles have no power or infinite focal length. Same is true for a transparent sheet but with the difference that here $R_1 = R_2 = \infty$.

(7) If the two radii of curvatures of a thin lens are not equal, the focal length remains unchanged whether the light is incident on first face or the other. This is because if we substitute R_1 and R_2 with proper sign in lens-makers formula, we always have



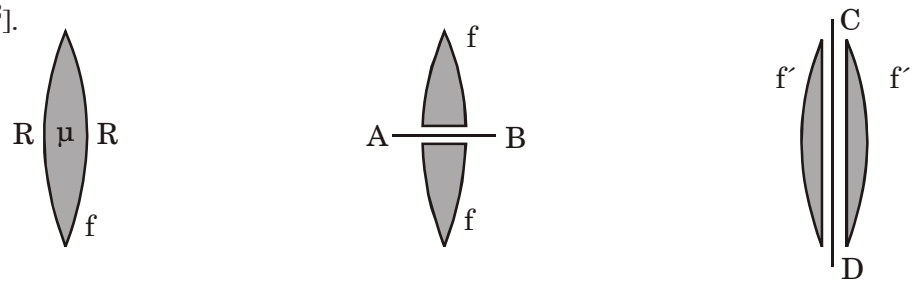
(A) (B)

$$\frac{1}{f} = (\mu - 1)\left[\frac{1}{R_1} + \frac{1}{R_2}\right]$$

(8) If an equi-convex lens of focal length f is cut into two equal parts by a horizontal plane AB Fig. B], then as none of μ, R_1 and R_2 will change, the focal length of each part will be equal to that of initial lens, i.e.,

$$\frac{1}{f} = (\mu - 1)\left[\frac{1}{R} - \frac{1}{R}\right] = \frac{2(\mu - 1)}{R}$$

However in this situation as light transmitting area of each part becomes half of initial, so intensity will be reduced to half and aperture to $\left(1/\sqrt{2}\right)$ times of its initial value [as I ∝ (Aperture)^2].



(A) (B) (C)

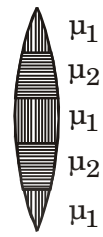
However, if the same lens is cut into two equal parts by a vertical plane CD [Fig. (C)], the focal length of each part will become

$$\frac{1}{f'} = (\mu - 1)\left[\frac{1}{R} - \frac{1}{\infty}\right] = \frac{(\mu - 1)}{R} = \frac{1}{2f}$$

i.e., $f' = 2f$

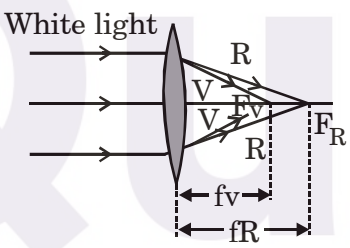
i.e., focal length of each part will be double of initial value. In this situation as the light transmitting area of each part of lens remains equal to initial, intensity and aperture will not change.

(9) If a lens is made of a number of layers of different refractive indices as shown, for a given wavelength of light will have as many focal lengths or will form as many images as there are μ's as,

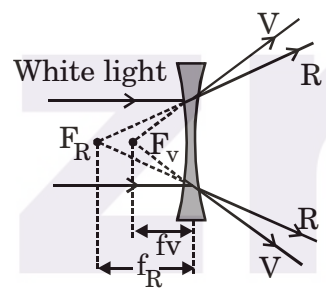


$$\frac{1}{f} \propto (\mu - 1)$$

(10) As focal length of a lens depends on μ, i.e., $(1/f) \propto (\mu - 1)$, the focal length of a given lens is different for different wavelengths and maximum for red and minimum for violet whatever be the nature of lens.



(A)



(B)

(11) If a lens of glass ($\mu = 3/2$) is shifted from air ($\mu = 1$) to water ($\mu = 4/3$) then as :

$$\frac{1}{f_A} = \left[\frac{3/2}{1} - 1\right]K \quad \text{and} \quad \frac{1}{f_W} = \left[\frac{(3/2)}{(4/3)} - 1\right]K$$

$$\text{with } K = \left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

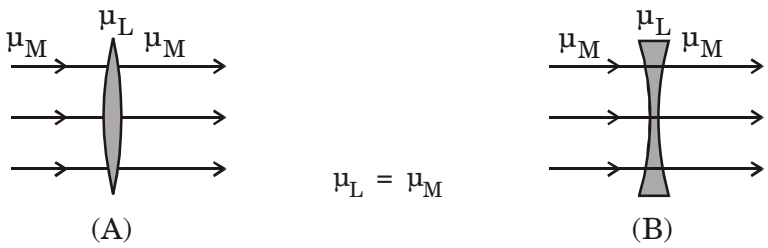
$$\therefore \quad \frac{f_W}{f_A} = \left[\frac{8}{K}\right] \times \left[\frac{K}{2}\right] \text{ i.e., } f_W = 4f_A$$

i.e., focal length of a lens in water becomes 4 times of its value in air and so power one-fourth [as P = 1 (1/f)].

(12) If a lens is shifted from one medium to the other, depending on the refractive index of the lens and medium, the following three situations are possible :

(a) $\mu_M < \mu_L$ but μ_M increases : In this situation $\mu = (\mu_L / \mu_M)$ will remain greater than unity but will decrease and as $(1/f) \propto (\mu - 1)$, $(1/f)$ will decrease, i.e., f will increase (without change in nature of lens) as explained in point (11).

(b) $\mu_M = \mu_L$. In this situation $\mu = (\mu_L/\mu_M) = 1$, so that $(1/f) \propto (\mu - 1) = 0$, i.e., $f = \infty$, lens will neither converge nor diverge but will behave as a plane glass plane.



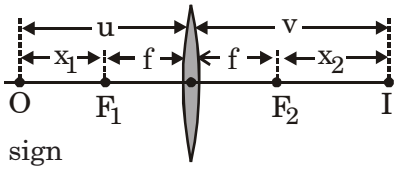
(c) $\mu_M > \mu_L$: In this situation $\mu = (\mu_L/\mu_M) < 1$, so in lens-maker's formula sign of f and hence nature of lens will change, i.e., a convergent lens will behave as divergent and vice-versa [Fig.].

Question I :

In case of a thin lens of focal length F if an object is placed at a distance x_1 from first focus and its image is formed at a distance x_2 from the second focus, show that $x_1x_2 = f^2$

Answer :

As in case of thin lens the distance of either foci from the optical centre is f,



$|u| = (f + x_1)$

and $|v| = (f + x_2)$

Substituting these values of u and v in lens-formula with proper sign

$$\frac{1}{(f + x_2)} - \frac{1}{-(f + x_1)} = \frac{1}{f} \text{ or } \frac{(x_1 + x_2 + 2f)}{(f + x_1)(f + x_2)} = \frac{1}{f}$$

i.e. $$fx_1 + fx_2 + 2f_2 = f^2 + fx_1 + fx_2 + x_1x_2$$

or $$x_1x_2 = f^2$$

Note : Formula $x_1x_2 = f_2$ is called **Newton's formula.**

Question II :

A thin converging lens of focal length f is placed between an object and a screen fixed at a distance D apart. Show that if D > 4f, there are two positions of the lens at which a sharp image of the object is formed on the screen.

Answer :

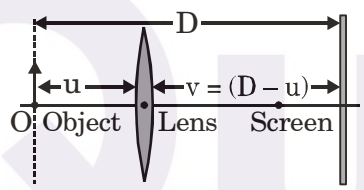
If the object is at a distance μ from the lens, the distance of image from the lens v = (D – u). So from lens-formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

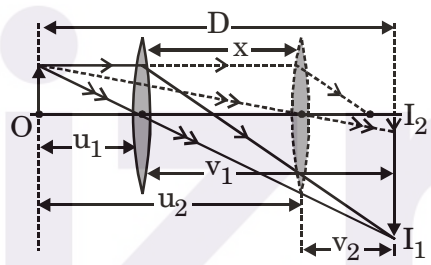
we have $$\frac{1}{(D-u)} - \frac{1}{-u} = \frac{1}{f}$$

i.e. $$u^2 - Du + Df = 0$$

so that $$u = \frac{1}{2}\left[D \pm \sqrt{D(D-4f)}\right] \quad ...(1)$$



(A)



(B)

Now there are three possibilities :

(a) If D < 4f; u will be imaginary, so physically no position of lens is possible.

(b) If D = 4f : In this situation u = D/2 = 2f. So only one position is possible and in this situation, v = D – u = 4f – 2f = 2f (= u)

(c) If D > 4f : In this situation both the roots of Eq. (1) will be real, i.e.,

$$u_1 = \frac{1}{2}\left[D - \sqrt{D(D-4f)}\right]$$

and $$u_2 = \frac{1}{2}\left[D - \sqrt{D(D-4f)}\right] \quad ...(2)$$

So if D > 4f : there are two positions of lens at distances u_1 and u_2 from the object for which real images is formed on the screen.

Note : This method is called **'Displacement method'** and is used to determine the focal length of convergent lens. In case of displacement method :

(i) If the distance between two positions of the lens is x

$$x = u_2 - u_1 = \sqrt{D(D-4f)} \text{ [From Eq. 2]}$$

i.e., $$x_2 = D^2 - 4Df \quad \text{so} \quad f = (D^2 - x^2)/4D \qquad ...(3)$$

(ii) The image distances corresponding to two positions of lens will be

$$v_1 = D - u_1 = D - \frac{1}{2}\left[D - \sqrt{D(D-4f)}\right]$$

$$= \frac{1}{2}\left[D + \sqrt{D(D-4f)}\right] = u_2$$

and $$v_2 = D - u_2 = D - \frac{1}{2}\left[D + \sqrt{D(D-4f)}\right]$$

$$= \frac{1}{2}\left[D - \sqrt{D(D+4f)}\right] = u_1$$

i.e., for two positions of the lens object and image distances are interchangeable.

(iii) As $$x = u_2 - u_1 \text{ and } D = v_1 + u_1 = u_2 + u_1 \quad [\text{as } v_1 = u_2]$$

i.e., $$u_1 (= v_2) = (D - x)/2 \text{ and } u_2 (= v_1) = (D + x)/2$$

So the magnification for two positions of the lens will be respectively,

$$\therefore \quad m_1 = \frac{I_1}{O} = \frac{v_1}{u_1} = \left[\frac{D+x}{D-x}\right]$$

and $$m_2 = \frac{I_2}{O} = \frac{v_2}{u_2} = \left[\frac{D-x}{D+x}\right] \qquad ...(4)$$

And hence :

(a) $$m_1 \quad m_2 = (I_1 I_2/O^2) = 1, \text{ i.e., } O = \sqrt{I_1 I_2} \qquad ...(5)$$

(b) $$\frac{m_1}{m_2} = \frac{I_1}{I_2} = \left[\frac{D+x}{D-x}\right] \times \left[\frac{D+x}{D-x}\right] = \left[\frac{D+x}{D-x}\right]^2 \qquad ...(6)$$

(c) $$m_1 - m_2 = \left[\frac{D+x}{D-x}\right] - \left[\frac{D-x}{D+x}\right] = \frac{4Dx}{D^2 - x^2}$$

which in the light of Eq. (3) yields

$$m_1 = m_2 = \frac{x}{f}, \text{ i.e., } f = \frac{x}{m_1 - m_2} \qquad ...(7)$$

Example 37

Diameter of a plano-convex lens is 6 cm and its thickness at the centre is 3 mm. What is the focal length of the lens if the speed of light in the material of lens is 2 10^8 m/s ?

Solution :

According to lens-maker's formula,

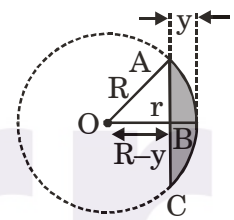
$$\frac{1}{f} = (\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right] \text{ with } \mu = \frac{\mu_L}{\mu_M}$$

As the lens is plano-convex, i.e., $R_1 = R$ and $R_2 = \infty$ [or $R_1 = \infty$ and $R_2 = -R$]

$$\frac{1}{f} = \frac{(\mu - 1)}{R} \text{ or } f = \frac{R}{(\mu - 1)} \qquad ...(1)$$

Now as speed of light in the medium of lens is 2 10^8 m/s.

$$\therefore \qquad \mu = \frac{c}{v} = \frac{3 \times 10^8}{2 \times 10^8} = 1.5 \qquad ...(2)$$



And if r is the radius and y is the thickness of lens (at the centre), the radius of curvature R of its curved surface in accordance with Fig. will be given by

$$R^2 = r^2 + (R - y)^2$$

$$\text{i.e.} \qquad R = \frac{r^2}{2y} = \frac{(6/2)^2}{2 \times 0.3} = 15 \text{ cm} \qquad ...(3)$$

So substituting the values of μ and R from Eqs. (2) and (3) in (1),

$$f = \frac{15}{(1.5 - 1)} = 30 \text{ cm}$$ **Ans.**

Example 38

A glass convex lens of refractive index (3/2) has got a focal length equal to 0.3 m. Find the focal length of the lens if it is immersed in water of refractive index (4/3).

Solution : As according to lens-maker's formula,

$$\frac{1}{f} = (\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right] \text{ with } \mu = \frac{\mu_L}{\mu_M}$$

$$\text{So} \qquad \frac{1}{f_A} = \left[\frac{(3/2)}{1} - 1\right]K$$

and $$\frac{1}{f_W} = \left[\frac{(3/2)}{(4/3)} - 1\right]K$$

with $$K = \left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

i.e., $$\frac{f_W}{f_A} = \left[\frac{8}{K}\right] \times \left[\frac{K}{2}\right] = 4$$

so $f_W = 4 \quad 0.3 = 1.2$ m **Ans.**

Example 39

A plano-convex lens has a thickness of 4 cm. When placed on a horizontal table with the curved surface in contact with it, the apparent depth of the bottom-most point of the lens is found to be 3 cm. If the lens is inverted such that the plane face is in contact with the table, the apparent depth of the centre of the plane face of the lens is found to be (25/8) cm. Find the focal length of the lens.

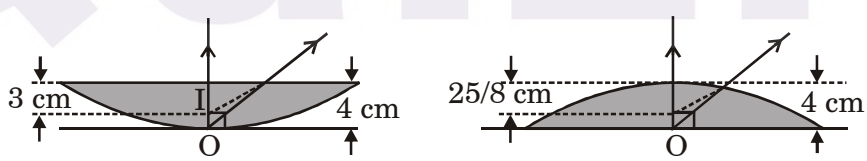
Solution :

In case of refraction from a curved surface,

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

So when curved surface is in contact with the table, refraction will take place at plane surface.

So $$\frac{1}{-3} - \frac{\mu}{-\infty} = \frac{1-\mu}{-\infty}, \quad \text{i.e.,} \quad \mu = \frac{4}{3} \quad ...(1)$$



(A) (B)

And when plane surface is on the table, refraction will take place at curved surface.

So $$\frac{1}{-(25/8)} - \frac{(4/3)}{(-4)} = \frac{1-(4/3)}{(-R)}$$

i.e., $$-\frac{8}{25} + \frac{1}{3} = \frac{1}{3R} \text{ or } R = 25 \text{ cm} \quad ...(2)$$

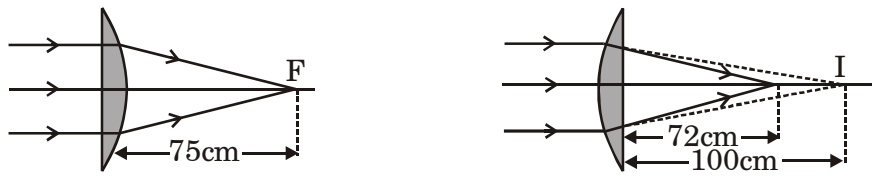
Now the lens is thick; as it has thickness 4 cm, there are two possibilities :

(a) If the plane surface faces the incident parallel beam of light [Fig. (A)] : In this situation refraction will take place only on the curved surface with

$u = -\infty$, $v = f$; $\mu_1 = (4/3)$; $\mu_2 = 1$ and $R = -25$ cm

So from $\dfrac{\mu_2}{v} - \dfrac{\mu_1}{u} = \dfrac{\mu_2 - \mu_1}{R}$

we have $\dfrac{1}{f} - \dfrac{(4/3)}{-\infty} = \dfrac{1-(4/3)}{25}$



(A) (B)

i.e., f = + 75 cm. So the focal length, i.e., distance of focus from plane surface = 4 + 75 = 79 cm

(b) If the curved surface faces the incident parallel beam of light [Fig. 3.35 (B)] : In this situation from refraction at curved surface, we have

$$\frac{(4/3)}{v_1} - \frac{1}{\infty} = \frac{(4/3)-1}{25}, \text{ i.e., } v_1 = 100 \text{ cm}$$

So the distance of I from plane surface will be 100 – 4 = 96 cm. The image I will act as virtual object for plane surface so that

$$\frac{1}{f} - \frac{(4/3)}{96} = \frac{1-(4/3)}{\infty}, \text{ i.e., } f = 72 \text{ cm}$$

i.e., in this situation focal length is 72 cm, i.e., focus will be at a distance of 72 cm from the plane surface.

Note : Calculating focal length of the lens by using lens-maker's formula.

$$\frac{1}{f} = (\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right] \text{ which gives } f = 75 \text{ cm}$$

is not proper as the lens is thick and so its focal length will depend on the fact that whether the plane or curved surface faces the incident parallel beam. Further more, if curved surface faces the incident light, u for plane surface will not be v of curved surface [but v – 4) as is taken here in case (b)]. So lens-maker's formula is not valid here.

Example 40

A converging beam of light forms a sharp image of a screen. A lens is placed in the path of the beam, the lens being 10 cm from the screen. It is found that the screen has to be moved 8 cm further away from the lens to obtain a sharp image. Find the focal length and nature of lens.

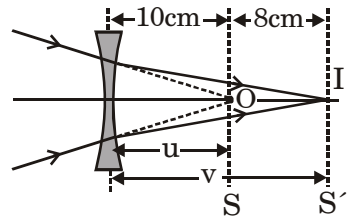
Solution :

As shown in Fig., in this problem object O is virtual while image I real, so that

u = 10 cm and v = (10 + 8) = 18 cm

and hence from lens formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$



we have $$\frac{1}{18} - \frac{1}{10} = \frac{1}{f}$$

which on simplification gives

$$f = -\frac{90}{4} = -22.5 \text{ cm}$$

i.e., the lens is diverging (i.e., concave) of focal length 22.5 cm. **Ans.**

Example 41

An object 25 cm high is placed in front of a convex lens of focal length 30 cm. If the height of image formed is 50 cm, find the distance between the object and the image.

Solution :

As object is in front of the lens, it is real and as

$$|m| = \frac{I}{O} = \frac{v}{u} = \frac{50}{25} = 2$$

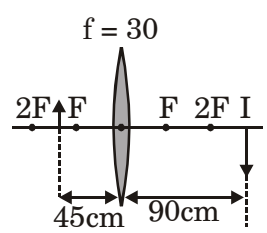
there are two possibilities :

(a) If the image is inverted (i.e., real)

$$m = \frac{v}{u} = -2, \quad \text{i.e.} \quad v = -2u$$

So from the lens-formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$



we have $$\frac{1}{-2u} - \frac{1}{u} = \frac{1}{30}$$

i.e., $$-\frac{3}{2u} = \frac{1}{30}$$

i.e., $u = -45$ cm

So $v = -2u$

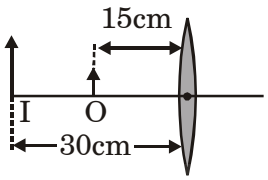
$= (-2)(-45)$

$= 90$ cm

As in this situation object and image are on opposite sides of lens, the distance between object and image $d_1 = u + v = 45 + 90 = 135$ cm. **Ans.**

(b) If the image is erect (i.e., virtual)

$$m = \frac{v}{u} = 2$$



i.e., $v = 2u$

So from lens-formula, we have

$$\frac{1}{2u} - \frac{1}{u} = \frac{1}{30}$$

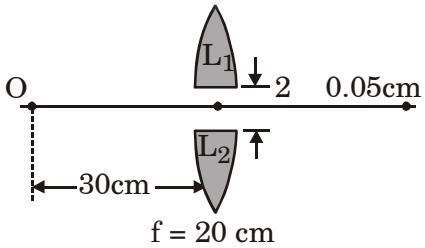
i.e., $u = -15$ cm

So $v = 2u = 2(-15) = -30$ cm

As in this situation both image and object are in front of the lens, the distance between object and image $d_2 = v - u = 30 - 15 = 15$ cm. **Ans.**

Example 42

A point object O is placed at a distance of 30 cm from a convex lens (focal length 20 cm) cut into two halves each of which is displaced by 0.05 cm as shown. Find the position of the image. If more than one image is is formed, find their number and distance between them.



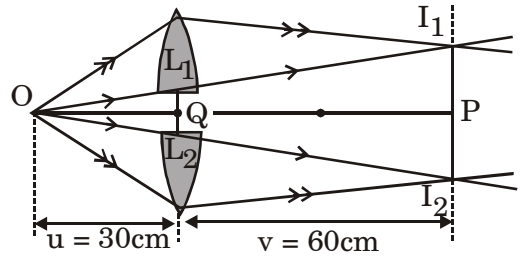
Solution :

Considering each part as separate lens with $u = -30$ cm and $f = 20$ cm, from lens-formula.

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

we have $$\frac{1}{v} - \frac{1}{-30} = \frac{1}{20} \quad \text{i.e., } v = 60 \text{ cm}$$

So each part will form a real image of the point object O at 60 cm from the lens as shown in Fig. (B). As there are two pieces, two images are formed. Now in similar triangles OI_1I_2 and OL_1L_2



$$\frac{I_1I_2}{L_1L_2} = \frac{OP}{OQ} = \frac{(u+v)}{u}$$

i.e., $$I_1I_2 = \frac{90}{30} \times (2 \times 0.05) = 0.3 \text{ cm}$$

So the images formed are 0.3 cm apart.

Example 43

A convex lens is kept 50 cm above the bottom of an empty tank. A coin is placed at the bottom of tank and its image is formed 25 cm above the lens. A liquid is now filled in the tank to a depth 40 cm. In this situation, image of the coin is formed $\frac{200}{7}$ cm above the lens. Find the refractive index of liquid.

Solution :

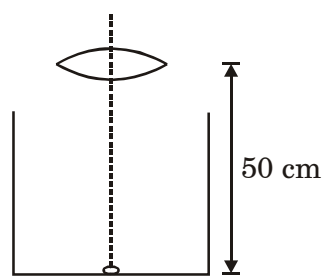
In Fig. , the tank is empty.

$$u = -50 \text{ cm}$$

$$v = 25 \text{ cm}$$

using lens-formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$



$$\frac{1}{25} + \frac{1}{50} = \frac{1}{f}$$

$$\therefore \quad f = \frac{50}{3} \text{ cm}$$

When a liquid is filled in the tank to a depth 40 cm, apparent position of the coin shifts towards the lens by

$$40\left(1-\frac{1}{\mu}\right) \text{ cm}$$

Hence distance of object from the lens now becomes

$$50 - 40\left(1-\frac{1}{\mu}\right) = 10 + \frac{40}{\mu}$$

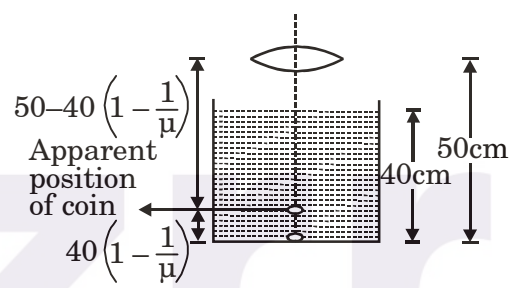
In this case,

$$u = -\left(10+\frac{40}{\mu}\right) \text{ cm}$$

$$v = \frac{200}{7} \text{ cm} \qquad \text{(given)}$$

$$f = \frac{50}{3} \text{ cm} \qquad \text{(calculated earlier)}$$

Using $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$



$$\frac{7}{200} + \frac{1}{\left(10+\frac{40}{\mu}\right)} = \frac{3}{50}$$

or $$\frac{1}{10+\frac{40}{\mu}} = \frac{3}{50} - \frac{7}{200} = \frac{1}{40}$$

$\therefore$ $$10 + \frac{40}{\mu} = 40$$

$$\mu = 1.33 \qquad \textbf{Ans.}$$

Example 44

(a) A screen is kept at a distance 1 m from an object. A converging lens between the object and the screen, when placed at any of the two positions which are 60 cm apart, forms a sharp image of the object on the screen. Find focal length of the lens.

(b) In the two positions of the lens, lateral size of the image is 4 cm and 9 cm. Find the size of the object.

Solution :

This problem is based on 'Displacement Method' which is commonly used to determine the focal length of a converging lens.

(a) $$f = \frac{D^2 - x^2}{4D}$$

D is distance between the object and the screen.

x is distance between the two positions of lens for which sharp image of the given object is obtained on the screen

here $D = 1 \text{ m} = 100 \text{ cm}$

$x = 60 \text{ cm}$

$$\therefore \quad f = \frac{(100)^2 - (60)^2}{4 \times 100}$$

or $f = 16 \text{ cm}$ **Ans.**

(b) Size of the object can be obtained from the relation

$O = \sqrt{I_1 I_2}$

I_1, I_2 lateral size of image in the two positions of lens

$\therefore \quad O = \sqrt{4 \times 9} = 6 \text{ cm}$ **Ans.**

8. COMBINATION OF LENSES AND MIRRORS

When several lenses or mirrors are used co-axially, the image formation is considered one after another in steps. The image formed by the lens facing the object serves as an object for the next lens or mirror, the image formed by the second lens (or mirror) acts as an object for the third, and so on. The total magnification in such situations will be given by

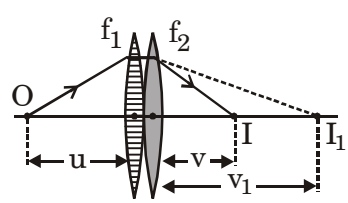
$$m = \frac{I}{O} = \frac{I_1}{O} \times \frac{I_2}{I_1} \times \ldots$$

i.e. $m = m_1 \quad m_2 \quad \ldots$

In case of two thin lenses in contact if the first lens of focal length f_1 forms the image I_1 (of an object O) at a distance v_1 from it,

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1} \qquad \ldots(1)$$

Now the image I_1 will act as an object for second lens and if the second lens forms image I at a distance v from it, then



$$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2} \quad ...(2)$$

So adding Eqs. (1) and (2), we have

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2}$$

or $$\frac{1}{v} - \frac{1}{u} = \frac{1}{F} \text{ with } \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

i.e., the combination behaves as a single lens of equivalent focal length F given by

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} \text{ or } P = P_1 + P_2 \quad ...(3)$$

Note : If the two thin lenses are separated by a distance d, F is given by

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2} \text{ so } P = P_1 + P_2 - P_1 P_2 d$$

Here it is worthy to note that :

(1) If two thin lenses of equal focal length but of opposite nature (i.e., one convergent and other divergent) are put in contact, the resultant focal length of the combination will be

$$\frac{1}{F} = \frac{1}{f} + \frac{1}{-f} = 0$$

i.e., $$F = \infty \text{ and } P = 0$$

i.e., the system will behave as a plane glass plate.

(2) If two thin lenses of same nature are put in contact, then as

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{F} > \frac{1}{f_1} \text{ and } \frac{1}{F} > \frac{1}{f_2}$$

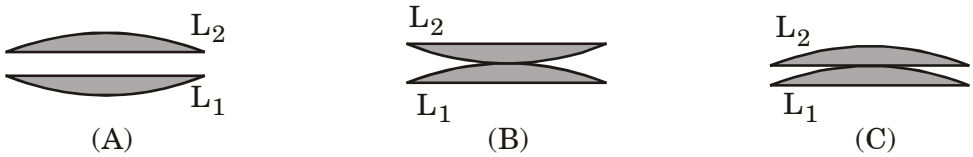
i.e. $$F < f_1 \text{ and } F < f_2$$

i.e., the resultant focal length will be lesser than smallest individual.

(3) If two thin lenses of opposite nature with different focal lengths are put in contact, the resultant focal length will be of same nature as that of the lens of shorter focal length but its magnitude will be more than that of shorter focal length.

(4) If a lens of focal length f is divided into two equal parts as shown in Fig. (A) and each part has a focal length f′ then as

$$\frac{1}{f} = \frac{1}{f'} + \frac{1}{f'} \text{ i.e., } f' = 2f$$

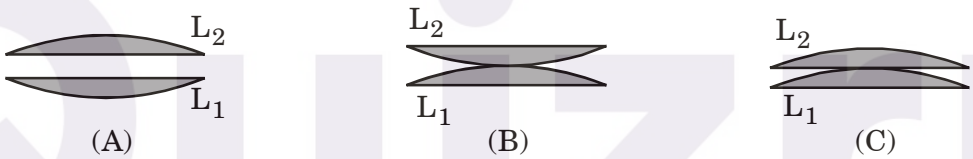


i.e., each part will have focal length 2f.

Now if these parts are put in contacts as in Fig.(B) or (C) the resultant focal length of the combination will be

$$\frac{1}{F} = \frac{1}{2f} + \frac{1}{2f}, \text{ i.e., } F = f \text{ (= initial value)}$$

(5) If a lens of focal length f is cut in two equal parts as shown in Fig. (A), each part will have focal length f. Now if these parts are put in contact as shown in Fig. (B), the resultant focal length will be



$$\frac{1}{F} = \frac{1}{f} + \frac{1}{f}, \text{ i.e., } F = (f/2)$$

However if the two parts are put in contact as shown in Fig. (C), first part will behave as convergent lens of focal length f while the other divergent of same focal length (being thinner near the axis); so in this situation

$$\frac{1}{F} = \frac{1}{f} + \frac{1}{-f}, \text{ i.e., } F = \infty \text{ or } P = 0$$

Question III :

A plane glass plate is constructed by combining a plano-convex and a plano-concave lens of different materials as shown in Fig. Will it act as a lens ? If so, what will be its focal length and nature ?

Answer :

As μ_C and μ_D are refractive indices of convergent and divergent lens respectively and R the radius of curvature of common interface, then by lens-maker's formula,

$$\frac{1}{f_C} = (\mu_C - 1)\left[\frac{1}{\infty} - \frac{1}{-R}\right] = \frac{(\mu_C - 1)}{R} \quad ...(1)$$

and

$$\frac{1}{f_D} = (\mu_D - 1)\left[\frac{1}{-R} - \frac{1}{\infty}\right]$$

$$= \frac{-(\mu_D - 1)}{R} \quad ...(2)$$

Now as the lenses are in contact,

$$\frac{1}{F} = \frac{1}{f_C} + \frac{1}{f_D} = \frac{(\mu_C - \mu_D)}{R}$$

i.e.,

$$F = \frac{R}{(\mu_C - \mu_D)}$$

As $\mu_C \neq \mu_D$, the system will act as a lens. The system will behave as convergent lens if $\mu_C > \mu_D$(as its focal length will be positive) and as divergent lens if $\mu_C < \mu_D$ (as F will be negative).

Example 45

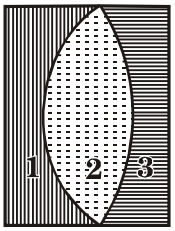
Two plano-concave lenses of glass of refractive index 1.5 have radii of curvature of 20 and 30 cm. They are placed in contact with curved surfaces towards each other and the space between them is filled with a liquid of refractive index (4/3). Find the focal length of the system.

Solution :

As shown in Fig, the system is equivalent to combination of three thin lenses in contact,

i.e.,

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$$



But by lens-maker's formula,

$$\frac{1}{f_1} = \left[\frac{3}{2} - 1\right]\left[\frac{1}{-\infty} - \frac{1}{20}\right]$$

$$= -\frac{1}{40}$$

$$\frac{1}{f_2} = \left[\frac{4}{3} - 1\right]\left[\frac{1}{20} + \frac{1}{30}\right]$$

$$= \frac{5}{180}$$

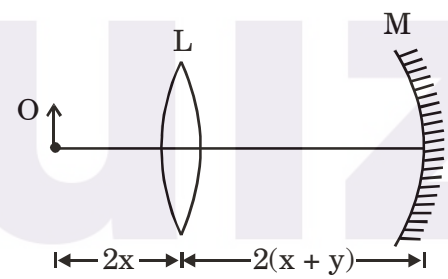
$$\frac{1}{f_3} = \left[\frac{3}{2} - 1\right]\left[\frac{1}{-30} - \frac{1}{\infty}\right] = -\frac{1}{60}$$

So $\frac{1}{F} = -\frac{1}{40} + \frac{5}{180} - \frac{1}{60}$, i.e. $\frac{1}{F} = \frac{-9+10-6}{360}$ or $F = -72$ cm.

i.e., the system will behave as a divergent lens of focal length 72 cm. **Ans.**

Example 46

In the given figure, L is a lens such that a parallel beam of light incident on it, after refraction, converges to a point at a distance 'x' from it and M is a mirror such that a parallel beam of light incident on it, after reflection, converges to a point at distance y from it. An object of size 2 cm is kept as shown. Find the nature, position and size of the image that will be seen by an observer looking towards the mirror through the lens.



Solution :

It is given that a parallel beam incident on the lens, after refraction, converges to a point at distance x from the lens. **This implies that focal length of the lens is 'x'.**

It is also given that a parallel beam incident on the mirror, after reflection, converges to a point at distance 'y' from it. **This implies that the mirror is a concave of focal length 'y'.**

Obviously, the object O is kept at 2F of lens L. Hence its image I_1 forms on the other side at 2F, i.e., at distance 2x right of the lens. The image will be real, inverted and magnification of this image

$$m_1 = \frac{v}{u} = \frac{2x}{-2x} = -1.$$

Image I_1 of O formed by lens L acts as object for the mirror. Its distance from the pole of mirror will be $2(x + y) - 2x = 2y$. This implies that it will be at the centre of curvature of the concave mirror (y being the focal length of mirror). Therefore, the concave mirror forms a real, inverted (relative to I_1)

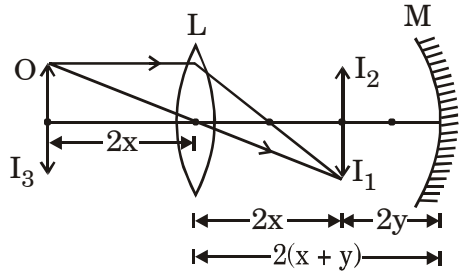
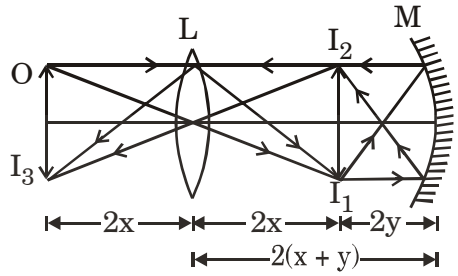


image I_2 at the same position. Magnification of I_2 (relative to I_1), $m_2 = -1$. Finally, I_2 acts as object for the lens. Its distance from the lens is 2x. Hence, it is at 2F of the lens (x being the focal length of lens). Therefore, the lens will form a real and inverted image of I_2 at 2F on the other side, i.e., to its left. Obviously, I_3 will be at the same position as of the object.



Magnification of I_3 (relative to I_2) $= -1$.

Total magnification $= m_1 \quad m_2 \quad m_3$

$= -1 \quad -1 \quad -1$

$= -1$

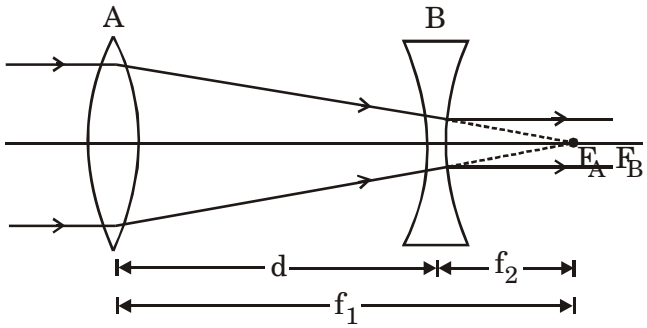
Thus the final image is real and inverted and of the same size (2 cm). **Ans.**

Example 47

A convex lens 'A' of focal length 'f_1' and a concave lens 'B' of focal length 'f_2' are kept along the same axis with a distance 'd' between them. For what value of 'd' does a parallel beam of light incident on A leave B as a parallel beam ?

Solution :

A parallel beam incident on the convex lens A, after refraction, will converge towards its focus. If the concave lens B is so placed that the point of convergence of rays, which is the focus of lens A,



also becomes the first principal focus of lens B, then the rays leaving B will form a parallel beam. In the other words, the lens B intercepts the rays which are converging towards its focus so that the rays after refraction become parallel. This is shown in the given figure.

Applying the relation $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ for relation at B.

The point F_A or F_B acts as virtual object for the image to form at ∞.

$$\therefore \quad u = f_1 - d$$
$$v = \infty$$
$$f = -f_2$$

$$\frac{1}{\infty} - \frac{1}{(f_1 - d)} = -\frac{1}{f_2}$$

$$\therefore \quad f_1 - d = f_2$$

or $$d = f_1 - f_2$$

Therefore, in the given situation, the two lenses have to be kept at a separation that is equal to the difference of their focal lengths.

Example 48

A concave lens of focal length 20 cm is placed 15 cm in front of a concave mirror of radius of curvature 26 cm and further 10 cm away from the lens is placed an object. The principal axis of the lens and the mirror are coincident and the object is on this axis. Find the position and nature of the image.

Solution :

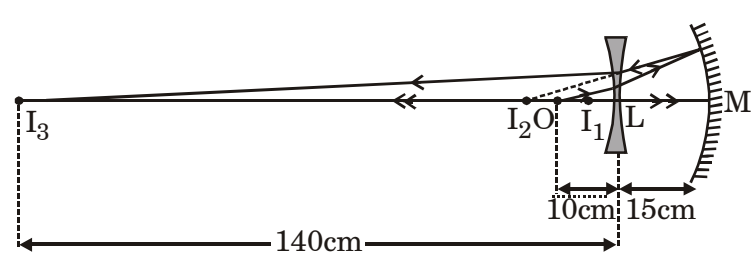
The lens will from the image I_1 of the object O at distance v from it such that

$$\frac{1}{v} - \frac{1}{-10} = \frac{1}{-20} \quad \text{i.e.,} \quad v = -\frac{20}{3} \text{ cm}$$

i.e., at a distance (20/3) cm in front of the lens. So the distance of this image I_1 from the mirror will be 15 + (20/3) = (65/3) cm.

The image I_1 will act as an object for the mirror and hence, the mirror will form image I_2 (of object I_1) such that

$$\frac{1}{v} + \frac{-3}{65} = \frac{1}{-13} \quad \text{i.e., } v = -32.5 \text{ cm}$$



i.e., the mirror will form image I_2 at a distance of 32.5 cm in front of it. However, as the lens at a distance of 15 cm from the mirror, the image I_2 will act as an object for the lens again u = 32.5 – 15 = 17.5 cm, so that

$$\frac{1}{v} - \frac{2}{+35} = \frac{1}{-20} \quad \text{i.e.,} \quad v = 140 \text{ cm}$$

i.e., final image I_3 is at a distance of 140 cm in front of the lens and as here

$$m = m_1 \quad m_2 \quad m_3$$

$$\text{i.e.,} \quad m = \left[\frac{(-20/3)}{(-10)} \times (-1)\frac{(-65/2)}{(-65/3)} \times \frac{(140)}{(35/2)}\right] = -8$$

Final image will be inverted, real and 8 times of the object.

Example 49

Consider a co-axial system of two thin convex lenses of focal length f each separated by a distance d. Draw ray diagrams for image formation corresponding to an object (a) d < f (b) (c) f < d < 2f (d) d = 2f and (e) d > 2f.

Indicate the nature of the combination (concave, convex or plane) in each case.

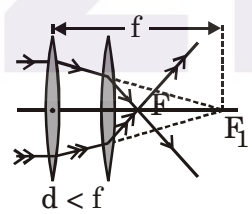
Solution :

In case of two thin lenses separated by a distance d, we have

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$$

For $f_1 = f, f_2 = f,$

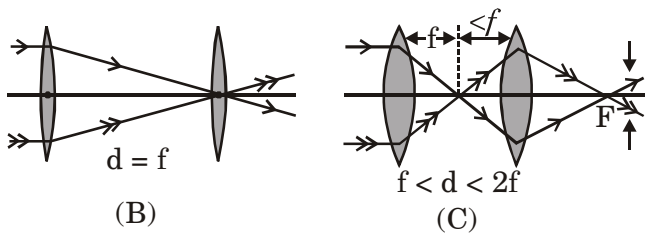
$$\frac{1}{F} = \frac{1}{f} + \frac{1}{f} - \frac{d}{f^2}$$



i.e., $$F = \frac{f^2}{2f - d}$$

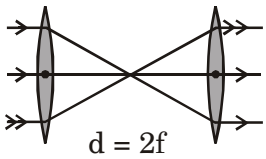
So (a) **If** d **<** f **:** F will be positive and < f, so the system will behave as convex lens of focal length < f

(b) **If** d **=** f **:** F =f, i.e., the system will behave as convex lens of focal length f

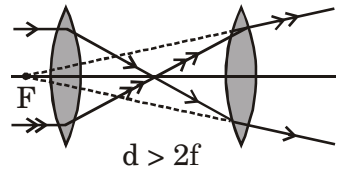


(c) **If** f **<** d **< 2**f **:** F will be positive and > f, so the system will behave as convex lens of focal length > f as shown in Fig.(C).

(d) **If** d **= 2**f **:** F will be infinite, i.e., the system will behave as a plane glass plate of infinite focal length [Fig.(D)].



(D)

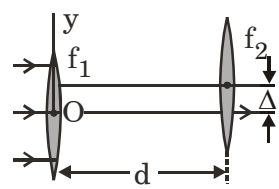


(E)

(e) **If** d **> 2**f **:** F will become negative, i.e., the system will behave as concave lens as shown in Fig. (E).

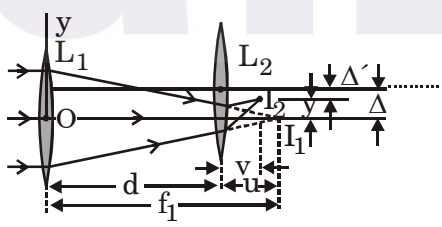
Example 50

Two thin convex lenses of focal lengths f_1 and f_2 are separated by a horizontal distance d ($d < f_1$ and $d < f_2$) and their centres are displaced by a vertical separation Δ as shown in Fig. 3.67 (A). Taking the origin of co-ordinates O at the centre of first lens, what would be the x and y co-ordinates of the focal point of this lens system for a parallel beam of rays coming from left ?



Solution :

As the incident beam is parallel, first lens L_1, in absence of second lens L_2, will form the image I_1 at its focus, i.e., at a distance f_1 from O. This image I_1 will act as an object for second lens and so for second lens $u = (f_1 - d)$. If the second lens forms image I_2 of I_1 at a distance v from it,



$$\frac{1}{v} - \frac{1}{(f_1 - d)} = \frac{1}{f_2}, \quad \text{i.e.,} \quad v = \frac{f_2(f_1 - d)}{(f_1 + f_2 - d)} \qquad ...(1)$$

So the x co-ordinate of final image I_2 from O,

$$x = d + v = d + \frac{f_2(f_1 - d)}{(f_1 + f_2 - d)}$$

$$= \frac{d(f_1 - d) + f_1 f_2}{(f_1 + f_2 - d)} \qquad ...(2)$$

Also as for the second lens

$$m = \frac{I}{O} = \frac{v}{u}$$

[here $I = \Delta'$; $O = \Delta$; $u = (f_1 - d)$ and v is given by Eq. (1)]

So $$\frac{\Delta'}{\Delta} = \frac{f_2(f_1 - d)}{(f_1 + f_2 - d)(f_1 - d)}$$

i.e., $$\Delta' = \Delta\frac{f_2}{(f_1 + f_2 - d)}$$

$\therefore$ $$y = \Delta - \Delta' = \Delta\left[1 - \frac{f_2}{(f_1 + f_2 - d)}\right]$$

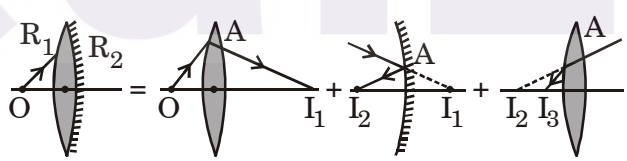
$$= \Delta\left[\frac{f_1 - d}{(f_1 + f_2 - d)}\right] \quad ...(3)$$

Eqs. (2) and (3) are the desired results.

9. LENS WITH ONE SILVERED SURFACE

If the back surface of a lens is silvered and an object is placed in front of it then :

(1) First, light will pass through the lens and it will form the image I_1.



(A) (B) (C) (D)

(2) The image I_1 will act as an object for silvered surface which acts as curved mirror and forms an image I_2 of object I_1.

(3) The light after reflection from silvered surface will again pass through the lens and lens will form final image I_3 of object I_2.

In such situation power of the silvered lens will be

$$P = P_L + P_M + P_L$$

with $$P_L = \frac{1}{f_L} \text{ where } \frac{1}{f_L} = (\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

and $\quad P_M = -\dfrac{1}{f_M}$ where $f_M = \dfrac{R_2}{2}$

So the system will behave as a curved mirror of focal length F given by

$$F = -\frac{1}{P}$$

To make this all clear we now consider the case of a silvered plano-convex lens under following circumstances.

(A) When the plane surface is silvered and the object is in front of curved surface : In this situation

$$\frac{1}{f_L} = (\mu - 1)\left[\frac{1}{R} - \frac{1}{\infty}\right] = \frac{(\mu - 1)}{R}$$

and $\quad F_M = \dfrac{\infty}{2} = \infty$

So $\quad P_L = \dfrac{1}{f_L} = \dfrac{(\mu - 1)}{R}$

and $\quad P_M = -\dfrac{1}{f_M} = \dfrac{1}{\infty} = 0$

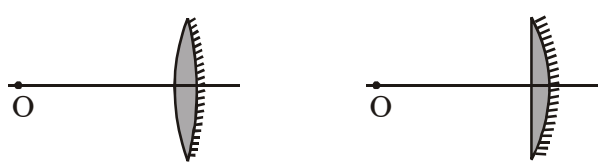
and hence power of system

$$P = P_L + P_M + P_L = 2P_L + P_M$$

i.e., $\quad P = 2\dfrac{(\mu - 1)}{R} + 0 = \dfrac{2(\mu - 1)}{R}$ ...(1)

So $\quad F = -\dfrac{1}{P} = -\dfrac{R}{2(\mu - 1)}$ ...(2)

i.e., the lens will behave as a concave mirror of focal length [R/2(μ – 1)].



(A) (B)

(B) When the curved surface is silvered and the object is in front of plane surface : In this situation

$$\frac{1}{f_L} = (\mu - 1)\left[\frac{1}{\infty} - \frac{1}{-R}\right] = \frac{(\mu - 1)}{R}$$

and $$F_M = \frac{(-R)}{2}$$

So $$P_L = \frac{1}{f_L} = \frac{(\mu - 1)}{R} \text{ and } P_M = -\frac{1}{f_M} = \frac{2}{R}$$

and hence power of system

$$P = P_L + P_M + P_L = 2P_L + P_M$$

i.e., $$P = \frac{2(\mu - 1)}{R} + \frac{2}{R} = \frac{2\mu}{R} \quad ...(3)$$

So $$F = -\frac{1}{P} = -\frac{R}{2\mu} \quad ...(4)$$

i.e., the lens will be equivalent to a converging mirror of focal length (R/2μ).

Example 51

A pin is placed 10 cm in front of a convex lens of focal length 20 cm, made of material having refractive index 1.5. The surface of the lens farther away from the pin is silvered and has a radius of curvature 22 cm. Determine the position of the final image. Is the image real or virtual?

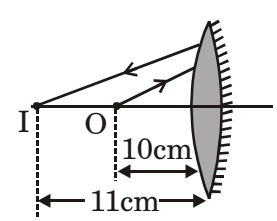
Solution :

As radius of curvature of silvered surface is 22 cm, so

$$f_M = \frac{R}{2} = \frac{-22}{2} = -11 \text{ cm} = -0.11 \text{ m}$$

and hence $$P_M = -\frac{1}{f_M} = -\frac{1}{-0.11} = \frac{1}{0.11} D$$

Further as the focal length of lens is 20 cm, i.e., 0.20 m, its power will be given by :



$$P_L = \frac{1}{f_L} = \frac{1}{0.20} D$$

Now as in image formation, light after passing through the lens will be reflected back by the curved mirror through the lens again

$$P = P_L + P_M + P_L = 2P_L + P_M$$

i.e.,
$$P = \frac{2}{0.20} + \frac{1}{0.11} = \frac{210}{11} D$$

So the focal length of equivalent mirror

$$F = -\frac{1}{P} = -\frac{11}{210} m = \frac{110}{21} cm$$

i.e., the silvered lens behaves as a concave mirror of focal length (110/21) cm. So for object at a distance 10 cm in front of it,

$$\frac{1}{v} + \frac{1}{-10} = -\frac{21}{110} \quad \text{i.e., } v = -11 \text{ cm}$$

i.e., image will be 11 cm in front of the silvered lens and will be real.

10. MICROSCOPE

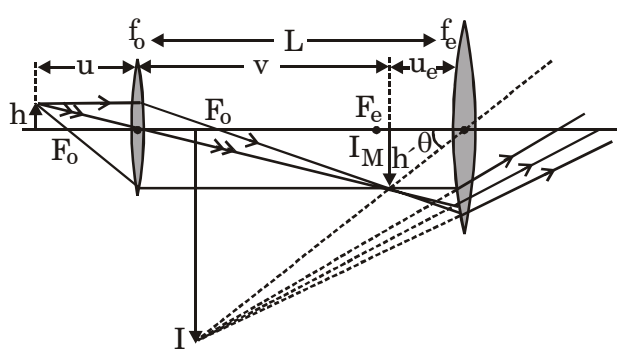
It is an optical instrument used to increase the visual angle of near objects which are too small to be seen by naked eye.

10.1 Construction

It consists of two convergent lenses of short focal lengths and apertures arranged co-axially. Lens (of focal length f_0) facing the object is called objective or field lens while the lens (of focal length f_e) facing the eye, eye-piece or ocular. The objective has a smaller aperture and smaller focal length than eye-piece. The separation between objective and eye-piece can be varied.

10.2 Image Formation

The object is placed between F and 2F of objective, so the image I_M formed by objective (called intermediate image) is inverted, real, enlarged and a distance greater than $2f_0$ on the other side of the lens. This image I_M acts as object for eye-piece and is within its focus. So eye-piece forms final image I which is erect, virtual and enlarged with respect to intermediate image I_M. So the final image I with respect to object is inverted, virtual enlarged and at a distance D to from eye on the same side of eye-piece as I_M.



10.3 Magnifying Power (MP)

Magnifying power of an optical instrument is defined as :

$$MP = \frac{\text{Visual angle with instrument}}{\text{Max. visual angle for unaided eye}} = \frac{\theta}{\theta_0}$$

If the size of object is h and least distance of distinct vision is D

$$\theta_0 = \frac{h}{D} \text{ and } \theta = \frac{h'}{u_e}$$

So $$MP = \left[\frac{h'}{u_e}\right] \times \left[\frac{D}{h}\right] = \left[\frac{h'}{h}\right]\left[\frac{D}{u_e}\right]$$

But for objective

$$m = \frac{I}{O} = -\frac{v}{u}$$

i.e., $$\frac{h'}{h} = -\frac{v}{u} \quad \text{[as u is – ive]}$$

So, $$MP = -\frac{v}{u}\left[\frac{D}{u_e}\right]$$

with length of tube $$L = v + u_e \quad ...(1)$$

Now there are two possibilities :

(b_1) If the final image is at infinity (Far point)

This situation is called normal adjustment as in this situation eye is least strained or relaxed. In this situation as for eye-piece $v = \infty$,

$$\frac{1}{-\infty} - \frac{1}{-u_e} = \frac{1}{f_e} \text{ i.e., } u_e = f_e = \text{maximum}$$

Substituting this value of ue in Eq. (1), we have

$$MP = -\frac{v}{u}\left[\frac{D}{f_e}\right] \text{ with } L = v + f_e \quad ...(2)$$

A microscope is usually considered to operate in this mode unless stated otherwise. In this mode as u_e is maximum, MP is minimum for a given microscope.

(b_2) If the final image is at D (Near point)

In this situation as for eye-piece v = D

$$\frac{1}{-D} - \frac{1}{-u_e} = \frac{1}{f_e}, \text{ i.e., } \frac{1}{u_e} = \frac{1}{D}\left[1 + \frac{D}{f_e}\right]$$

Substituting this value of u_e in Eq. (1), we have

$$MP = -\frac{v}{u}\left[1 + \frac{D}{f_e}\right]$$

with $$L = v + \frac{f_e D}{(f_e + D)} \quad ...(3)$$

In this situation as u_e is minimum MP is maximum and eye is most strained.

In case of microscope as f_0 is small and object is close to objective $u \simeq f_0$. Also as intermediate image is in front of eye lens which has very short focal length, length of tube.

$$L = v + u_e \approx v \quad [\text{as } u_e << v]$$

So for normal adjustment, i.e., from Eq. (2)

$$|MP| \simeq \frac{L}{f_0} \times \frac{D}{f_e} \quad ...(4)$$

10.4 Important Points :

(1) As magnifying power is negative, the image seen in a microscope is always truly inverted, i.e., left is turned right with upside down simultaneously.

(2) As intermediate image is between the two lenses, crosswire (a measuring device) can be used. So with microscope measurements can be made along with observations.

(3) As $m = \frac{v}{-u}$ and $m_\theta = \frac{D}{u_e}$, Eq. (1) gives MP = m m_θ

i.e., magnifying power of a microscope is equal to the product of longitudinal magnification of objective and angular magnification of eye-piece.

(4) If final image is at least distance of distinct vision

$$\frac{1}{-D} - \frac{1}{-u_e} = \frac{1}{f_e} \text{ i.e., } m_e = \frac{v_e}{u_e} = \frac{D}{u_e} = \left[1 + \frac{D}{f_e}\right] = m_\theta$$

So MP = m m_θ = m m_e = Linear magnification

(5) For a microscope MP is minimum when final image is at ∞ and maximum when final image is at least distance of distinct vision D, i.e., $(MP)_{min} = -\frac{v}{u}\frac{D}{f_e}$ and $(MP)_{max} = -\frac{v}{u}\left[1+\frac{D}{f_e}\right]$

(6) For a given microscope MP for normal setting remains practically unchanged if field and eye lens are interchanged as MP $\simeq$ [LD/f_0f_e].

(7) As the aperture of both the lenses in a microscope is small, the defects of images particularly spherical aberration is minimum.

(8) As MP for normal setting is (LD/f_0f_e), so to have large MP, f_0 and f_e must be as small as practically possible. This is why in a microscope both the lenses have small focal lengths.

(9) Out of f_0 and f_e, f_0 is taken to be smaller, so that field of view may be increased and objective being closer to object may collect more light to increase brightness of the image. This also ensures that intermediate image is in front of eye lens within its focus.

(10) While working with a microscope the eye must be close to eye-piece, i.e, persons wearing specks should take off their specks (if possible) otherwise MP will be adversly affected. This is because if eye is at a distance d from the eye-piece, the distance of final image from eye-piece will be D´ = (D – d) and hence MP will become

$$MP^1 \frac{LD'}{f_0 f_e} = \frac{L(D-d)}{f_0 f_e} = MP\left[1-\frac{d}{D}\right] < MP$$

(11) In reference to a microscope, the minimum distance between two lines at which they are just distinct is called limit of resolution and reciprocal of limit of resolution is called **resolving power** which for a microscope varies inversely with wavelength, i.e.,

$$RP^* = \frac{1}{\Delta x} \propto \frac{1}{\lambda}$$

Smaller the limit of resolution of higher the resolving power, more details of object will be visible in the image. In case of microscope RP is increased by decreasing the wavelength of light used and we have ultra-microscope and electron-microscope.

Note : In case of electron-microscope it has been shown that wavelength of electron $\lambda = \sqrt{(150/V)}$ Å where V is the accelerating voltage; so RP of electron microscope will be proportional to $\sqrt{V}$ and can be upto about 10^5 times that of optical microscope.

Example 52

A compound microscope has a magnifying power 30. The focal length of its eye-piece is 5 cm. Assuming the final image to be at the least distance of distinct vision (25 cm), calculate the magnification produced by objective.

Solution :

In case of compound microscope,

$$MP = m \quad m_\theta \qquad ...(1)$$

And in case of final image at least distance of distinct vision,

$$m_\theta = \left[1 + \frac{D}{f_e}\right] \qquad ...(2)$$

So from Eqs. (1) and (2),

$$MP = m\left[1 + \frac{D}{f_e}\right]$$

Here, $MP = -30$; $D = 25$ cm and $f_e = 5$ cm

So $-30 = m\left[1 + \frac{25}{5}\right]$, i.e., $m = \frac{-30}{6} = -5$ **Ans.**

Negative sign implies that image formed by objective is inverted.

Example 53

A compound microscope has an objective of focal length 2 cm and an eye-piece of focal length 5 cm. If an object is placed at a distance of 2.4 cm in front of the field lens, find the magnifying power of the instrument and length of the tube if (a) final image is at infinity (b) final image is at least distance of distinct vision (= 25 cm).

Solution :

As object is at a distance of 2.4 cm in front of field lens of focal length 2 cm, field lens will form its image at distance v such that

$$\frac{1}{v} - \frac{1}{-2.4} = \frac{1}{2} \quad \text{i.e., } v = 13 \text{ cm}$$

so that $m = \frac{v}{u} = \frac{12}{-2.4} = -5$

(a) If final image is at infinity (Far point)

For eye-piece, $\frac{1}{\infty} - \frac{1}{u_e} = \frac{1}{5}$ i.e., $u_e = -5$ cm

and $m_\theta = (D/f_e) = (25/5) = 5$

So, $MP = m \quad m_\theta = -5 \quad 5 = -25$

and $L = v + u_e = 12 + 5 = 17$ cm

In this situation eye is said to be relaxed and for a given microscope MP is minimum while length of tube maximum.

(b) If final image is at D (Near point)

For eye-piece, $\frac{1}{-25} - \frac{1}{u_e} = \frac{1}{5}$, i.e., $u_e = \frac{-25}{6} = -4.17$ cm

and $m_\theta = [1 + (D/fe)] = [1 + (25/5)] = 6$

So $MP = m \quad m_\theta = -5 \quad 6 = -30$

and $L = v + u_e = 12 + 4.17 = 16.17$ cm

In this situation eye is said to be strained and for a given microscope MP is maximum while length of tube is minimum.

Note : In case (b) $m_e = \frac{D}{u_e} = \frac{-25}{(-25/6)} = 6 = m_\theta$

So here $MP = m \quad m_\theta = m \quad m_e$ = linear magnification

Example 54

A compound microscope is used to enlarge an object kept at a distance 0.03 m from its objective which consists of several convex lenses in contact and has focal length 0.02 m. If a lens of focal length 0.1 m is removed from the objective, find out the distance by which the eye-piece of the microscope must be moved to refocus the image.

Solution :

If initially the objective forms the image at distance v_1,

$$\frac{1}{v_1} - \frac{1}{-3} = \frac{1}{2} \quad \text{i.e., } v_1 = 6 \text{ cm}$$

Now as in case of lenses in contact

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} + \dots \text{ or } \frac{1}{F} = \frac{1}{f_1} + \frac{1}{F'}$$

with $\frac{1}{F} = \frac{1}{f_2} + \frac{1}{f_3} + \dots$

So if one of the lenses is removed, the focal length of the remaining lens system

$$\frac{1}{F'} = \frac{1}{F} - \frac{1}{f_1} = \frac{1}{2} - \frac{1}{10} \quad \text{i.e., } F' = 2.5 \text{ cm}$$

This lens will form the image of same object at a distance v^2 such that

$$\frac{1}{v_2} - \frac{1}{-3} = \frac{1}{25} \qquad \text{i.e.,} \qquad v_2 = 15 \text{ cm}$$

So to refocus the image, eye-piece must be moved by the same distance through which the image formed by the objective has shifted, i.e., 15 – 6 = 9 cm away from the objective.

Example 55

The focal lengths of the objective and the eye-piece of a compound microscope are 2.0 cm and 3.0 cm respectively. The distance between the objective and the eye-piece is 15.0 cm. The final image formed by the eye-piece is at infinity. Find the distance of object and image produced by the objective, from the objective lens.

Solution :

As final image is at infinity, the distance of intermediate image from eye lens u_e will be given by

$$\frac{1}{-\infty} - \frac{1}{u_e} = \frac{1}{f_e} \quad \text{i.e., } u_e = -f_e = -3 \text{ cm}$$

and as the distance between the lenses is 15.0 cm, the distance of intermediate image (formed by objective) from the objective will be

$$v = L - u_e = L - f_e = 15 - 3$$
$$= 12 \text{ cm}$$

and if u is the distance of object from objective,

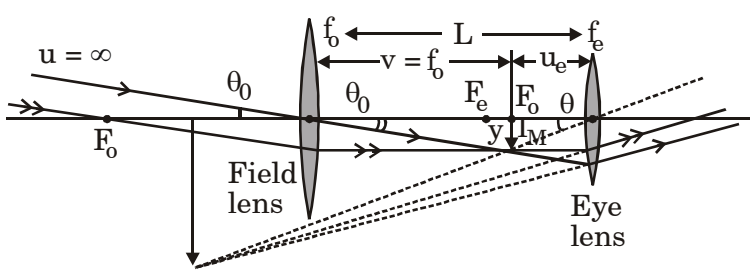
$$\frac{1}{12} - \frac{1}{u} = \frac{1}{2} \quad \text{i.e., } u = -2.4 \text{ cm}$$

So object is at a distance of 2.4 cm in front of objective.

11. TELESCOPE

It is an optical instrument used to increase the visual angle of distant large objects such as a star, a planet or a cliff, etc.

Astronomical telescope consists of two converging lenses. The one facing the object is called objective or field lens and has large focal length and aperture while the other facing the eye called eye-piece or ocular has small focal length and aperture. The distance between the two lenses is adjustable.



As a telescope is used to see distant objects, in it object is between ∞ and 2F of objective and hence image formed by objective is real, inverted and diminished and is between F and 2F on the other side of it. This image (called intermediate image) acts as an object for eye-piece and by shifting the position of eye-piece, it is brought within its focus. So final image I, with respect to intermediate image, is erect, virtual, enlarged and at a distance D to ∞ from the eye. This is turn implies that final image with respect to object is inverted, enlarged and a distance D to ∞ from the eye.

11.1 Magnifying Power (MP)

Magnifying power of a telescope is defined as,

$$MP = \frac{\text{Visual angle with instrument}}{\text{Visual angle for unaided eye}} = \frac{\theta}{\theta_0}$$

But from Fig.

$$\theta_0 = \frac{y}{f_0} \text{ and } \theta = \frac{y}{-u_e}$$

So, $$MP = \frac{\theta}{\theta_0} = -\left[\frac{f_0}{u_e}\right]$$

with length of tube $L = (f_0 + u_e)$ ...(1)

Now there are two possibilities :

(d_1) If the final image is at infinity (Far point)

This situation is called normal adjustment as in this situation eye is least strained or relaxed. In this situation as for eye-piece v = ∞,

$$\frac{1}{-\infty} - \frac{1}{-u_e} = \frac{1}{f_e} \quad \text{i.e., } u_e = f_e$$

So substituting this value of u_e in Eq. (1), we have

$MP = -(f_0/f_e)$ and $L = (f_0 + f_e)$ ...(2)

Usually, a telescope operates in this mode unless stated otherwise. In this mode as u_e is maximum, for a given telescope MP is minimum while length of tube maximum.

(d_2) If the final image is at D (Near point)

In this situation as for eye-piece, v = D

$$\frac{1}{-D} - \frac{1}{-u_e} = \frac{1}{f_e} \quad \text{i.e., } \frac{1}{u_e} = \frac{1}{f_e}\left[1 + \frac{f_e}{D}\right]$$

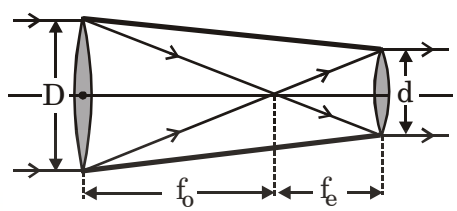
So substituting this value of ue in Eq. (1), we have

$$MP = -\frac{f_0}{f_e}\left[1 + \frac{f_e}{D}\right]$$

with $$L = f_0 + \frac{f_e D}{(f_e + D)} \quad ...(3)$$

In this situation u_e is minimum, so for a given telescope MP is maximum while length of tube minimum and eye is most strained.

In case of a telescope, if object and final image are at infinity and total light entering the telescope leaves it parallel to its axis as shown in Fig.,



$$\frac{f_0}{f_e} = \frac{\text{Aperture of objective}}{\text{Aperture of eye-piece}}$$

i.e., $$MP = \frac{f_0}{f_e} = \frac{D}{d} \quad ...(4)$$

11.2 Important Points

(1) As magnifying power is negative, the image seen in astronomical telescope is truly inverted, i.e., left is turned right with upside down simultaneously. However, as most of the astronomical objects are symmetrical this inversion does not affect the observations.

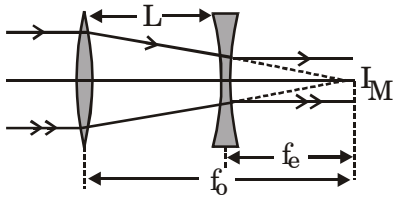
(2) For given telescope, magnifying power is minimum when final image is at infinity (Far point) and maximum when it is at distance of distinct vision (Near point), i.e.,

$$(MP)_{min} = -\left[\frac{f_0}{f_e}\right] \text{ and } (MP)_{max} = -\frac{f_0}{f_e}\left[1 + \frac{f_e}{D}\right]$$

(3) In a telescope, if field and eye lenses are interchanged MP will change from (f_0/f_e) to (f_e/f_0), i.e., it will change from m to (1/m), i.e., will become $(1/m^2)$ times of its initial value.

(4) As MP for normal setting as (f_0/f_e), so to have large MP, f_0 must be as large as practically possible and f_e small.

(5) If an astronomical telescope, the convergent eye-piece is replaced by a divergent lens which is placed in such a way that rays from objective are directed towards its focus final image



Galilean telescope

will be erect, enlarged and virtual. This telescope is also used to see distant terrestial objects and is called **Galilean telescope** and for it

$$MP = \frac{f_0}{f_e} \text{ with } L = f_0 = f_e$$

In this telescope as intermediate image is outside the tube, the telescope cannot be used for making measurements.

Question V :

How does magnifying power vary with change in length of tube for a given telescope ?

Answer : For a telescope,

$|MP| = (f_0/u_e)$ with $L = (f_0 + u_e)$

and as for a given telescope f_0 is constant, with increase in length of tube L, u_e (distance of intermediate image from eye-piece) will increase [from a minimum value $Df_e/(D + f_e)$ to the maximum f_e] and hence magnifying power will decrease.

Note : Similarly, in case of microscope as

$$|MP| = \frac{v}{u} \times \frac{D}{u_e} \text{ with } L = v + u_e$$

and as for a given microscope, with increase and hence MP will decrease.

Example 56

An astronomical telescope has an angular magnification of magnitude 5 for distant objects. The separation between the objective and eye-piece is 36 cm and the final image is formed at infinity. Determine the focal length of objective and eye-piece.

Solution :

In case of astronomical telescope if object and final image both are infinity,

$MP = -(f_0/f_e)$ and $L = f_0 + f_e$

So here $-(f_0/f_e) = -5$ and $f_0 + f_e = 36$

Solving these for f_0 and f_e, we get

So here $f_0 = 30$ cm and $f_e = 6$ cm **Ans.**

Example 57

A telescope has an objective of focal length 50 cm and an eye-piece of focal length 5 cm. The least distance of distinct vision is 25 cm. The telescope is focused for distinct vision on a scale 2 m away from the objective. Calculate (a) magnification produced and (b) separation between objective and eye-piece.

Solution :

As objective has focal length 50 cm and object is 2 m from it, it will form the image of object at a distance v such that

$$\frac{1}{v} - \frac{1}{-200} = \frac{1}{50} \qquad \text{i.e., } v = \frac{200}{3} \text{ cm}$$

with

$$m_0 = \frac{v}{u} = \frac{(200/3)}{-200} = -\frac{1}{3}$$

and as focal length of eye-piece is 5 cm and it forms as image 25 cm in front of it, the distance of object (image formed by objective) from it will be

$$\frac{1}{-25} - \frac{1}{u_e} = \frac{1}{5} \qquad \text{i.e., } u_e = -\frac{25}{6} \text{ cm}$$

with

$$m_e = -\frac{25}{-(25/6)} = 6$$

and hence

(a) Magnification $m = m_0 \quad m_e = (-1/3) \quad 6 = -2$, i.e., final image is inverted, virtual, double of object and is at a distance of 25 cm in front of eye lens.

(b) As distance of intermediate image (which is between the two lenses), from objective is (200/3) cm while from the eye-lens is (25/6) cm, so separation between the objective and eye-piece,

$$L = \frac{200}{3} + \frac{25}{6} = \frac{425}{6} = 70.83 \text{ cm} \qquad \textbf{Ans.}$$

Example 58

A Galilean telescope consists of an objective of focal length 12 cm and eye-piece of focal length 4 cm. What should be the separation of the two lenses when the virtual image of a distant object is formed at a distance of 24 cm from the eye-piece ? What is the magnifying power of telescope under this condition ?

Solution :

As object is distant, i.e., u = – ∞, so

$$\frac{1}{v} - \frac{1}{-\infty} = \frac{1}{f_0} \text{ i.e., } v = f_0 = 12 \text{ cm}$$

i.e., objective will form the image I_M at its focus which is at a distance of 12 cm from O.

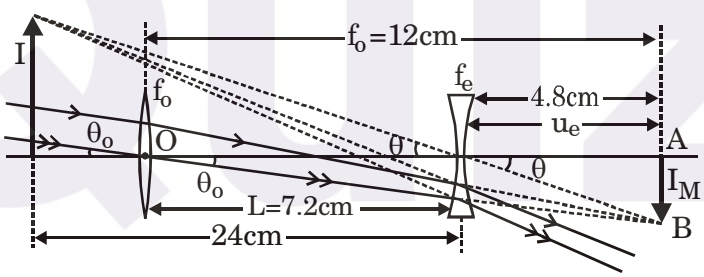
Now as eye-piece of focal length – 4 cm forms image I at a distance of 24 cm from it,

$$\frac{1}{-24} - \frac{1}{u_e} = \frac{1}{-4} \text{ i.e., } u_e = \frac{24}{5} = 4.8 \text{ cm}$$

i.e., the distance of I_M from eye lens EA is 4.8 cm. So the length of tube

L = OA – EA = 12 – 4.8 = 7.2 cm

Now by definition :

$$MP = \frac{\theta}{\theta_0} \simeq \frac{\tan\theta}{\tan\theta_0} = \frac{(AB/EA)}{(AB/OA)} = \frac{OA}{EA}$$

So, $MP = \frac{f_0}{u_e} = \frac{12}{4.8} = \frac{10}{4} = 2.5$ **Ans.**